

Inside AI

Over 150 billion purchases per year use this author's AI

Akli Adjaoute

Foreword by Raymond Kendall



 MANNING

Praise for Inside AI

“In this book, my dear friend Akli draws upon two decades of hands-on experience as the founder and CEO of Brighterion, a successful AI company (acquired by Mastercard), to bring clarity to the intricate world of artificial intelligence. Akli explores the core of AI and its optimal application, providing valuable insights grounded in a profound understanding and mastery of the AI field gained from applying it in high-stakes, mission-critical applications.”

—Raymond Kendall, Honorary Secretary General of INTERPOL

“It’s like everything you do, a labor of love, and the readers would love it.”

—Ajay Bhalla, President of Cyber & Intelligence Solutions, Mastercard

“Akli Adjaoute has spent several decades at the heart of artificial intelligence. In this book, he vividly tells us about his journey and that of a technology that is starting to profoundly change our societies. With clarity and generosity, he makes one understand what AI is and is not. Not only is this a delightful read, but also an invaluable one.”

—Patrick Pérez, CEO, Kyutai

“Having experienced the real-world impact of Brighterion, an AI company founded by Akli Adjaoute, in crucial applications, I can attest to its incredible power. This book explores applied AI through the lens of an expert in mission-critical tasks. Coupled with his academic background as an AI professor, the author is an unparalleled source for education on a transformative technology shaping our world.”

—Ian Whyte, Former Chief Risk Officer, WorldPay

“There are many books written on AI, but few that actually give readers a framework for how to think about AI and its transformational impact on everyone in the world. This is that book. It is powerful in its simplicity and admirable for its accessibility—the kind of book that will have readers posting sticky notes and highlighting passages throughout to refer to, again and again.”

—Karen Webster, CEO, PYMNTS

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AKLI ADJAOUTE

FOREWORD BY RAYMOND KENDALL



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*To my beloved wife, Nathalie,
and our wonderful children, Ghislene and Eddy,
as well as to my parents, brothers, and sisters.
I want to express my profound gratitude for your constant encouragement
and belief in my aspirations.*

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foreword

In 1999, I was in the last year of my five-year mandate as Secretary-General of the International Criminal Police Organization (INTERPOL). An American friend who thought we might have mutual interests introduced me to Akli. We met for lunch in Paris, and after the usual introductory niceties, we began to discuss the possible connections between what Akli was doing professionally and the essence of my own role at the Interpol headquarters in Lyon, France. In my previous function at Scotland Yard in London, I had worked for some period in the Criminal Intelligence Division and was naturally interested in any developments where the exploitation of intelligence analysis could be applied to Interpol databases.

I should perhaps, at this stage, indicate why my first meeting with Akli was an “experience.” During my functions at Interpol, I have met many important and impressive people, including two presidents of the United States, but Akli impressed me in a different way. His physical presence, no doubt because of his Kabyle origins, of which he is particularly proud, gives the impression of someone who is sure of himself, can be relied upon to tell you the truth, and will give authoritative opinions. Akli’s impression immediately strikes you and sets the climate in which discussions will take place. Unfortunately, circumstances did not make it possible for me to develop a project for Interpol at the time, so my continued interest in Akli’s activities became more personal than professional. However, by this time, he was establishing himself in the United States.

Akli earned a master’s degree at the University of Technology of Compiègne and a doctorate in Artificial Intelligence at Pierre and Marie Curie University. For his thesis,

Akli developed software at Necker Hospital in Paris to enable doctors to diagnose emergency patients rapidly. Artificial intelligence was known at the time but was in its infancy. The work that Akli accomplished in his five-year thesis was recognized by the press. This proved to Akli that he could build a viable future career using his special brand of AI technology.

By the time of his departure to the United States, Akli had already founded his own software company and worked with a number of well-known French companies and governmental organizations. During this time, he also taught at EPITA, a school of engineering, where he led the Department of Cognitive Science. For his last year in France, Akli had achieved certain notoriety but had already decided that the best chance for the effective development of his professional talents lay in the United States. In 1999, he took the steps that would eventually lead to his present successes in AI.

Beginning his new enterprise in San Francisco was not easy, and to a certain extent, I shared a part of the trials and tribulations with him. He was living alone without his wife and family, who were still in France. He still had a teaching obligation in Paris with EPITA, but he was trying at the same time to establish his own company, which he called Brighterion. Resources were limited, and anyone other than Akli would have thrown in the towel. It was in such difficult circumstances that Akli's strength of character came to the fore.

Akli will say that he owes persistence to his Kabyle origins. Akli is a man of principle who lives by the application of certain basic rules in his day-to-day activities, both in public and in private. Some would refer to these values as old-fashioned, including me, because we feel that they are lacking in our modern society: the notions of friendship, loyalty, and honesty seem to be lacking, particularly in the world of affairs. I am convinced that Akli owes his success to the fact that he has always applied these rules to his relations with other people. There is no doubt that Brighterion owes its present status and its future development—albeit now sold to Mastercard—to Akli and a key group of people who have been loyal to him, particularly when times have been difficult.

I have often asked Akli for explanations of what I see in the media, which never corresponds to my understanding of what artificial intelligence means. I am sure that there are many others in my situation. Therefore, I am particularly grateful to Akli for having written such a book.

—Raymond Kendall, Honorary Secretary General of INTERPOL

preface

Welcome to the world of artificial intelligence (AI), a domain where the boundaries between science fiction and reality often become indistinct. AI has captivated our collective imagination, particularly in 2022 and 2023, thanks to the release of ChatGPT. This groundbreaking product has played a pivotal role in democratizing AI usage by offering a user-friendly interface, empowering individuals without technical expertise to harness its benefits. ChatGPT boasts impressive capabilities, including answering questions, crafting narratives, composing music and poetry, and generating computer code.

For more than three decades, I've been passionate about artificial intelligence, dedicating my adult life to teaching and applying AI to address real-world challenges. In 1987, I established my first company, Conception en Intelligence Artificielle, in Paris before completing my PhD. We developed the MINDsuite platform, which seamlessly combines various AI techniques and has found successful applications in defense, insurance, finance, healthcare, and network performance. While leading this company, I also shared my expertise with students at the École Pour l'Informatique et les Techniques Avancées (EPITA), where I served as the head of the AI department and chaired the scientific committee.

In April 2000, I launched my second venture, Brighterion (acquired by Mastercard), in San Francisco. This company was founded to address the pervasive issues of payment fraud and cybersecurity, which pose significant challenges across various industries, leading to annual losses amounting to billions of dollars. Brighterion-powered software is now used by over 2,000 clients worldwide, with 74 of the largest

U.S. banks relying on its technology to safeguard against fraud and risk. Annually, more than 150 billion transactions are processed through Brighterion software.

In this book, we embark on a transformative journey to educate readers about the fascinating world of AI. Whether you're new to the field or a seasoned enthusiast, my aim is to equip you with a clear and comprehensive understanding of what AI truly is and what it can and cannot achieve. Throughout this exploration, we will discuss the expansive and multifaceted landscape of AI, marked by a diverse range of techniques and methodologies aimed at simulating human cognition.

Our journey will take us to the very heart of AI, where we'll dissect these techniques and methodologies. From the early days of expert systems to the cutting-edge advancements in deep learning algorithms, you'll gain a thorough comprehension of the full spectrum of AI techniques that drive AI applications. Along the way, we'll also explore various aspects of human cognition, including imagination, intuition, curiosity, common sense, and creativity, to illustrate that current AI techniques still fall short of replicating these qualities.

Insights from both successful and unsuccessful AI projects will demonstrate that many human jobs remain beyond the capabilities of AI and refute the notion of technological singularity, which envisions a future where intelligent robots can replicate themselves, potentially leading to the end of human civilization. As we progress, we'll also address ethical questions surrounding bias, fairness, privacy, and accountability. Drawing from my three decades of experience in developing and deploying mission-critical AI systems, I will outline the characteristics that, in my perspective, will define the next generation of AI platforms.

I firmly believe that it is crucial for every citizen to acquire knowledge about AI, given its pervasive effect on our modern world. Whether you are an aspiring AI developer, a business professional, an investor, a policymaker, or simply a concerned citizen, I welcome you to embark on this journey to discover the true essence of AI and its profound effect on our world. My hope is that, by the time you turn the final page of this book, you will not only possess the ability to discern AI reality from its illusions but also have the capacity to engage thoughtfully with the imminent AI-driven future that awaits us all.

Let the voyage begin.

acknowledgments

Numerous individuals generously dedicated their time to reviewing this book, and I am sincerely grateful for the valuable comments and suggestions received. I extend my gratitude to Raymond Kendall, a dear friend who insisted I write this book. Lucien Bourelly, a friend and former partner in my company, deserves special mention for his consistently valuable insights and encouragement.

My deepest appreciation goes to the great team of technical editors I was fortunate to have, namely Patrick Perez, Raymond Pettit, James T. Deiotte, Dick Sini, Shawn Nevalainen, François Stehlin, Florent Gastoud, Philippe Hallouin, and Philippe Perez. These exceptional experts invested their time and expertise in thoroughly reviewing my work. Through numerous discussions, their insights, suggestions, and meticulous inputs significantly contributed to refining the manuscript. Each editor brought a unique perspective and a wealth of knowledge to the table, enhancing the overall quality of the content. Their dedication to precision, helpful critiques, and collaborative approach played a pivotal role in shaping the content of the book. Throughout the different iterations leading to the final edition, their feedback acted as a guiding force, ensuring the narrative remained engaging, informative, and accessible to readers from diverse backgrounds.

The content of this book is profoundly influenced by my journey in applying AI to mission-critical applications. We encountered numerous challenges that required refining our AI algorithms, finding an efficient way to design models, and creating a storage technique suitable for storing intelligence while providing real-time responses in milliseconds to adhere to our stringent service level agreements. These agreements

demanded scalability, resilience, adaptability, explicability, and compliance. I express my heartfelt gratitude once again to François Stehlin, Florent Gastoud, and Philippe Hallouin; an extraordinarily talented team that not only believed in my venture but also stood steadfast with me through every hurdle we encountered. Their intelligence and unwavering support were instrumental in turning challenges into triumphs, and for that, I am sincerely thankful.

I would like to express my sincere appreciation and gratitude to the entire Manning team for their invaluable contributions to this project. Special thanks go to Doug Rudder, India Hackle, Alisa Larson, Jason Everett, and many others who reviewed the book at the request of Manning Publications. Their insightful contributions, made throughout various iterations, played a pivotal role in enhancing every aspect of the content.

Special thanks to Richard Vaughan, a CTO at Purple Monkey Collective, a research focused startup delivering machine learning and cloud guidance services. Richard is a highly experienced engineer who has worked across many different industry verticals and countries in a highly varied career, and worked as my technical editor on this book.

A special acknowledgment is reserved for Daniel Zingaro, whose compelling arguments and persuasive influence were crucial in deciding to incorporate a dedicated chapter on generative AI. This addition holds particular significance given the current prominence and extensive discussions surrounding generative AI in the broader field of artificial intelligence.

Finally, a big thank you to all the reviewers who provided feedback: To Alain Couniot, Alfons Muñoz, Andre Weiner, Andres Damian Sacco, Arnaldo Gabriel Ayala Meyer, Arturo Geigel, PhD, Arun Saha, Bill LeBorgne, Bonnie Malec, Clifford Thurber, Conor Redmond, Dinesh Ghanta, Georgerobert Freeman, James Black, Jamie Shaffer, Jeelani Shaik, Jereme Allen, Jesús Juárez, Kay Engelhardt, Lucian-Paul Torje, Marc-Anthony Taylor, Mario Solomou, Milorad Imbra, Mirna Huhoja-Dóczy, Piotr Pindel, Ranjit Sahai, Rolando Madonna, Salil Athalye, Satej Sahu, Shawn Bolan, Shivakumar Swaminathan, Simon Verhoeven, Stephanie Chloupek, Steve Grey-Wilson, Tandeep Minhas, and Tomislav Kotnik, your insight and suggestions helped make this book what it is.

about the book

In this book, the primary goal is to provide a comprehensive understanding of both the capabilities and limitations of artificial intelligence. We'll explore a diverse range of AI techniques, spanning from expert systems to deep learning, and emphasize the distinctions between AI and human cognition. Insights drawn from real-world AI projects not only question the notion of machines taking over the majority of human jobs but also underscore the implausibility of the technological singularity concept. Ethical considerations, including issues like bias and privacy, will be addressed. Drawing on three decades of experience in applying AI to mission-critical applications, I outline the characteristics that define the next generation of AI platforms.

Who should read this book?

This book is a comprehensive guide for anyone interested in learning about artificial intelligence, an ever-evolving field that profoundly shapes our future, influencing how we learn, work, and live.

How this book is organized

Embark on an extensive exploration of the field of artificial intelligence within the 11 chapters of this insightful book. The journey begins with an introduction to fundamental principles, encompassing algorithms and programming languages, laying a solid foundation for understanding AI. Moving beyond, chapters 2 to 4 explore various AI techniques, covering expert systems, business rules, fuzzy logic, genetic algorithms, case-based reasoning, classical neural networks, deep learning, Bayesian

networks, unsupervised learning, and smart agents. Chapters 5 and 6 shift focus to the advancements in generative AI and the comparison between human cognition and artificial intelligence. Subsequent chapters tackle diverse topics, including the limitations of AI, its impact on human jobs, and a critical examination of technological singularity. The book concludes with valuable insights from past AI projects, providing guidance for future endeavors and a visionary perspective on the next generation of AI platforms. Additionally, an insightful appendix complements the narrative by exploring the historical evolution of AI technology. Each chapter offers a unique lens into the multifaceted landscape of AI, making this book an essential read for both enthusiasts and those seeking a deeper understanding of this transformative field:

- *Chapter 1*—In the introductory chapter, we explore a range of real-world examples to showcase how AI is emerging as a pivotal force that propels positive transformations across diverse fields by enhancing efficiency and fostering innovation. Additionally, we also highlight the challenges that stem from the inherent inclination of AI algorithms and models towards errors.
- *Chapter 2*—In this chapter, we provide an overview of multiple AI techniques, accompanied by practical examples. We will explain expert systems, which rely on human expertise and inference procedures to solve problems, as well as case-based reasoning, a method that uses past experiences to tackle new challenges. Additionally, we will explore fuzzy logic as an elegant means of representing and capturing the approximate and imprecise nature of the real world. Finally, we'll conclude this chapter with an examination of genetic algorithms, which offer a powerful, straightforward, and efficient approach to solving non-linear optimization problems.
- *Chapter 3*—In this chapter, we will continue to explore various AI techniques. We'll begin with data mining, a powerful AI technique used to extract valuable information, patterns, and associations from data. Following that, we'll introduce artificial neural networks and deep learning, powerful algorithms for pattern recognition that have yielded impressive results in computer vision, natural language processing, and audio analysis. Next, we'll briefly touch on Bayesian networks, a technique that encodes probabilistic relationships among variables of interest. To wrap up the chapter, we'll explore unsupervised learning, a collection of algorithms designed to analyze unlabeled datasets and uncover similarities and differences within them.
- *Chapter 4*—In this chapter, we will introduce smart agents, a powerful artificial intelligence technique centered on the use of adaptive, autonomous, and goal-oriented entities to address complex problems. We will specifically focus on a proprietary smart agent approach, providing an illustrative example to elucidate how each agent possesses the capability to assess inputs as either beneficial or detrimental with respect to its objectives. Furthermore, we will explore the adaptability of these agents, draw comparisons with more conventional approaches,

and examine instances where this technique has been effectively employed to solve real-world challenges.

NOTE Chapters 2 to 4 contain a high-level explanation of some of the technical underpinnings of AI and can be skipped by those who want to dive into the discussion of the reality and illusion of current AI.

- *Chapter 5*—AI has witnessed numerous ups and downs, but the emergence of ChatGPT, OpenAI’s impressive chatbot, capable of composing poems, college-level essays, computer code, and even jokes, represents a pivotal moment. In this chapter, we will introduce generative AI, an impressive technology that offers a multitude of benefits across various domains and holds great potential for revolutionizing many industries. We will also examine its advantages, limitations, and the potential risks associated with the use of this technology.
- *Chapter 6*—In this chapter, we will explore various aspects of human cognition to illustrate what it means to be imaginative, intuitive, curious, and creative. We’ll show that current AI falls short in emulating these traits. We’ll compare human reasoning to AI to examine whether machines can replicate human-like thinking. Additionally, we’ll reflect on our limited understanding of the human mind. Furthermore, we will highlight that genuine comprehension is a prerequisite for vision, revealing the current limitations of AI algorithms in recognizing objects and their substantial gap in achieving human-like object and scene perception.
- *Chapter 7*—In this chapter, we will highlight that no matter how extensive the dataset or advanced the algorithms, AI programs ultimately fall short of attaining genuine intelligence. We will elaborate on the challenge AI encounters when attempting to extract true intelligence from data, as even with current AI techniques excelling in data processing, they continue to grapple with comprehending its deeper nuances.
- *Chapter 8*—We demonstrate that despite concern about AI taking our jobs, most human tasks are still out of the reach of AI.
- *Chapter 9*—The prevailing narrative often suggests that AI’s evolution will result in intelligent robots capable of replicating themselves, ultimately leading to the downfall of human civilization. While this scenario might be the stuff of compelling fiction, it doesn’t align with reality. In this chapter, we aim to debunk the notion of technological singularity as baseless and argue that our concerns should focus less on AI and more on the potential pitfalls of artificial stupidity.
- *Chapter 10*—Each AI project, whether it meets with success or faces hurdles, offers a wealth of valuable lessons. Drawing insights from these experiences empowers us to make informed decisions, steering our AI projects toward favorable outcomes while steering clear of common pitfalls. In this chapter, we will discuss insights gained from both the missteps and achievements of past AI

projects. Furthermore, we will provide valuable guidance on assembling the right team, cultivating the necessary mindset, and crafting a promising strategy for your AI project.

- *Chapter 11*—In this chapter, I use my three decades of experience in the development and deployment of mission-critical AI systems where reliability, precision, and effect are not mere goals but absolute necessities. I will describe a set of characteristics that, in my perspective, will define the next generation of AI platforms.

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about the author



AKLI ADJAOUTE is the founder of Exponion, a venture capital firm that provides cutting-edge startup companies with financial resources and expertise. Prior to Exponion, he was the founder and CEO of Brighterion, which was acquired in 2017 by Mastercard. Brighterion provides enterprise AI applications for payment service providers and financial institutions. Additionally, Dr. Adjaoute founded and led Conception Intelligence Artificielle, a company focused on AI technology. Alongside his entrepreneurial pursuits, he has shared his real-world AI experience as an adjunct professor in both France and the United States. Dr. Adjaoute has been awarded 32 patents with over 2,000 citations.

about the cover illustration

The figure on the cover of *Inside AI* is “Femme Tschouwache,” or “Chuvache Woman,” taken from a collection by Jacques Grasset de Saint-Sauveur, published in 1797. Each illustration is finely drawn and colored by hand.

In those days, it was easy to identify where people lived and what their trade or station in life was just by their dress. Manning celebrates the inventiveness and initiative of the computer business with book covers based on the rich diversity of regional culture centuries ago, brought back to life by pictures from collections such as this one.

The rise of machine intelligence

This chapter covers

- How AI is transforming industries and empowering lives
- The challenges associated with AI
- Some of the remarkable contributions of early innovators

Few ideas have captured the imagination like artificial intelligence (AI). The notion of “thinking machines” has been a theme in scientific literature and popular culture for decades. Now, AI is poised to be a transformative economic force that will amplify productivity and give birth to entirely new industries and markets. Recently, generative AI tools like ChatGPT have attracted exceptional attention and substantial investments.

While there is no doubt that AI has made incredible advancements, it is important to temper this enthusiasm with a dose of reality. Many AI applications are still in their infancy, and the road to achieving true artificial general intelligence remains long and uncertain. Separating genuine advancements from exaggerated claims is essential to harness the true potential of AI while navigating the challenges it presents. In this book, we’ll look at the current state of AI and consider where AI may be going next.

1.1 **What is artificial intelligence?**

Warren S. McCulloch and Walter H. Pitt's 1943 paper "A Logical Calculus of the Ideas Immanent in Nervous Activity" is often cited as the starting point in practical AI research. Their paper introduces a mathematical procedure for inferring relationships between propositions, and they suggested that neurons and their connections could be modeled in terms of true-or-false logical statements. Their technique led to the design of circuits that can compute Boolean functions, which are the fundamental operations of all digital computers.

In October 1950, Alan Turing published "Computing Machinery and Intelligence," which has become one of the most referenced papers in the field. Turing proposes that a machine could be considered intelligent if it passed the so-called Turing test. If the observer cannot discern a human's answers from a computer's answers, the computer would pass the test and be considered capable of thinking like a person.

The birth of the term *artificial intelligence* occurred during a workshop organized at Dartmouth College in the summer of 1956. John McCarthy, then an assistant professor of mathematics at Dartmouth, used the term for its marketing effect. He expected the catchy phrase to boost the visibility of this new research field. It would seem that he succeeded since he later became the first director of the AI labs at MIT.

Following that revolutionary workshop, the AI field experienced cycles of excitement and disenchantment. One might question whether inflated expectations and the subsequent disillusionment are, in part, attributable to the field's name. So, what precisely is the definition of AI? At its core, AI involves a set of technologies that enable computers to perform tasks traditionally associated with human intelligence. Since its inception, AI has relied heavily on two fundamental methods for algorithms to learn from data. In supervised learning, imagine yourself wanting to teach an AI system about fruits by providing labeled examples like "This is an apple; that's a banana," allowing the system to learn how to identify new fruits based on prior examples. On the other hand, unsupervised learning takes a more exploratory approach, operating without labels or guides. The AI system independently analyzes the data, grouping similar items together, such as placing all red apples in one category without being explicitly told they are apples.

AI pioneer Joseph Weizenbaum describes AI as machines that appear to behave intelligently, whether they actually employ the same mechanisms required for human intelligence. Weizenbaum sees AI in machines capable of mundane tasks like assembling nuts and bolts with the aid of synthetic vision and robotic arms as well as in computer programs that could aid in synthesizing chemical compounds, translating languages, composing music, and even writing software—all tasks possible now with current generative AI tools.

I think of intelligence as a wonderful faculty found at the root of virtually all the decisions we make. The challenge of AI is simply knowing to what extent a computer program can simulate certain intelligence reflexes and how a computer program behaves when faced with a problem for which we often ignore our own mental

operations to resolve. However, no matter how we define AI, we are increasingly able to see its effects in everyday life, shaping the way we work, communicate, and navigate the world around us.

1.2 The AI revolution

AI has undergone significant evolution over several decades, driven by a combination of human innovation in AI algorithms, advancements in hardware technology, and the increasing accessibility of data. In the early days, the concept of machines emulating human intelligence seemed like a distant dream, relegated to the world of science fiction. Little did we know that we were on the verge of a revolution that is reshaping and enhancing nearly every facet of our modern world. Indeed, today, AI brings numerous advantages to virtually every field, boosting efficiency and driving innovation in the business environment, while also making positive contributions to various human endeavors.

For example, over the last decade, we've witnessed a significant enhancement in our daily interactions, thanks to AI-driven chatbots that engage in natural conversations, providing real-time support and information. Meanwhile, smart wearables, like smartwatches, act as advanced companions, ensuring connectivity and contributing to our well-being by offering specific health benefits. In the healthcare sector, AI applications are helping clinicians in early disease detection and diagnosis. Additionally, AI-powered robotic surgery systems contribute to precision in medical procedures, reducing recovery times and improving patient outcomes. Telemedicine platforms utilize AI for remote patient monitoring and personalized health recommendations, making healthcare services more accessible and tailored to individual needs. The transformative effect of AI extends to individuals with disabilities, leading to a new era of inclusivity and empowerment. Voice recognition software empowers those with mobility impairments to control devices and communicate more effectively, while AI-driven prosthetics enhance mobility and independence for individuals with limb disabilities.

In the business world, AI stands as a cornerstone of modern business strategies, offering a plethora of benefits that drive efficiency, innovation, and competitiveness. AI enables businesses to extract valuable insights from massive datasets, informing strategic decision-making and forecasting trends. Automation powered by AI streamlines routine tasks, reducing operational costs and freeing up human resources for more complex and creative endeavors. In supply chain management, AI optimizes inventory and logistics, minimizing disruptions and improving overall efficiency. In the financial industry, AI emerges as a potent asset in the continuous combat against financial crimes, with a particular emphasis on fortifying efforts in fraud prevention and money laundering detection.

A breakthrough and impressive milestone in the evolution of AI is the advent of generative AI, with powerful products such as ChatGPT playing a pivotal role in democratizing access to AI. Generative AI has emerged as a transformative force across various

industries, offering a multitude of benefits that extend creativity, efficiency, and innovation. The release of ChatGPT has empowered billions of nontechnical individuals, providing them with the ability to harness the power of AI without requiring specialized technical knowledge. It is also becoming instrumental in facilitating global business expansion by empowering organizations to generate relevant and effective content without the immediate need for human support as these advanced language models excel at generating coherent text and adapting to specific communication styles.

The advent of AI-powered automatic translation has become a catalyst for global connectivity, breaking down language barriers, fostering cross-cultural communication and creating a more interconnected world. With sophisticated natural language processing algorithms, AI-driven translation tools enable real-time language interpretation, facilitating seamless interactions between individuals and businesses from diverse linguistic backgrounds. This advancement not only enhances international collaboration but also promotes a deeper understanding of different cultures and perspectives.

In the field of education, AI is revolutionizing learning experiences through personalized tutoring systems and adaptive learning platforms. Virtual tutors and AI-based educational assistants will offer additional resources, answering questions and providing guidance to students both inside and outside the classroom. AI is poised to bridge educational gaps and democratize access to quality learning resources, ensuring that education becomes a dynamic and inclusive process for learners worldwide.

AI technologies have greatly strengthened global security, making the world safer in ways we've never seen before. They play a vital role in detecting and preventing threats, especially in cybersecurity and public safety. In law enforcement, AI-powered surveillance quickly identifies and captures individuals involved in criminal activities, improving overall public safety. In homeland security, AI is a crucial asset for reinforcing defenses and protecting nations, creating a safer environment for citizens, and ensuring border integrity.

The previously mentioned instances represent only a portion of the domains where AI is delivering significant benefits. The list of sectors experiencing positive effects from AI is continually broadening, and I firmly believe that virtually every field stands to gain from its capabilities. Nevertheless, it's essential to recognize the paradox—while AI offers opportunities for substantial advancements, it also introduces the risk of misuse and challenges within those same areas.

1.3 *Error-prone intelligence*

The deployment of AI facial recognition technology raises significant concerns, especially regarding biases and ethical considerations. Instances of racial and gender bias in facial recognition systems have been well-documented, raising serious ethical questions about how biased technology might perpetuate societal inequalities. Similar biases have been observed in AI used in automated hiring systems, inadvertently favoring certain characteristics and contributing to disparities that further marginalize underrepresented

groups. Furthermore, incorporating AI into the criminal justice system raises concerns about potential biases, fairness, and the transparency of the sentencing process. AI algorithms that analyze various factors to predict recidivism or assess the likelihood of reoffending may inadvertently sustain existing biases in the system, presenting challenges in ensuring a fair and just legal system.

In the healthcare sector, AI is susceptible to errors, raising substantial concerns about possible misdiagnoses or inaccurate treatment recommendations. The opacity of AI models, as we will discuss in chapter 3, adds an extra layer of concern, making it challenging for users to understand the process behind arriving at specific conclusions.

The use of AI in self-driving cars raises worries about errors causing accidents, especially due to algorithmic issues or malfunctions. The ever-changing and unpredictable nature of real-world traffic situations makes it challenging for AI systems to accurately understand and respond to complex scenarios. A recent tragic incident in San Francisco in October 2023 serves as a devastating example, where a self-driving car failed to detect a pedestrian trapped beneath it, dragging her about 20 feet.

The rapid advancement of AI in social media platforms could worsen its negative effect by fueling the spread of false information, an increase in cyberbullying, and an elevated risk to individual mental well-being. Additionally, the growing use of deepfake technology, allowing the creation of convincingly fabricated content like realistic videos and audio recordings, has the potential to deceive and manipulate individuals.

While large language models (LLMs) achieved an impressive AI milestone by democratizing AI for the larger public, they also introduced potential drawbacks, particularly the risk of overreliance and blind trust by nontechnical users in every output produced by these AI models. In chapter 5, we will demonstrate that LLMs can generate text that is seemingly coherent but factually inaccurate and sometimes entirely fictional. As AI becomes increasingly prevalent in various sectors, the likelihood of misuse and accompanying challenges rises when the technology is applied improperly.

Finally, let's consider another issue stemming from AI. This concern is not linked to errors in AI algorithms but paradoxically arises from the technology's efficiency in specific tasks. Corporate executives and economists anticipate the potential loss of numerous jobs due to the influence of generative AI, causing an effect across various industries. Even sectors like software and AI engineering are at risk, as generative AI speeds up productivity, potentially reducing the demand for other engineering roles.

As AI rapidly evolves, it doesn't just change how we work but transforms how we live, connect, and shape our societal identity. To understand this effect, let's look at how AI is transforming one of the most common human-to-human interactions—the conversation.

1.4 Chatbots

Chatbots, also known as “conversational agents,” are programs that use AI and machine learning to provide technology for human–machine interaction. This approach was introduced in 1966 by Joseph Weizenbaum with his program ELIZA, which was created to provide person-centered therapy. ELIZA is the ancestor of ChatGPT, Siri, Alexa, and Google Assistant, which are now omnipresent in smartphones and home devices.

ELIZA parsed sentences looking for words such as *I*, *you*, *like*, and *mother*. Once ELIZA identified one of these words, it responded by changing the sentence in accordance with a rule associated with that word. ELIZA would then give a content-free response or reference the previous topic for input text such as “Why do you think that?” which contained no keywords. In his paper, Weizenbaum gave a slightly contrived example of a conversation between ELIZA and a female test subject:

Subject: Men are all alike.

ELIZA: IN WHAT WAY?

Subject: They’re always bugging us about something or other.

ELIZA: CAN YOU THINK OF A SPECIFIC EXAMPLE?

Subject: Well, my boyfriend made me come here.

ELIZA: YOUR BOYFRIEND MADE YOU COME HERE?

Subject: He says I’m depressed much of the time.

ELIZA: I AM SORRY TO HEAR YOU ARE DEPRESSED.

To Weizenbaum’s dismay, some users came to believe that ELIZA actually understood what it was saying. A few psychotherapists even thought it might be a helpful tool for their practice. In reality, all the program could do was decompose the user’s input into its constituent parts of speech and repeat them back in a manner that sustained the conversation. Weizenbaum was so disturbed that users could put their faith in a relatively simple program that he became an advocate for social responsibility. He became a vocal critic of any AI that gave the illusion of intelligence.

The current generation of AI-driven chatbots is much more sophisticated. Powered by LLMs trained on massive amounts of text and other data, tools like OpenAI’s ChatGPT, Google’s Bard (now named Gemini), and others are capable of providing very human-like responses when prompted in natural language. For example, consider this brief exchange with ChatGPT:

User: Give me a short definition of artificial intelligence.

ChatGPT: I think this will look better if the text is left-aligned under the first line of spoken text. Not under Chat GPT, but under Artificial Intelligence (AI) refers to. Same instructions for the ChatGPT paragraph on the following page.

User: Based on this definition, are you intelligent?

ChatGPT: No, I am not intelligent in the way humans are. I am a machine learning model developed by OpenAI. While I can process and generate human-like text based on patterns learned from vast amounts of data, I don't possess true consciousness, self-awareness, or subjective experiences. My responses are generated based on patterns and information present in the data on which I was trained, without true understanding or awareness.

AI chatbots have already become a frontline approach to customer service operations in a number of major corporations, a trend that's likely to expand rapidly. It seems inevitable that AI will transform other traditionally human-to-human communications such as email, phone-based voice interactions, and ordering lunch at the local drive-thru restaurant.

1.5 Looking ahead

As we stand on the brink of a new era, it is crucial for business leaders to not only recognize the significance of AI but also to understand that embracing AI is not merely a choice; it is an imperative for those who seek to lead in an era defined by innovation, efficiency, and intelligent decision-making. The AI journey is far from complete; it is an ongoing expedition marked by continual exploration, refinement, and adaptation to the intricate interplay between technology and humanity.

Drawing on my firsthand insights from real-world business challenges, the goal in the chapters ahead is to provide you with practical insights into the applications of AI within a business context. As we advance through the next 10 chapters, gaining insights into various AI approaches, we will learn how to seamlessly integrate their continually expanding capabilities. This will help you navigate the complexities of integrating AI into your own enterprises, ensuring that you are well-equipped for the transformative journey that lies ahead.

Summary

- AI positively influences diverse fields, enhancing efficiency, fostering innovation, and positively affecting human endeavors.
- Generative AI, as exemplified by ChatGPT, marks a historic moment in the field of AI by democratizing its use. It empowers nontechnical individuals and small businesses, allowing them to harness the power of AI.
- Deployment of AI raises concerns about biases in various fields.
- AI is susceptible to making errors that may cause harm.
- The origins of AI can be traced back to 1956 when researchers convened at the Dartmouth Conference to explore the possibilities of creating intelligent machines.
- AI techniques can be categorized into two main types: supervised learning, where the algorithm is trained with labeled data to associate inputs with specific

targets, and unsupervised learning, which works with unlabeled data to discover patterns.

- The first chatbot, ELIZA, was created in 1966 with a primary focus on therapeutic interactions.



AI mastery: Essential techniques, Part 1

This chapter covers

- An introduction to expert systems
- An introduction to business rules management system
- An introduction to case-based reasoning system
- An introduction to fuzzy logic
- An introduction to genetic algorithms

This chapter and the next explore various AI techniques that empower computers to emulate human cognition—from expert systems that capture, store, and allow us to reuse valuable expertise, mastered only by a limited number of experts, to the business rules that systematize decision-making. We will learn about case-based reasoning (CBR), which uses analogy to solve problems by reusing knowledge learned from past experience, and fuzzy logic, which handles uncertainty and navigates ambiguity. We will learn how genetic algorithms solve complex problems when the solution space is large and how data mining, like mining for gold, mines data to uncover hidden and valuable insights. We will also explore the awe-inspiring world of neural networks and deep learning, where machines learn from large sets of data. Lastly, we'll examine unsupervised learning, where algorithms discover hidden

patterns from unlabeled data. Each technique has its own strengths and weaknesses, which is why real-world applications combine the power of multiple techniques.

I do my best to avoid jargon and formulas, but I will still introduce a few technical terms you will likely encounter when reading about the subject. You can skip this chapter and the next without loss of continuity if you are only interested in learning the story of AI or developing an educated perspective on its potential.

2.1 **Expert systems**

In many fields, experts are rare, as it takes years of study and real-world experience to become an expert in any given field. It would be highly beneficial if we could capture, store, and share the valuable expertise possessed by this limited number of experts. Furthermore, a large number of experts retire regularly, taking with them their extensive knowledge and critical business experiences. For instance, when 25 experts retire, they take over 1,000 years of collective experience with them.

Wouldn't it be great if we could find an intelligent way to create virtual experts in various fields to efficiently solve problems without necessarily involving the actual experts? For example, imagine you want to do some gardening, and you're trying to select the right plants for your climate, soil, and sunlight conditions. You may ask someone who is a gardening expert to suggest the best plants for your situation. Now, suppose we could extract the expert's knowledge about plant types, soil conditions, weather patterns, and everything related to gardening to design a "gardening digital expert" software. This example illustrates the purpose of expert systems, which are computer-based systems designed to mimic the decision-making abilities of human experts in specific domains.

While traditional procedural systems combine algorithms and knowledge in a single program, expert systems separate the domain-specific know-how from the procedural methods used to execute the program. An expert system comprises three main modules (figure 2.1):

- A knowledge base that contains the domain expertise and is stored as a collection of simple `if-then` statements rules. This module acts as a repository of the knowledge accumulated by many experts.
- A working memory that contains the data inputs and tracks the progress of what has been deduced by the inference engine.
- An inference engine, which is a computer program that runs in a loop. At each cycle, it evaluates the facts in the working memory against the rules. When the conditions of a rule are met, that rule may generate new facts, modify existing ones, and/or stop the loop. The loop automatically stops if no more rules can be triggered with the current facts.

This separation of components brings many benefits. It allows people without a programming background, such as doctors, traders, underwriters, and compliance experts, to share their expertise in the knowledge base. This setup also facilitates the

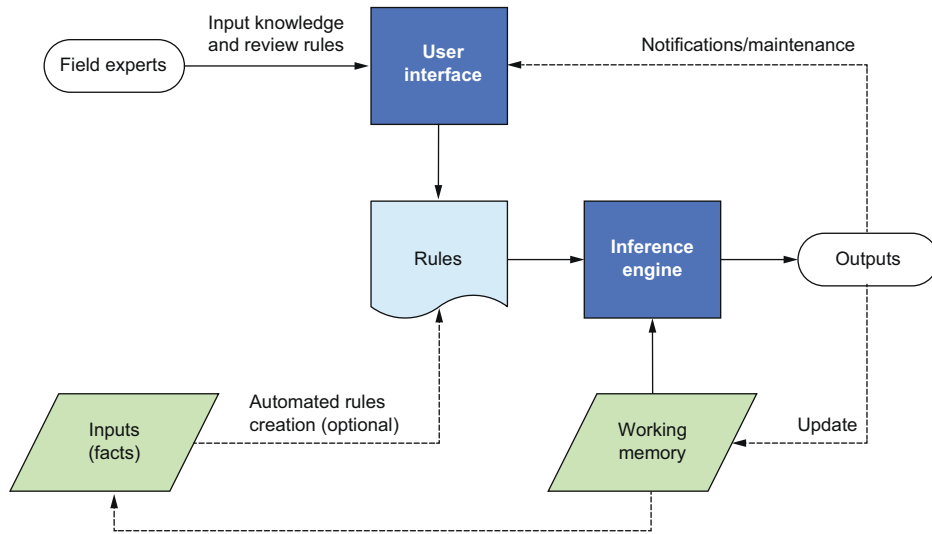


Figure 2.1 Flowchart of an expert system

expert system's enrichment and maintenance since knowledge can easily be added, updated, or removed from the knowledge base without modifying the underlying code.

In 1972, Edward H. Shortliffe created MYCIN [1], one of the first expert systems. It was used experimentally to diagnose bacterial infections, and it made inferences from input information regarding bacterial organisms, infection sites, and the presence of specific clinical signs such as fever or headache. There are more than 500 if-then rules contained in the MYCIN knowledge base. The following is one of the rules used by Mycin:

```

Rule 20
IF substrate of culture is blood
AND stain of organism is gram negative
AND morphology is rod
AND patient has been seriously burned
THEN likelihood is 40% that organism is pseudomonas
  
```

As you might deduce, the straightforwardness of the if-then-else format is vital. You can likely envision experts crafting such rules without requiring specialized training. The following example is a rule from a different expert system that aimed to enhance plant safety and diagnose system faults [2]:

```

Rule 27
IF Condenser Temperature Increase
AND Steam Temperature Increase
AND Condenser Pressure Rapid Increase
AND Condensate Level Increase
THEN Emergency Pump is Idle
AND DISPLAY Condensate Pump Break Down and Emergency Pipe Idle
  
```

One of the pioneering commercial expert systems was eXpert CONfigurer (XCON). Digital Equipment Corporation (DEC), a computer supplier, faced the challenge of managing a diverse range of components that experienced frequent changes and could be configured in numerous ways. In the early 1980s, before the assembly and configuration of computer systems commenced, orders from customers were manually verified to ensure they contained all the required components and nothing extra. Despite these manual checks, errors in the configuration process occasionally persisted, necessitating corrective actions. Therefore, computers were reevaluated at DEC's final assembly station before being shipped to customers. XCON was introduced, utilizing its knowledge base and a set of rules to validate the layout of each computer. This innovation enabled DEC to ship most components directly to the customer's site for final assembly, streamlining the process.

Another notable system emerged in 1983 from General Electric: the expert system for Diesel Electric Locomotive Repair (DELTA) [3]. DELTA was designed to encapsulate the extensive knowledge of David I. Smith, one of GE's senior field service engineers. Smith was renowned as a foremost expert in locomotive engines and repair, often traveling across the country to mentor small groups of apprentices. DELTA is comprised of over 500 rules tailored for troubleshooting and assisting in locomotive maintenance, making it a valuable tool in the field.

In conclusion, it's essential to understand that developing an expert system in a specific field is a continuous journey involving multiple design stages and thorough performance testing. This process is akin to crafting a fine-tuned instrument, where each iteration refines its capabilities. During this dynamic development process, several crucial activities come into play. First, formulating new rules may be necessary to encompass a broader range of scenarios and enhance the system's accuracy. These fresh insights enable the system to adapt to evolving conditions and challenges. Second, constant review and revision of existing rules is imperative. As the expert system interacts with real-world data and encounters diverse situations, it gains the capacity to fine-tune its responses. Review and revision entail regularly updating rules to maintain alignment with current practices and knowledge in the field. It's also critical to recognize that not all rules remain relevant indefinitely. Over time, some rules may become outdated or less effective due to changes in the field's dynamics or technological advancements. Consequently, part of this iterative process involves identifying and removing outdated rules to maintain the system's efficiency and accuracy.

2.2 Business rules management system

If you have ever used a system such as TurboTax for your income tax return, you have used an application powered by a *business rules management system* (BRMS). BRMS represents an advancement over the concept of an expert system and is widely used in many fields. For instance, a financial institution can use this technique to determine the eligibility of loan applicants based on factors such as credit scores, income levels,

and debt ratios. This ability enables them to streamline and automate the decision-making process.

One example of a rule is “If the applicant has a credit score above a specified threshold and maintains a stable income, then the applicant should be automatically approved.” In the healthcare sector, a BRMS is crucial in helping hospitals manage patient admissions efficiently by automating decision-making processes based on, for example, bed availability, patient conditions, and medical protocols.

Business rules can also be applied to self-driving vehicles to ensure safety. A set of rules can be designed to comply with traffic rules and regulations, including speed limits, stop signs, traffic lights, and right-of-way rules. Additionally, rules can be established to dictate how self-driving vehicles respond to specific behaviors and situations, such as prioritizing pedestrians and cyclists over other vehicles and handling crosswalks, lane changes, merging, road closures, detours, weather and road conditions, visibility, traffic congestion, and unexpected stops.

BRMSs enable organizations to easily centralize, manage, and automate complex decision-making. In technical terms, a BRMS consists of several key components:

- *A rule repository*—This component serves as the repository for business rules, where rules are defined, organized, and efficiently stored.
- *A rule-authoring environment*—The rule-authoring environment provides user-friendly interfaces that facilitate rules creation, modification, and testing. It supports rule development by business users or analysts.
- *The rule coherence module*—This crucial component is designed to prevent contradictions between rules within the system. Its primary purpose is to assess newly added or modified rules to ensure they do not introduce conflicts or inconsistencies with existing rules, which could lead to confusion or unintended consequences in decision-making processes.
- *An inference engine*—The inference engine is the computational module responsible for executing the defined rules. It processes and applies the rules to make decisions or automate actions within business processes.
- *Rule governance*—This system is crucial for managing rule versions, tracking changes, and ensuring compliance with regulatory requirements and internal standards. It helps maintain a history of rule modifications and their effects.
- *An analytic tool*—An analytic tool is used to monitor and analyze the effects of rules on business processes. It provides insights and data-driven information to organizations, enabling them to refine and optimize their rule sets. This optimization enhances decision-making and overall operational efficiency. This module includes “what if” and “what if not” simulation capabilities that offer valuable insights into the potential benefits or drawbacks of rule adjustments, helping organizations make informed decisions and optimize their business rules to achieve desired outcomes while mitigating unintended consequences.

The advantages of this technique include its flexibility and simplicity so that a non-technical person can easily add, update, or remove any rules. It's like having a rule-book that adapts to your business needs, making it easier to maintain and modify your rules as circumstances change.

The inference logic for business rules is essentially similar to that of early expert systems, and it possesses several limitations. For instance, a significant portion of human expertise cannot be adequately encapsulated within concise snippets of logic that can be transformed into `if-then-else` rules. Even in cases where it appears feasible, it remains exceedingly difficult to fully extract the comprehensive knowledge of a proficient expert and effectively transfer their expertise to a computer.

I faced this challenge during my PhD research at the Centre Informatique Necker Enfant Malade (CINEM) at Necker Hospital in Paris. My objective was to develop a simulation program to instruct medical students in clinical reasoning. The program was designed to assist them in preparing for their clinical experiences and avoiding errors when providing patient care.

Clinical reasoning is a skill that requires years of practice to master, and much of the expertise and knowledge is challenging to extract through interviews. A significant portion of a doctor's expertise and skill is implicit and challenging to articulate. Even when medical experts can explain their clinical reasoning, it remains a formidable task to outline most of the underlying logic and causality. Rules-based medical decision-making can involve numerous complex questions:

- How should we categorize contextual factors such as urgency, age, or pre-existing conditions, and how do they affect the consequences of a particular course of action?
- How can we establish a coherent, noncontradictory, and meaningful set of rules that addresses diverse contexts and scenarios?
- How do we represent clinical experience and connect it to factual information about anatomy, pathology, and organ systems, which form essential background knowledge for a given case?
- How can we adapt to changing behavior or shifts in the distribution of data?

2.3 **Case-based reasoning**

Wouldn't it be great if we could find an intelligent way to store and reuse the expertise gained in any particular area? Reusing expertise is important because it affects our ability to remember, reason, and solve problems. For instance, suppose we can write a program that reuses the expertise gained by a doctor in medical diagnosis by reusing a library that contains past patient cases to treat new patients with similar symptoms. The program will first retrieve similar cases in which patients had similar symptoms and conditions by comparing the important symptoms, the medical history, age, gender, and other contextual factors. The program will also be able to adapt and

personalize the treatment to the unique characteristics of the new patient. Based on the outcome, we will add the knowledge and specificities of this new patient to our “expertise library.” Our expertise should also contain failed outcomes to avoid making the same errors. The more we update our library, the more efficient it becomes in treating patients.

This example describes the CBR approach, an AI technique that involves solving new problems by referencing and adapting solutions from similar past experiences. For each new problem, the first step is to search, in the case library, for cases similar to the new problem. Once one or more similar cases are found, the system will assess whether any existing solutions or their combination can be used directly or whether adjustments need to be made to fit the new problem based on the similarities and differences between the new problem and the similar cases. Once the new problem is solved, the adapted solution and the data specific to the new problem are added to the library for future use. This case enrichment improves the efficiency of the CBR performance over time.

CBR is an AI technique that uses past experiences to address new and similar problems. It involves a series of three steps:

- 1 *Retrieve*—In this step, the CBR identifies and retrieves relevant cases from its case library that are similar to the current problem.
- 2 *Reuse*—Once relevant cases are retrieved, the CBR system will assess the applicability of the knowledge stored in these cases. This step often involves adaptations and adjustments to make the solution fit the current problem, as the resolutions used in past cases may not directly apply to the current problem.
- 3 *Revise*—After solving the current problem, the revised solution is added to the case library. As more cases are solved and inserted into the case library, the CBR system becomes more proficient at solving problems within its domain. This adaptive and knowledge-driven methodology makes CBR particularly valuable in technical support, troubleshooting, and maintenance tasks.

To illustrate these steps, let’s contemplate the following scenario. Say you take your vehicle to an automotive repair shop for maintenance. Upon arrival, the mechanic initiates a conversation to gather information about the problems you’ve been experiencing. You might mention that your car exhibits vibrations, emits a loud exhaust noise, and carries a noticeable odor of burning oil. At that moment, the mechanic promptly taps into their recollection of similar cases from the past to identify the underlying problems affecting your vehicle. Each task they tackle contributes to strengthening their expertise, enhancing their understanding, and refining their skills along the way.

An important advantage of CBR over expert systems and BRMSs is the absence of the need to write explicit rules. Unlike expert systems, which require time-consuming rule formulation, CBR leverages real-world cases and their associated solutions to

solve new problems. This feature substantially reduces the knowledge engineering effort and makes CBR an efficient technique in the various domains where codifying expert knowledge into explicit rules is challenging. Moreover, CBR's capacity to learn directly from experience not only simplifies knowledge acquisition but also enhances its adaptability, making it a valuable technique for dealing with constantly changing situations.

Since CBR relies on past cases to solve problems, it may face challenges when confronted with issues lacking close analogies in the case base. Furthermore, storing and retrieving cases can become computationally intensive, particularly as the case base expands over time. These factors can significantly affect the real-time performance of CBR systems.

2.4 Fuzzy logic

In real life, situations often lack straightforward, easily identifiable answers that are definitively right or wrong. Instead, they usually encompass a variety of potential outcomes and factors to consider. Take medical diagnosis, for example. Not only do patients experience and describe symptoms differently, but when doctors inquire about issues like headaches or fatigue, the answers aren't always a simple *yes* or *no*. Patients often use terms like *sometimes*, *usually*, *rarely*, and *regularly* to convey the varying degrees of their symptoms. Furthermore, the perception and labeling of symptoms, using words such as *none*, *mild*, *moderate*, *severe*, and *intense*, can vary widely among patients due to unique physiology, psychology, past experiences, pain thresholds, and tolerances. These variabilities lead to diverse descriptions and ratings of pain, and the resulting complexities cannot be adequately represented using a binary logic system limited to *present* or *absent*. Fuzzy logic, on the other hand, provides a more adaptable and efficient approach to capturing these nuances and subtleties.

Fuzzy logic, which might have been given the better name “flexible logic,” is a mathematical framework for dealing with uncertainty and imprecision in decision-making and control systems. Lotfi Zadeh [4] pioneered this innovative approach in the 1960s to surmount the constraints of classical logic, permitting a nuanced measure of truth between the binary states of “completely true” and “completely false.” Fuzzy logic relies on the concept of fuzzy sets to represent partial membership and captures the gradual transition between sets. For example, when defining tall and short people, fuzzy sets recognize that height is a continuum, and individuals can belong to both sets simultaneously to varying degrees.

To say that a person is tall, classical logic would require that we specify a height h , and the statement that they are tall would be true or false depending on whether their height was greater than or less than h . For example, suppose we decide that a person is tall if they are at least 6 feet, 2 inches in height. Statements about hypothetical people's heights can then be evaluated using classical logic by assigning binary truth values (yes or no, 1 or 0, true or false) as demonstrated in table 2.1.

Table 2.1 Determining the values of tall and short based on height

Name	Height	Tall	Short
Gary	6'1"	0	1
Joe	6'3"	1	0
Sandy	5'6"	0	1
Sue	6'1"	0	1

The membership of each person in the set of tall people and the set of short people is indicated in the third and fourth columns, respectively. Given our benchmark of 6 feet, 2 inches, it is evident that Joe is tall and not short and that Sandy is short and not tall. The 1s and 0s in the table indicate as much, and these cases seem clear-cut. However, it doesn't seem right intuitively to classify Sue and Gary as short when they are almost the same height as Joe, especially when their heights are compared to Sandy's. We might feel that what applies to Joe should also apply to Gary and Sue, but traditional logic requires that a person be either tall or short, with no option in between.

Let's consider another example. Suppose that a bank is using a set of rules to decide whether an applicant will receive a loan, and one of the rules states:

```
IF "X has a bachelor's degree" is TRUE
AND Number of years employed ≥ 2
AND income ≥ $100,000
THEN "Loan approved" = TRUE
```

The rule essentially says that applications for customers with a college degree who have had a job for two or more years and who have a high income (more than \$100,000) are approved. Let's suppose that an applicant has a master's degree and their income is more than \$200,000, but they have only been employed for one year, 11 months, and 27 days. In this case, the applicant would be declined. Of course, this loan rejection does not make sense in real life, and such strict thresholding works against the bank and the borrower. A decision based on fuzzy logic would allow the bank to build a relationship with a great potential customer since it would recognize that one year, 11 months, and 27 days is close enough to two years.

Fuzzy logic assists in handling data that is uncertain, imprecise, vague, incomplete, and noisy. It became the basis for modern technology used in a variety of situations, including facial recognition, air conditioning, washing machines, car transmissions, weather forecasting, and stock trading. As humans, we use fuzzy logic all the time without knowing it. When we park our car, we do not think about turning the wheel 20 degrees to the left and backing up at two miles per hour. Instead, we think about turning the wheel slightly to the left, then perhaps a little more to the right, and backing up slowly. A fraud-prevention expert might formulate a rule that says, "When the number of cross-border transactions is high and a transaction occurs in the evening, then

that transaction might be suspicious,” or a manager at a chemical plant might say, “If the temperature in the storage tank is getting too high, lower the pressure.”

None of these scenarios can be properly described with binary distinctions like true/false, yes/no, or 0/1. Fuzzy logic benefits many domains by dealing with nuanced inputs instead of binary.

Fuzzy logic offers several advantages in various fields thanks to its capacity to address uncertainty and imprecision, making it well-suited for systems characterized by ambiguity. Additionally, fuzzy logic excels in managing noisy data and remains efficient even when dealing with incomplete or inaccurate information. This technique further simplifies complex control systems, enhancing their intuitiveness in both design and comprehension. This renders fuzzy logic valuable in numerous applications, ranging from industrial control and robotics to medical diagnosis and natural language processing.

Fuzzy logic, while powerful in many applications, has its limitations. One key challenge is its potential to generate results that are challenging to understand or explain in everyday language. This lack of clarity can be particularly problematic in critical areas like healthcare, where transparent explanations are essential. Another limitation lies in the complexity of selecting the appropriate membership functions and parameters. Making incorrect choices in this regard can significantly affect the performance of the fuzzy logic system. Consider, for example, determining the hotness or coldness of something within a fuzzy logic system; these decisions can be subjective and often rely heavily on expert knowledge.

2.5 **Genetic algorithms**

Suppose we want to find the most efficient way to deliver products from a large manufacturer to retailers. We need to optimize resource allocations, select the timing, the best routes to take, and many other factors that need to be optimized. Solving this supply chain problem with a genetic algorithm involves many potential solutions (the population) that include different routes, distribution plans, and scheduling methods. We should select a method to evaluate (the fitness evaluation) these potential solutions based on delivery time, cost, and other factors. The next step is to sort the solutions based on how they performed in terms of speed, cost, and other factors and select (the selection) a group of these solutions that need to be combined (crossover) to find the best solution.

The idea is to combine parts of two good solutions to create a better one. For instance, we could combine the faster route from one solution with the resource allocation of another to create a new, better solution. Occasionally, making a random change to a solution (mutation) can lead to entirely new strategies and solutions. Once the crossover and mutation have been performed, we will have a new population that should be better, as they are the results of combining the best potential solutions. We repeat this process over multiple generations until no improvement is created or after a specific number of cycles.

This example describes the genetic algorithm approach. This technique starts with a pool of potential solutions, evaluates them, combines the better ones, introduces randomness to search new options, and repeats this process until we find the most efficient solution to our problem.

Genetic algorithms are optimization algorithms. Invented by John Holland in 1960 and improved over the years, they are inspired by the Darwinian principle of natural selection that suggests that all species share a common ancestor and that the species evolve over time. Darwin also introduced the concept of natural selection, which states that the members of a population best adapted to their environment are more likely to survive and generate offspring. The beauty of a genetic algorithm lies in its simplicity. Rather than solving a problem by exhaustively analyzing a possibly astronomical number of possibilities, it generates potential solutions to the problem, evaluates their performance on a scoring system, and evolves them toward some ideal. The worst-performing candidate solutions are discarded, the best ones are kept, and new ones are created by slightly modifying the top performers. The new variants are evaluated against the existing ones, and this process continues until a solution with suitable performance is obtained.

In his more formal description, Holland related his algorithms to natural selection by employing apt terms such as *crossover*, *fitness*, and *mutation*. For the analogy to hold up in a particular situation, though, we must find a good objective function to evaluate the randomly generated solutions, and we need an adequate population size, a suitable mutation rate, and an effective crossover procedure to create new candidate solutions from existing ones.

A genetic algorithm generally involves the following steps, as shown in figure 2.2:

- 1 Choose an initial population. Each population member represents a possible solution to our problem.
- 2 Evaluate each individual in the population with the chosen objective function and assign it a fitness score.
- 3 Eliminate individuals with low scores.
- 4 Create new individuals by mutating or combining copies of the ones with the highest scores.
- 5 Add the new individuals to the population.

Repeat steps 2 through 5 until a prescribed amount of time has passed, a predetermined number of generations has been tested, or the process stops producing individuals with higher fitness. The member of the population with the highest fitness score when the process ends is the solution to the problem.

Note that, in theory, better solutions might still exist, so genetic algorithms are most suitable for problems admitting good-enough solutions, for which it suffices to attain a satisfactory but not perfect performance.

Genetic algorithms have many advantages over classical optimization techniques. First, they are relatively easy to understand, yet they can be used to address extremely

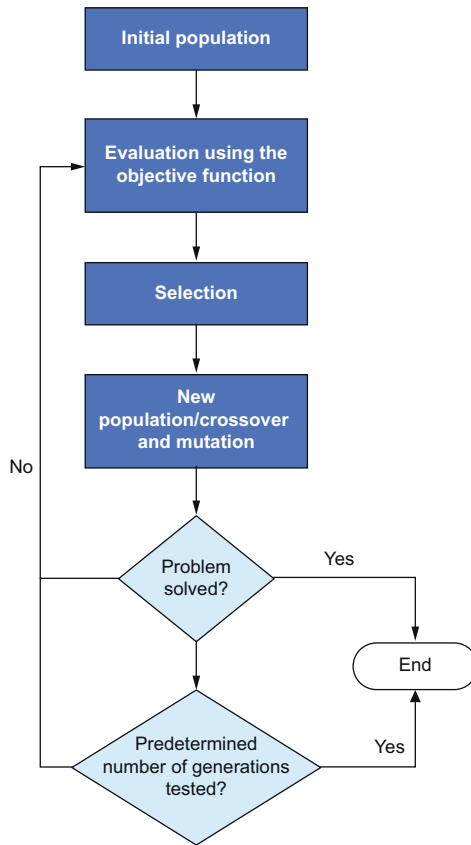


Figure 2.2 Flowchart of a genetic algorithm

complex problems. Second, given enough time, they will always provide a near-optimal solution, even when the perfect solution is unavailable. Third, they are well suited to parallel computing, an approach in which many computations are performed simultaneously, which allows for rapid solving.

In 1992, Holland summarized the concept of a genetic algorithm and touted its potential in a *Scientific American* article [5]:

Pragmatic researchers see evolution's remarkable power as something to be emulated rather than envied. Natural selection eliminates one of the greatest hurdles in software design: specifying in advance all the features of a problem and the actions a program should take to deal with them. By harnessing the mechanisms of evolution, researchers may be able to "breed" programs that solve problems even when no person can fully understand their structure. Indeed, these so-called genetic algorithms have already demonstrated the ability to make breakthroughs in the design of such complex systems as jet engines. Genetic algorithms make it possible to explore a far greater range of potential solutions to a problem than do conventional programs.

To better appreciate the power of genetic algorithms, let's examine the well-known traveling salesman problem (TSP) with its notorious *combinatorial explosion*, a term used to describe problems for which a small increase in the complexity of the input is associated with a dramatic increase in the difficulty of finding a solution. The TSP asks us to find the best order to visit a collection of cities, where the best route could be the shortest, the most cost-effective, or perhaps the most scenic. Because of the potentially astronomical number of routes to be compared, making a list and checking the length of each is not a viable option.

A genetic algorithm solves the TSP as follows:

- 1 *Initial population*—The genetic algorithm starts with an initial population of potential solutions randomly generated or developed using heuristics. Each potential solution is a sequence of cities where each city is visited only once, and the sequence ends in the first city where it started.
- 2 *Fitness function*—The fitness function evaluates the quality of a solution. In the TSP, it could be, for instance, a function that calculates the total distance traveled for a given solution. We can also choose to use different metrics for the fitness function.
- 3 *Selection*—The solutions with lower total distances will be kept for the next steps. Other methods can be used for the selection.
- 4 *Crossover*—The crossover consists of combining two solutions to create a new one. In the TSP, this step could involve selecting a portion of one solution and completing it with cities from the other solution while preserving the order.
- 5 *Mutation*—The mutation involves introducing minor random changes in the solutions to introduce variety in the population of potential solutions. In the TSP, the mutation could involve the swap of two cities.

The solutions resulting from crossover and mutations will become our new population. The genetic algorithm iterates through the previous steps until the best solution is found or until a termination predefined criterion, such as a maximum number of generations, is reached.

Genetic algorithms have broad applications across various optimization problems, spanning supply chains, financial models, stock exchanges, production schedules, automotive manufacturing, and robotics. Consider a school district with a fixed fleet of buses aiming to efficiently pick up each child at their residence and safely deliver them to school. Their goal may be to discover a set of routes that minimizes the total distance traveled by the buses, all while ensuring that no child boards a bus before 7 a.m. and that every child reaches school by 8 a.m.

As the number of locations to be visited increases, the complexity of the problem grows significantly. This complexity is vividly illustrated in table 2.2, which outlines the exponential growth in the number of possible routes for a single bus when visiting various numbers of locations, assuming that there is a direct path between any two of them.

Table 2.2 Complexity of the genetic algorithm for the school bus route example

Number of locations to visit	Number of possible routes to compare
1	1
2	2
3	6
4	24
10	3,628,000
20	2,432,902,008,176,640,000

As we can observe, the numbers become extremely large. Solving similar combinatorial problems can be exceedingly challenging, and in some cases, it is impossible with classical algorithms. In science, as in many other fields, the most efficient ideas are often simple, and genetic algorithms provide a prime example of this principle, offering a straightforward yet remarkably effective approach to solving complex problems.

Genetic algorithms are a powerful optimization technique, but like all other AI methods, they come with certain limitations. First, there is no guarantee that they will find the best solution, and their performance relies heavily on the initial population and parameter tuning. They also require substantial computing resources and can be slow for complex problems. Additionally, the solutions they provide can be challenging to interpret. Still, despite these constraints, genetic algorithms excel at solving various types of problems, especially in cases of complexity where other methods may prove less effective.

Summary

- Expert systems are software programs that assess data using *if-then* rules crafted by domain experts. Over time, they evolved with the addition of various management and deployment modules, giving rise to business rules management systems.
- Case-based reasoning is a technique that involves translating human knowledge into generalized cases, which can then be applied to solve similar problems.
- Fuzzy logic was introduced to address limitations associated with the use of strict true/false distinctions in real-world scenarios. It's particularly valuable when dealing with questions that require a nuanced understanding.
- Genetic algorithms draw inspiration from biology to find optimal solutions to problems. They achieve this by testing, combining, and modifying potential solutions to retain those that perform well while discarding those that do not.

AI mastery: Essential techniques, Part 2

This chapter covers

- An introduction to data mining
- An overview of the artificial neural networks
- A description of deep learning
- An introduction to Bayesian networks
- An overview of unsupervised learning

AI expert Arthur Samuel, introduced in chapter 1 for the success of his 1959 checkers program, defined machine learning as the field of study that gives computers the ability to learn without being explicitly programmed. “Without being explicitly programmed” can be misleading, as learning is achieved with techniques such as data mining and neural networks, which rely on algorithms explicitly programmed by engineers.

In this chapter, we will explore data mining, a technique used to extract valuable information, patterns, and associations from data. I briefly mention Bayesian networks, a method that encodes probabilistic relationships between variables of interest. I then introduce artificial neural networks and deep learning, powerful pattern recognition algorithms that have achieved impressive results in computer vision, natural language, and audio processing. We finish this chapter with unsupervised

learning, a set of algorithms that can analyze unlabeled datasets to discover similarities and differences. I'll provide enough detail to allow you to understand what these machine learning techniques entail and how they're applied, but we won't get caught up in the theory.

3.1 *Data mining*

Imagine a program that helps a grocery store ensure that its shelves are consistently stocked with fresh produce precisely when needed, eliminating the problems of overstocking or running out of popular items. The program also excels at optimizing store layouts, strategically placing complementary items to boost sales, fine-tuning prices for maximum profit, and crafting personalized promotions and discounts based on individual customers' past purchases to enhance customer loyalty, increase sales, and optimize profits. This example perfectly illustrates one of the numerous benefits data mining techniques can bring to the retail industry. Data mining is an artificial intelligence approach encompassing a range of techniques and algorithms to discover hidden patterns, relationships, and valuable insights from vast and complex data sources. Its applications are vast and continually evolving as organizations increasingly recognize the immense value of extracting actionable insights from the ever-expanding volumes of data at their disposal. Indeed, the amount of data available has increased exponentially over recent decades due to the near-universal adoption of the internet, the popularization of e-commerce, the use of barcodes on most commercial products, the popularity of social media, and ubiquitous web tracking. Exacerbated by low-cost data storage that promotes accumulation, the proliferation of data has created the need for automated techniques to extract knowledge and insight from it. It is obviously impossible for individuals to process or analyze even a minuscule fraction of what's available.

Much like gold mining, which extracts gold from piles of rock and sand, data mining is carried out to uncover meaningful correlations, patterns, anomalies, or rules hidden within extensive data sets. Formally, data mining refers to a collection of algorithms used for tasks such as classification, prediction, clustering, and market basket analysis. These algorithms make use of statistical, probabilistic, and mathematical techniques to identify data patterns, with some of their implementations having names like SLIQ [1] CART [2], C4.5 [3], and CHAID [4].

Data mining algorithms are applied across various industries. For instance, the Walt Disney MyMagic+ project utilizes data mining to enhance the efficiency of its operations and infrastructure. One notable application is its use in minimizing wait times for attractions and restaurants.

The food industry employs data mining for demand forecasting and competitive pricing of products. For instance, franchise companies like McDonald's utilize data mining to identify optimal locations for new stores. Local governments apply data mining to predict traffic volumes, especially during peak hours, while utility companies utilize data mining to forecast electricity demand and maintain a reliable supply.

A typical data mining task involves classification, which is the process of categorizing labeled data into meaningful groups. The knowledge derived from analyzing the data is often represented in a decision tree. A decision tree is a flowchart used to associate input data with the appropriate category through a series of questions or tests represented by the nodes in the tree. Each node evaluates a specific attribute of the data, and each distinct attribute value corresponds to a branch emanating from that node. An output node, also known as a leaf or terminal node, signifies a category or decision. The nodes situated between the input nodes and the terminal nodes are commonly referred to as test nodes.

The structure of a decision tree is inferred from the data. Mathematical formulas are employed to assess the potential contribution of each node in reaching a decision efficiently, and the most discriminative nodes are strategically positioned at the beginning of the tree. For instance, if you wish to determine whether an animal is a bird, the initial question you might consider is whether it has feathers or perhaps whether it can fly. On the other hand, asking whether it resides in a forest would not lead you to a decision as swiftly.

3.2 *Decision trees for fraud prevention*

Banks bear a substantial responsibility for effectively managing the risks associated with credit and payment transactions. Given the substantial sums of money involved, any security breach in banking operations can have a severe detrimental effect on an institution's reputation. When a credit card is used at a merchant's location, a highly efficient AI-powered system must swiftly provide authorization decisions, typically within milliseconds. To identify potential fraudulent activities, this system needs to assess numerous parameters linked to the card, all while processing 10s of thousands of transactions per second without causing any delays. Consider the sheer volume of queries that inundate the system on Black Friday, the day following Thanksgiving, which traditionally marks the commencement of the Christmas shopping season in the United States.

To build a decision tree that a card-issuing bank could use for real-time fraud prevention, we might analyze 18 months of their credit card transactions. Each transaction record will contain many attributes such as purchase amount, purchase time and date, international or domestic merchant, merchant category code (indicating merchant business type), geographic area, and whether the card was present during the transaction. Each fraudulent transaction would have to have been labeled as such by a human.

The decision tree will be constructed by weighing each attribute's potential to help the risk system decide in real time whether to accept or decline a credit card transaction. The space of attributes will be recursively partitioned based on importance, and the attributes most useful for forming an assessment will be placed earliest in the decision tree. In the fraud-prevention example, the data may show that fraud is significantly more common in international transactions than domestic ones, and therefore, this question should be asked first. Thus, the node associated with this question will be the first in the tree.

After creating this initial node, we create two branches, one for domestic and one for international. We then repeat the procedure to find the most discriminative question to ask about the transactions associated with each branch. For domestic transactions, perhaps the data shows that the likelihood of fraud is significantly higher for transactions done online or over the phone than transactions done by physically swiping a card.

In this case, the most significant question we could ask next might be whether the card was present for the transaction, and the node following the domestic branch in the decision tree might address this question. We would create a “card present” branch and a “card not present” branch in the domestic path, and we would repeat this process until the available attributes are all represented in an efficient decision tree. Figure 3.1 illustrates the first few nodes of the decision tree we’re discussing.

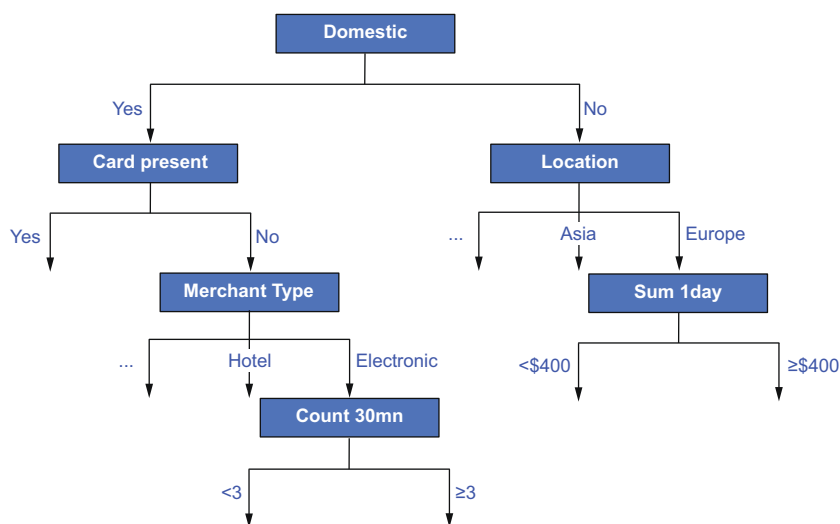


Figure 3.1 The first few levels of a decision tree. In a real-world application, a full tree may contain thousands of nodes.

We don’t always rely solely on the raw attributes available from the transaction records for a business application. We might also try to enrich the data with attributes gleaned from further analysis. For our credit card authorization problem, we might realize the importance of questions about the number of transactions performed in the last 15 minutes or the last hour, or maybe research identifies suspicious purchasing patterns involving restaurants and gas stations. The logic represented in a decision tree is sometimes translated into a set of `if-then-else` rules, making it easier to understand, especially if the tree is very large.

To summarize, a data mining project generally follows an iterative process:

- 1 Understanding the application domain and the goals of the data mining project
- 2 Gathering the data, which often involves a costly labeling step

- 3 Integrating the data gathered from various sources
- 4 Cleaning the data to remove inconsistencies
- 5 Performing analysis to identify new attributes that enrich the data
- 6 Dividing the data into at least two sets, one for training and one for testing
- 7 Selecting suitable data mining algorithms
- 8 Building the system using the designated training data
- 9 Pruning the decision tree to keep the model sufficiently general
- 10 Testing the model using the designated testing and evaluating its performance
- 11 Testing the scalability and resilience of the model
- 12 Repeating steps 2 to 11 until you achieve the desired performance
- 13 Deploying the model and integrating the system into operations

Figure 3.2 shows a process diagram outlining the creation and deployment of a data mining model.

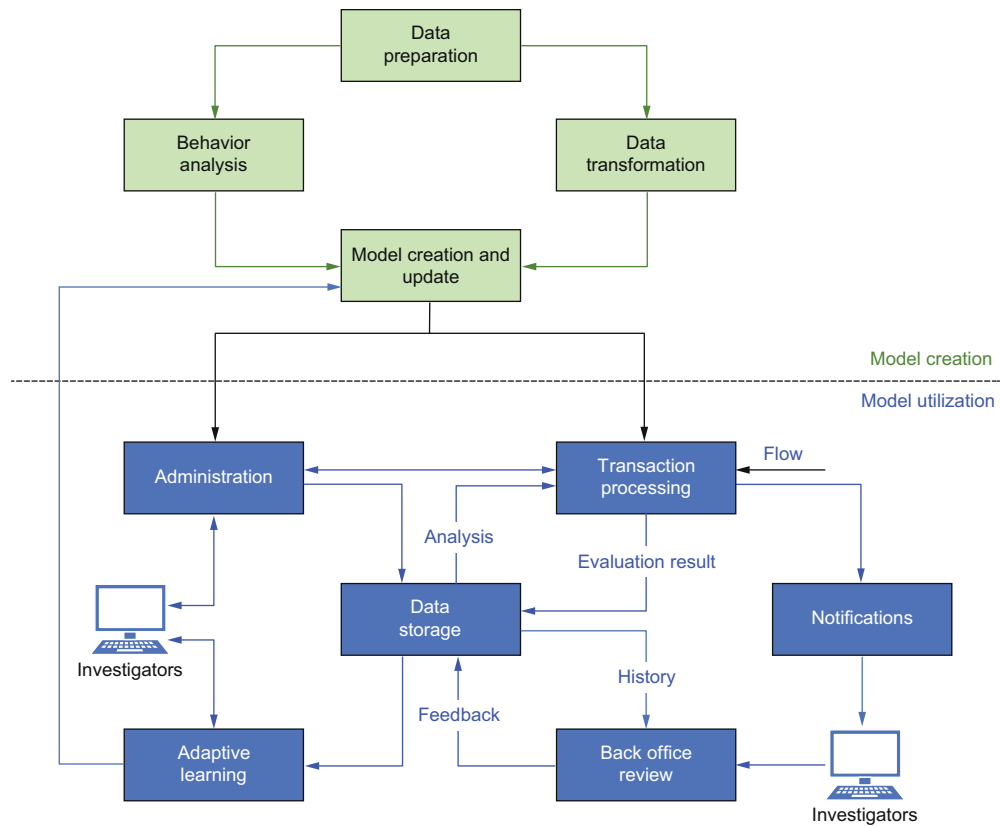


Figure 3.2 An example of a flowchart that depicts the various steps to design and deploy a data-mining model

Although decision-tree algorithms are the most popular, other data-mining techniques are also used. For example, association analysis is often used for market-basket studies, which attempt to identify sets of products that tend to be purchased together. This straightforward approach is based on addition, intersection, and union. For example, suppose we have one million receipts, of which 20,000 include the purchase of bread and cheese and 5,000 include bread, cheese, and olives. We could infer from this data that a customer who buys both bread and cheese has a 25% likelihood of also purchasing olives. Information about customer buying habits gleaned from association analysis can be used to develop cross-selling strategies, provide relevant coupons, and even optimize how products are displayed on store shelves. Information obtained through related approaches can be used to predict the effects of rebate offers or to develop strategies to improve customer retention.

The value of the predictions and guidance provided by data mining heavily depends on the quality of the input, and the adage “garbage in, garbage out” (GIGO) is very apt. Unreliable data leads to unreliable models, and the inconsistencies that arise when compiling data in different formats from multiple sources present significant problems. It can be almost impossible for software or even people to recognize when data has been labeled incorrectly, and it can be a challenge to mitigate the biases and other subjective effects that individual people can have when data is first recorded. Practical challenges arise, too, when applying data mining in real situations. For example, the structure and logic of a decision tree can’t be updated incrementally in light of new information, so decision trees aren’t effective for adapting to changes in data and behaviors.

3.3 **Artificial neural networks**

Imagine we want to teach a computer how to recognize handwritten numbers from 0 to 9. At the beginning, we show our program lots of pictures of handwritten numbers (training examples) to train the program. Our program relies on different layers to learn. The first layer, for instance, looks at the picture and tries to recognize things like edges and curves. The next layer will use the results of the first layer to look for shapes, like loops and lines. For instance, the number 8 has a shape that resembles two circles stacked on top of each other. We can continue this process by going deeper into more layers. After passing through all these layers, our program guesses which number is in the picture. We compare the guess made by our program to the correct number in the picture. If the program guessed wrong—for example, it guessed 1 when the number was actually a 7—we tell the program that it made an error. The program will then adjust its parameters (numerical values) that serve as the secret sauce behind the program’s classification. These numbers reflect the importance of certain shapes, inputs, or features and how strongly they influence the program’s response. During training, the program fine-tunes these parameters (referred to as weights) through a process known as backpropagation, which learns to recognize patterns and make correct predictions by discovering the optimal combination of weights

that minimizes errors. These weights are how our program learns to assign significance to different pieces of information. The more pictures we provide, the better our program will learn how to recognize numbers. In this example, we described the method of training a neural network.

Artificial neural networks are conceptualized as algorithmic models of brain cells that transform input data into output data. The introduction of this concept is attributed to McCulloch and Pitts in 1943 when they demonstrated that Boolean operations could be executed using “neural” elements that mimic living neurons. Since 1950, this field has witnessed significant advancements.

In 1958, Frank Rosenblatt introduced the Perceptron, which marked a pivotal moment in the field of artificial neural networks as it was able to learn and adapt its decision-making based on training data. Even though the Perceptron was a simple program with many limitations, it laid the foundation for further advancements in neural networks. In 1960, Bernard Widrow and Marcian Hoff developed the first neural network systems applied to real-world problems. They designed ADALINE (Adaptive Linear Neuron), which identified binary patterns, allowing it to predict the next bit when reading streaming bits from a phone line. MADALINE (Multiple ADALINE) was developed to eliminate echoes on phone lines.

In his 1974 PhD thesis, reprinted in 1994 [5], Paul Werbos proposed the development of reinforcement learning systems by using neural networks to approximate dynamic programming. Dynamic programming is an optimization approach that transforms a complex problem into a sequence of simpler problems. In 1986, Rumelhart, Hinton, and Williams rediscovered the backpropagation technique and made this fundamental technique broadly known with the publication of the backpropagation training algorithm [6].

The backpropagation algorithm structure comprises an input layer, one or more hidden layers, and an output layer. Each node, or artificial neuron, connects to another and has a weight and threshold. If the output of any node is more than the specified threshold value, that node is activated, transmitting data to the next layer of the network. Otherwise, no data is passed to the next layer of the network. Figure 3.3 depicts a backpropagation algorithm structure with three layers (input, hidden, and output).

- *Input layer*—This layer receives the input data fed into the network.
- *Hidden layer*—Neural networks encode the information learned from the training data using the value of the weights for the connections between the layers.
- *Output layer*—The output layer collects the predictions made in the hidden layers and computes the model’s prediction.

The input values in figure 3.3 are 0.5, 0.6, and 0.2. Each node in the input and hidden layers is connected to all the nodes in the next layer, and there are no connections between the nodes within a particular layer. Each connection between nodes has a weighting factor associated with it. Initially, the nodes are connected with random weights. The training consists of modifying the values of these weights by iteratively

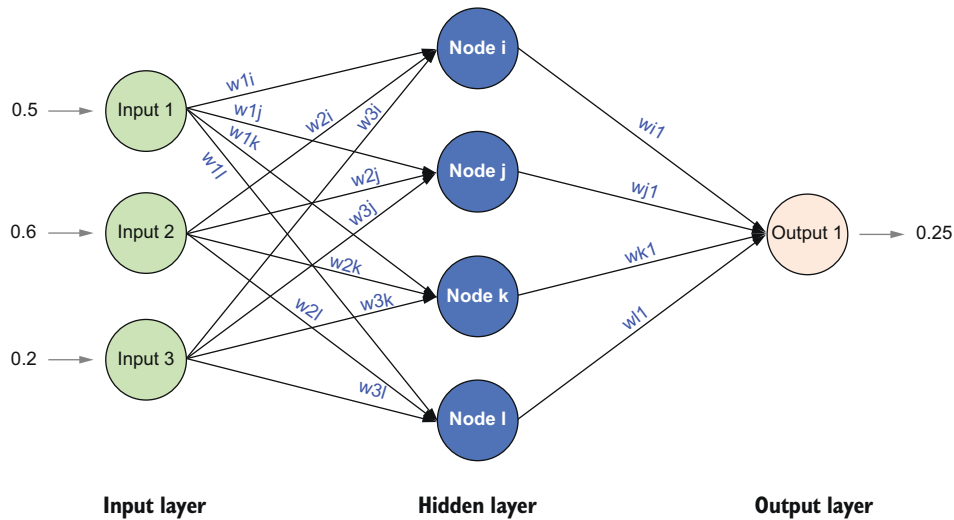


Figure 3.3 A backpropagation model with three layers

processing a set of training examples and comparing its prediction to each example's correct label. When the results are different, the weights are adjusted. These weight modifications are made backward (i.e., from the output layer through each hidden layer down to the input layer), hence the name backpropagation algorithm. Although it is not guaranteed, the weights will eventually converge, and the learning process ends. These modified weights, which are numbers between 0 and 1 or -1 and 1, represent what the neural network learns.

The input to individual neural network nodes must be numeric and fall in the closed interval range of $[0,1]$ or $[-1,1]$, which requires normalizing the inputs to values between 0 and 1 or -1 and 1 for each attribute from the training examples. Discrete-valued attributes may be encoded such that there is one input unit per domain value. To illustrate, suppose we have the attribute marital status with the values single, married, widowed, and divorced. One possible method is to represent the four values as single = (1, 0, 0, 0), married = (0, 1, 0, 0), widowed = (0, 0, 1, 0), and divorced = (0, 0, 0, 1).

For numerical data, we can apply the simple formula

$$\text{Normalized Value} = (\text{Value} - \text{MIN}) / (\text{MAX} - \text{MIN})$$

where MIN represents the smallest value in the dataset and MAX represents the highest value in the dataset.

For example, consider the numbers 2, 4, 5, 6, 20, 56, and 62. The minimum value is 2, the maximum value is 62, and the range is 60. Following the normalization procedure and rounding to the nearest hundredth yields the results in table 3.1.

Table 3.1 Normalization procedure

Original number	Normalized value
2	$(2 - 2)/60 = 0$
4	$(4 - 2)/60 = 0.03$
5	$(5 - 2)/60 = 0.05$
6	$(6 - 2)/60 = 0.07$
20	$(20 - 2)/60 = 0.3$
56	$(56 - 2)/60 = 0.9$
62	$(62 - 2)/60 = 1$

The backpropagation algorithm remains a widely used method for training supervised artificial neural networks. Initially, the neural network’s connections are established with randomly generated weights, typically between 0 and 1, connecting different nodes. The training process is iterative and involves presenting training examples to the network.

During each iteration, a labeled example is fed into the network’s input layer. The algorithm then computes the network’s output through a process known as forward propagation, which includes calculations through the hidden layers to produce the final output. Subsequently, the algorithm compares this output to the expected results or target values. When the computed output values differ from the expected results, the backpropagation algorithm comes into play. It applies an error-correction procedure by tracing back through the hidden layers toward the input layer, adjusting the network’s weights to minimize the error. This iterative process continues until the network’s performance improves and the desired accuracy is achieved.

While success is not assured, this process is reiterated through numerous cycles until one of two conditions is met: either the weights converge, allowing the neural network to correctly evaluate all test samples, or the neural network’s error falls within an acceptable threshold. Essentially, what a neural network “learns” boils down to a collection of numeric values between 0 and 1 (the adjusted weights). These adjusted weights encapsulate the essence of what an artificial neural network represents.

While neural networks trained using the backpropagation algorithm have demonstrated their utility for various problems, they come with several inherent limitations. First, these networks tend to operate as enigmatic black boxes because the inner workings of the trained model, including the critical features it identifies, can be challenging to examine. This opacity stems from the complex relationships between the hidden layers, which consist primarily of numerical weights (typically between 0 and 1). Consequently, neural network models struggle to explain their decisions, which can be a significant drawback. In contexts where accountability and transparency are paramount, this limitation becomes particularly concerning.

Second, the design of a neural network is not a straightforward task. It involves making crucial decisions, such as determining the optimal number of hidden layers, configuring the connections between nodes within these layers, setting the learning rate for weight adjustments, selecting appropriate training data, and establishing robust testing and validation procedures. These design choices significantly affect the network's overall performance and effectiveness.

Finally, while neural networks demand substantial computational resources, there is no guarantee that the training process will yield a highly effective model. Achieving convergence to an optimal solution is not assured, and issues like over- or underfitting can arise during training.

3.4 Deep learning

Deep learning, often hailed as the poster child of artificial intelligence, has become nearly synonymous with AI itself. Anthony W. Kosner, in a 2015 *Forbes* article titled “Deep Learning and Machine Intelligence Will Eat The World” [7], and Apurv Mishra, in a 2017 statement published in *Scientific American* [8], asserted that deep learning had propelled AI to a point where it could match or even surpass human experts in fields like medicine when it came to interpreting visual data. In 2018, CNN reported that deep neural networks developed by industry giants Alibaba (BABA) and Microsoft (MSFT) had surpassed human performance on a Stanford University reading comprehension test [9].

However, despite its impressive moniker, it's worth noting that many aspects of what we label as “deep learning” may already be familiar to us. What distinguishes a neural network as “deep,” as opposed to traditional backpropagation, lies primarily in the number of hidden layers and, occasionally, in how nodes are interconnected. The primary advantage of incorporating more hidden layers lies in their proven ability to construct a hierarchy of complex concepts from simpler ones, making them highly effective at discerning various characteristics. Figure 3.4 offers a visual representation of a typical deep neural network architecture.

The roots of deep learning algorithms can be traced back to the work of Ivakhnenko and Lapa in the mid-1960s. In their 1965 report “Cybernetic Predicting Devices” [10], they explore approaches to pattern recognition utilizing artificial neural networks with hierarchical layers of nodes between the input and output layers. They argue that deep networks hold computational advantages over classical networks, particularly when dealing with problems exhibiting nonlinear characteristics.

In contemporary deep learning, the fundamental approach remains anchored in the concept of the backpropagation algorithm. Data, regardless of its type, is typically transformed into numerical vectors within the range of 0 to 1. The core of deep learning still revolves around the adjustment of weights, represented by hundreds of thousands of values ranging between 0 and 1. While deep learning primarily operates in a supervised manner, where training data consists of numerous samples meticulously labeled, it's worth noting that the field has expanded to encompass various learning

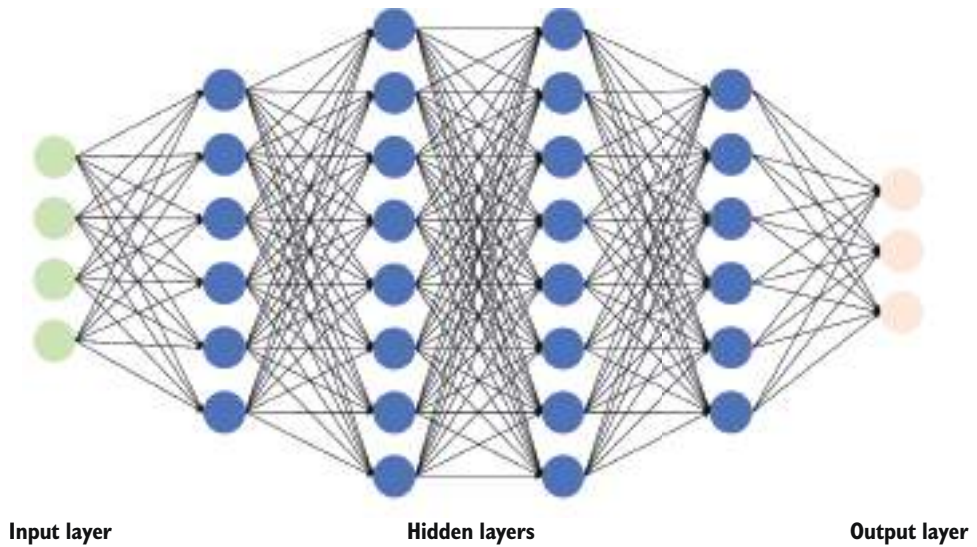


Figure 3.4 A deep neural network architecture

paradigms beyond traditional supervised learning, including unsupervised learning, reinforcement learning, and semi-supervised learning. This broader spectrum of techniques empowers deep learning to tackle a wide array of tasks across different domains.

To illustrate, let's embark on a straightforward project: creating a deep learning system capable of classifying color photographs of cats and dogs. When designing such a model, we may explore two prevailing architectures: convolutional neural networks [11] or vision transformers [12], which currently dominate the landscape of solutions for computer vision tasks.

Thanks to the insights of English physician Thomas Young in 1802, we understand how to translate color images into a series of numbers ranging from 0 to 1. Young's work elucidated the RGB (red, green, blue) color system, which enables us to represent the colors in an image as combinations of these three primary colors. Consequently, each pixel in our images can be described by a set of three numbers, representing its color in the RGB spectrum. In deep learning, images are treated as vectors of numbers, where each number corresponds to the RGB value for a specific pixel within the image.

The training process follows a similar pattern of weight adjustments, which is a hallmark of neural networks. With sufficient high-quality training data and a well-designed network, our deep neural network should be capable of accurately identifying most of the images earmarked for testing. It's important to note that we maintain separate sets for training and testing to evaluate the efficacy of the training process. This division allows us to validate the model's performance on previously unseen data.

In summary, the process of designing a deep learning project can be outlined with the following steps:

- 1 *Defining the objective*—Clearly articulate the neural network’s intended purpose and the problem it seeks to address.
- 2 *Data gathering*—Accumulate a substantial volume of high-quality data relevant to the project’s objectives.
- 3 *Data labeling*—Assign appropriate labels to the data. This may entail manual labeling or other labeling methods, depending on the project’s requirements.
- 4 *Data partitioning*—Segregate the labeled data into distinct sets, typically at least two: one for training the model and another for unbiased testing.
- 5 *Data preprocessing*—Prepare the data for input into the neural network by normalizing and converting it into vectorized form.
- 6 *Performance evaluation*—Develop a robust evaluation framework to assess the network’s performance, selecting pertinent metrics tailored to the specific task.
- 7 *Model refinement*—Continuously improve the model by fine-tuning parameters and adjusting architecture to mitigate errors and prevent overfitting.

3.4.1 *The benefits of deep learning*

Enumerating the advantages of deep learning is a multifaceted endeavor owing to its remarkable adaptability. Its core strength lies in its ability to extract patterns and glean insights from vast, complex datasets. This ability holds significant value across a wide spectrum of industries. It finds utility in healthcare, facilitating disease diagnosis, and in finance, where it enhances risk assessment and fraud detection. In natural language processing, deep learning models enable machines to understand and generate human language, leading to breakthroughs in virtual assistants and language translation.

Deep learning also excels in image recognition, where it powers technologies like facial recognition and autonomous vehicles. These models have the capacity to automate tasks, resulting in heightened productivity and substantial cost savings. Moreover, deep learning enables businesses to provide highly personalized experiences to customers, tailoring recommendations, content, and services to individual preferences.

Another notable strength lies in predictive analytics, where deep learning models analyze historical and real-time data to predict sales, anticipate market demand, and make data-driven decisions. Similarly, deep learning’s impressive computer vision capabilities automate tasks such as image recognition, object detection, and quality control in manufacturing, thereby enhancing operational efficiency and product quality.

By overcoming data challenges once deemed insurmountable, deep learning stands as one of the most transformative technologies, driving innovation across various sectors.

3.4.2 Limitations of deep learning

Let's begin by considering the insights of Francois Chollet, the creator of the Keras deep-learning library and a key contributor to the TensorFlow machine-learning framework. Chollet points out:

[The] deep learning model is “just” a chain of simple, continuous geometric transformations mapping one vector space into another. All it can do is map one data manifold X into another manifold Y , assuming the existence of a learnable continuous transform from X to Y , and the availability of a dense sampling of X : Y to use as training data. Most of the programs that one may wish to learn cannot be expressed as a continuous geometric morphing of a data manifold.

For a deeper exploration of deep learning, I recommend reading Chollet's excellent book, *Deep Learning with Python* [13], from which this quote is taken.

The coupling of the terms *deep* and *learning* with computers might lead one to believe that computers are truly learning in a profound sense. However, as we've explored in this book, deep-learning algorithms are essentially mathematical formulas. Using formulas alone, we cannot define or create true intelligence. This is why, when *PYMNTS* magazine asked for my views on deep learning in October 2017 [14], my response was, “Before you use the word ‘deep,’ show me the learning.” We will delve into this topic further in chapter 6.

In many business applications I've encountered, applying deep learning networks can be challenging due to the substantial requirement for labeled data. Numerous businesses not only lack the necessary amount of data but also labeled data. Even when the data is available, designing and training a deep network can be time-consuming, and such a network often underperforms when applied to data that deviates from its training data.

For example, consider AlphaGo, which required millions of training scenarios and years of meticulous engineering by a team of experts. While it excelled in the game of Go, adapting it for other purposes would necessitate extensive work by a different team of data scientists and engineers. Expensive retraining is required every time you make changes to the objectives of a deep learning project.

To be fair, deep learning has benefited in recent years from advances in computational speed, thanks to specialized hardware designed for it. In a few specific domains, there is an abundance of labeled data available for training. For instance, the ImageNet project provides access to over 14 million images annotated by humans, making it a valuable resource for object recognition testing. However, for most problems, the challenges associated with deep learning persist.

Additionally, deep neural networks are susceptible to hacking through adversarial examples. By making subtle pixel alterations, one can disrupt the training of a deep-learning surveillance system designed to detect intruders. Even though creating adversarial attacks might not be straightforward, the question remains: should we trust a technique with such a vulnerability when it comes to mission-critical tasks?

Even in fields where deep-learning algorithms demonstrate efficiency, a critical aspect they often lack is trustworthiness. How can one trust a system that cannot provide justifications for its conclusions? Deep-learning algorithms, like the traditional neural networks, are black boxes. Consider a scenario where a military operator must justify a life-and-death decision made by such a system. How can we permit the use of such systems in applications with significant societal implications, such as criminal justice or lending?

These concerns have motivated the European Commission to introduce a regulatory framework designed to ensure that all AI programs used in Europe can be relied upon to safeguard the safety and fundamental rights of individuals and businesses. Margrethe Vestager, Executive Vice President of the European Commission for a Europe Fit for the Digital Age, emphasized the importance of trust in AI, stating, “On Artificial Intelligence, trust is a must, not a nice to have. With these landmark rules, the EU is spearheading the development of new global norms to make sure AI can be trusted” [15].

This regulatory framework categorizes AI systems based on their level of risk and imposes corresponding restrictions. These range from an outright ban on programs that could pose threats to safety or livelihoods to stringent checks on those used in critical infrastructures, such as transportation, education, recruitment, credit scoring, law enforcement, criminal justice, or elections. AI systems used in Europe must adhere to the following requirements:

- Appropriate datasets must be used to minimize the risk of bias and discriminatory outcomes.
- Full traceability must be ensured to allow for oversight.
- Detailed documentation must be readily available to explain the system’s functioning.
- Clear and adequate information must be provided to individuals deciding whether to use such a system.
- Effective human oversight and monitoring mechanisms must be in place.
- The technology must be purposeful, robust, secure, accurate, and unbiased.

It’s important to note that while deep learning is a prevalent AI approach, not all deep-learning solutions inherently fail to meet these requirements. Nevertheless, addressing the issues related to trust and transparency remains a significant challenge in the field.

As for why this field consistently garners hype, one obvious explanation is the use of terms like “neurons” and the presentation of neural networks as being biologically inspired with structures similar to that of humans. In chapter 6, we will elaborate on why this claim is ludicrous.

Reinforcement learning, a machine learning approach dating back to the early days of cybernetics, involves learning behavior through trial and error. Its core principle revolves around using a system of rewards and punishments as a reinforcement

signal. Arthur Samuel pioneered the use of reinforcement learning in his 1956 checkers program. This program played against itself, making random moves and assessing the outcomes using the checkers rules to determine the best strategies. Reinforcement learning can be particularly beneficial in environments where you can clearly define rewards for positive actions, although such clarity is not commonly found in real-world applications.

3.5 Bayesian networks

In contrast to the previous AI techniques I've employed in the business world, I have limited experience with Bayesian networks. Nevertheless, I'll provide a brief overview since they offer a well-established approach that can be valuable in projects involving probabilistic descriptions.

Bayesian networks serve as maps illustrating the interplay between various events, enabling us to comprehend how changes in one event can have cascading effects on others. In more technical terms, Bayesian networks belong to the category of probabilistic models employing directed acyclic graphs to depict relationships between variables and their probabilistic dependencies. Their roots trace back to Thomas Bayes and his posthumously published manuscripts in 1763. Bayes' theorem introduced the concept of conditional probability. The theorem provides a formula for adjusting beliefs when presented with new evidence, elucidating the likelihood of an event based on information concerning conditions related to that event. One way to express Bayes' theorem is that the probability of event B happening, given that event A has occurred, multiplied by the probability of event A occurring, is equivalent to the probability of event A occurring, given that event B has occurred, multiplied by the probability of event B happening.

A Bayesian network encapsulates probabilistic relationships among variables of interest. It comprises a structure resembling a directed acyclic graph or belief network, with nodes interconnected by edges. Each node represents a variable, and the directed edges symbolize the conditional dependence between these variables.

For instance, consider a Bayesian network for weather prediction, where a node representing clouds is linked to a node representing rain with an arrow pointing from clouds to rain. If the clouds node is set to 40%, signifying a 40% likelihood of cloudy conditions, the value of the rain variable would be contingent on the cloud's variable. Perhaps, if it's cloudy, there's a 30% chance of rain, whereas if it's not cloudy, there's only a 5% chance. The model can handle these probabilities and dependencies by computing the likelihood of various scenarios: cloudy and not raining, cloudy and raining, not cloudy and raining, or not cloudy and not raining. In more complex Bayesian networks with hundreds of nodes and dependencies, they can be used to infer the overall significance of a particular variable in determining outcomes.

Figure 3.5 is a suite of Bayesian networks developed by researchers at the St. Petersburg Coastal and Marine Science Center [16] to (a) generate scenarios of total water level, (b) forecast storm effects, and (c) predict magnitudes of beach recovery. The

Bayesian networks incorporate topographic, bathymetric, and shoreline data available from the historical and post-Hurricane Sandy research programs at Fire Island. The Bayesian networks generate predictions in the form of probability of coastal change.

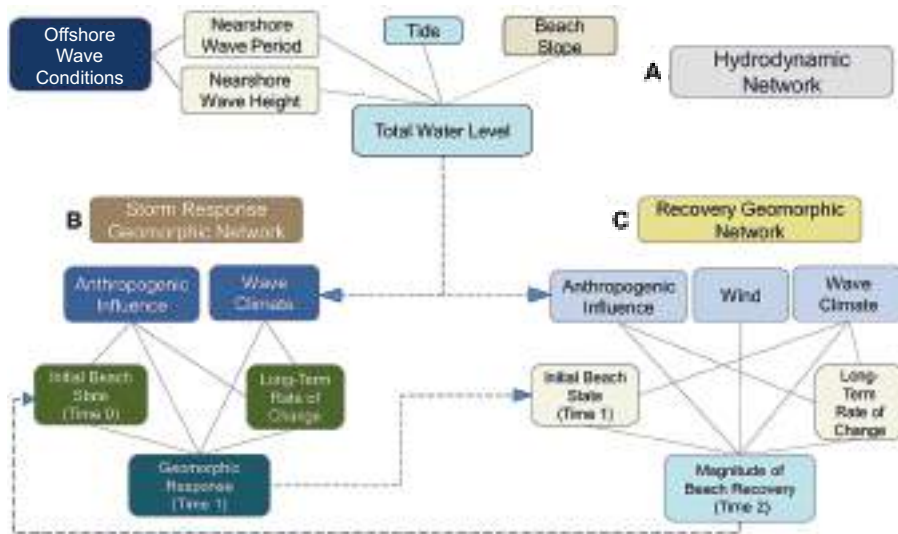


Figure 3.5 An example of a Bayesian network

Bayesian networks made their debut in the late 1970s, serving as a means to model distributed processing in reading comprehension. This modeling approach aimed to amalgamate semantical expectations with perceptual evidence, forging coherent interpretations [17]. Often hailed as the progenitor of Bayesian networks, Judea Pearl elucidates in *The Book of Why* [18] how one should design software for reasoning under uncertainty:

I was firmly convinced that any artificial intelligence system should emulate what we comprehend about human neural information processing. Machine reasoning under uncertainty, I believed, should be built upon a similar message-passing architecture. The question that vexed me was: What should these messages be? It took several months, but I eventually discerned that the messages took two forms—conditional probabilities in one direction and likelihood ratios in the other.

Pearl envisioned a hierarchical network, with each node transmitting its belief to neighboring nodes. The recipient node processed the incoming information in one of two ways: updating its beliefs using conditional probabilities when the message flowed from parent to child or adjusting beliefs by multiplication with a likelihood ratio when the message moved from child to parent.

Despite their merits, Bayesian networks have limitations, akin to other machine learning methods. They operate on precise probabilities, which may not align with

real-world scenarios characterized by imprecise knowledge necessitating confidence intervals, such as between 56% and 62%. Another substantial constraint emerges when attempting to create Bayesian network structures in complex domains. While experts can define simple Bayesian networks, in many fields, constructing such networks proves a formidable task for human operators. The ideal scenario involves computers autonomously learning network structures, parameters, and conditional probabilities from data, yet this remains a significant challenge in most practical applications. The absence of a universally effective method for constructing Bayesian networks from data implies that their creation often demands extensive time and human involvement.

3.6 ***Unsupervised learning***

Unsupervised learning is an approach within the field of artificial intelligence where algorithms are employed to discern patterns and structures within data without relying on labeled outputs. The primary objective is to group data based on inherent similarities, differences, or concealed structures. For instance, unsupervised learning techniques like clustering prove invaluable in the retail sector, as they enable the grouping of customers with similar purchasing behaviors. This categorization can unveil distinct customer segments, such as budget-conscious shoppers, luxury enthusiasts, and occasional buyers, empowering retailers to fine-tune marketing strategies, provide tailored product recommendations, and optimize inventory management more effectively to meet the diverse needs of their customer base.

Unsupervised learning excels at extracting meaningful insights and knowledge from unorganized or unlabeled data, aiding organizations in making informed decisions. To illustrate the potency of unsupervised learning, consider its role in combating money laundering—a critical societal issue that demands the application of this technique.

Despite the implementation of stringent legislation, criminals and terrorists have proven to be resourceful and persistent in their illicit endeavors. According to data from the United Nations Office on Drugs and Crime, the estimated global amount of money laundered in a single year range from 2% to 5% of the world's gross domestic product, which translates to a staggering \$800 billion to \$2 trillion in current US dollars as of 2022 [19].

Money laundering operations are primarily designed to conceal unlawfully acquired funds and obscure their origins. Typically, this is achieved through a series of complex financial transactions intentionally structured to be challenging to trace. Money launderers often employ ingenious tactics, frequently commingling illegal transactions with those of legitimate enterprises. They further complicate matters by concealing ownership structures, such as the creation of trusts and offshore companies in jurisdictions with lax regulatory oversight. Additionally, they use modern tools and technologies, including cryptocurrencies, currency exchanges, international money transfers, and even cash smuggling, to facilitate their activities. The proliferation of

internet technology has only exacerbated these challenges, as online auctions, gambling, banking, and peer-to-peer payment apps provide additional avenues for them to safeguard their anonymity.

The following money laundering case [20] study portrays a scenario that developed over several years and involved numerous individuals:

Tom works as a government employee. He was able to support his family until he became an addict to gambling. In his role, he had the power to propose and approve projects for private companies competing in the public sector. His gambling habit pushed to corruption, and large businessmen were willing to pay him well for the guarantee of government business, and Tom became rich very quickly through his corrupt activities. Tom's friend Gina, who owned an exchange and tourism company, was willing to help him launder the bribes that he was receiving. She used her employees as "straw men" to create a number of different bank accounts through which funds could be laundered—more than US\$4,000,000 was laundered in total through such accounts. However, the cash payments and subsequent transfer offshore risked attracting attention, and so Tom developed a more sophisticated laundering method—a fruit-delivery company. This company, which was owned by Gina's husband, laundered US\$2,700,000 in three months, disguising the transactions by creating false invoices which were settled by the businessmen on Tom's instructions. The total amount of money Tom laundered was estimated to be in the range of US\$1,000,000,000. It is worth noting that the disclosures by the institutions took place because of the simple initial laundering scheme, whereas the later scheme involving an established company appeared to have had little risk of disclosure.

Supervised learning methods prove inadequate when combating money laundering for several compelling reasons. One significant challenge arises from the difficulty, and sometimes impossibility, of creating effective training and testing datasets. Money laundering schemes are not isolated, discrete events. Criminals operate meticulously, using complex strategies to conceal their activities. These elaborate scenarios cannot be accurately represented through labeled samples typically used in supervised learning.

Moreover, supervised techniques lack adaptability, whereas criminal organizations are highly agile, constantly inventing complex tactics to elude detection. They operate under the assumption that their financial transactions are under scrutiny, meticulously crafting each transaction to mimic legitimate activities. For instance, businesses like restaurants and nightclubs, known for dealing primarily in cash, can deposit large sums without arousing suspicion from financial institutions. This makes them well-suited to money laundering since they can handle substantial cash volumes without triggering red flags.

As a result of supervised learning's ineffectiveness in the fight against money laundering, current approaches rely heavily on predefined rules set by anti-money laundering experts to flag suspicious transactions. However, as previously discussed, business rules have significant limitations in dynamic environments. Consequently, current anti-money laundering solutions suffer from a high rate of false positives. Some estimates even suggest that as much as 90% of compliance analysts' time is spent investigating the numerous false alarms generated by such systems.

Effectively combating money laundering necessitates the application of unsupervised learning techniques. One of the primary unsupervised learning methods is clustering, which endeavors to categorize unlabeled data into groups or clusters. These clusters are defined by the similarity among examples within each group, enabling the identification of anomalies. For instance, a collection of wire transfers might be grouped based on factors like frequency, dollar amount, and the type of beneficiary. Analysis could reveal connections between transfers originating from related brokerage houses, industrial firms, or money transmitters. These clustered transfers might also exhibit common financial traits, such as shared accounts or types of accounts and involvement with the same financial organizations and individuals. When investigators detect irregular patterns within the activities of a manufacturing firm or an insurance company, it prompts a closer examination to determine whether these businesses might be fronts for money laundering.

Unsupervised learning, while a powerful tool in machine learning, comes with its own limitations. The most critical limitation is the evaluation of unsupervised models. Unlike supervised learning, where we can rely on labeled output for performance assessment, unsupervised learning lacks such clear metrics or guidance for evaluating the quality of learned representations. Additionally, unsupervised learning models can be quite sensitive to the choice of hyperparameters and initializations, making it challenging to determine the optimal settings for a particular problem.

3.7 So, what is artificial intelligence?

After exploring AI and its diverse techniques, it's natural to inquire about the contemporary landscape of AI. Remarkably, I still adhere to the same definition I articulated in my 1988 PhD thesis [21]:

Artificial Intelligence is essentially a set of human-coded mathematical algorithms, primarily rooted in probabilities and statistics. These algorithms serve the purpose of scrutinizing data and deriving insightful patterns and interconnections among attributes and concepts.

Summary

- Data mining stands out as a prominent AI technique for analyzing vast datasets and uncovering patterns. Its popularity spans various domains due to its ability to provide human-readable explanations of the knowledge it extracts.
- Artificial neural networks represent another essential AI technique capable of learning from extensive datasets. These networks consist of interconnected layers of nodes, and during training, the weights connecting these nodes are adjusted to encode information about patterns.
- Deep learning, an extension of neural networks, introduces networks with significantly more layers than earlier models. However, it's important to note that the increased number of layers doesn't necessarily equate to higher intelligence.

Instead, it allows deep learning networks to capture complex hierarchical patterns and representations.

- Bayesian networks, with origins dating back over 260 years, are a type of statistical model. They prove particularly valuable for unsupervised learning techniques like clustering, making them a practical choice for handling unlabeled data.
- Unsupervised learning is a powerful AI technique where algorithms are utilized to identify patterns and structures within data without the need for labeled outputs.

Smart agent technology



This chapter covers

- An introduction to smart agents
- How attributes are represented
- How smart agents communicate with each other
- Examples of how smart agents solved a real-world problem

When you expand your imagination to new spaces, “Impossible” becomes “I M Possible.”

Imagine a group of security experts working together to protect sensitive financial information. Each expert specializes in a specific area, such as monitoring incoming network traffic for unusual activities, checking the security of customer accounts, overseeing employee access and permissions, including actions like copying or downloading sensitive data, and analyzing digital communications for any suspicious conversations or data transfers. While each expert has their unique tasks, they all share the common goal of safeguarding the financial institution. To achieve this, they collaborate by sharing insights and data. For example, if there’s unusual activity in a customer’s account, it triggers an alert that is immediately analyzed by the network traffic expert, who cross-references it with employee access logs. If something seems suspicious, the collaborative experts can take prompt action, such as

blocking the activity or raising an alert. In this example, we described the Smart Agents approach, a powerful artificial intelligence technique based on using adaptive, autonomous, and goal-oriented entities to solve a problem. Smart agents are designed to interact with their environment and other agents in a goal-oriented manner, using their knowledge and abilities to achieve the best outcome. One of the greatest strengths of this technology is that smart agents are not merely programmed to follow rules; they adapt and continually learn from activities.

The smart agent paradigm is based on reactive, autonomous, goal-oriented entities collaborating to solve a problem. A smart agent platform should provide the following capacities:

- *Autonomy*—Agents operate by communicating with each other.
- *Reactivity*—Agents perceive their environment and react to changes that occur.
- *Goal-oriented*—Agents have goals, and each agent will act to achieve them.
- *Emergence*—Solving a problem will emerge as a side effect of the communication between the agents.

Smart agent platforms are adaptive as they learn from their interactions with the environment and other smart agents to improve their performance over time.

Smart agent technology is applied to the design of large-scale commercial software systems in areas such as fraud prevention, data breach detection, air traffic control, network management, and many other mission-critical applications that require adjusting to the changes in a problem's parameters. In the context of fraud prevention, think of smart agents as a team of detectives collaborating. Each detective (smart agent) is responsible for monitoring all the transactions of the customer it is tasked with protecting. These smart agents learn the spending patterns of the customer, including how much they spend, what they buy, where they make purchases, and how often they do so.

4.1 *Principles of smart agents*

A thorough discussion of smart agent technology would require an entire book. For our purposes here, I will aim to give an overview of how this approach works and to provide a general sense of what makes it different from more traditional approaches. Some of the characteristics that distinguish a smart agent approach are collective reasoning, distributed architecture, personalization, adaptability, and self-learning. To highlight these features and to contrast smart agent technology with more conventional methods, we'll first examine how the latter typically address problems.

Legacy AI relies on four main concepts for the presentation and algorithmic solution of a problem:

- An *initial state* represents the circumstances of the problem to be solved.
- A *goal state* or *final state* represents a solution to the given problem.
- A set of *operators* and *constraints* describes possible changes in state.
- A *control strategy* governs the transition between states.

Under this conventional paradigm, the way to solve a problem is to explore the allowed state transformations in an organized way so that some optimal sequence of changes is found that leads from the initial state to the goal state. Potential solutions are generated and tested until the objective is attained or until no more possible solutions exist to evaluate. For problems that suffer from a combinatorial explosion, this exhaustive, brute-force approach is obviously impractical.

A similar but slightly different approach involves the definition of some sort of metric that expresses the distance between two states. From a given state, the system could evaluate all legal transitions and choose the one that results in a state that is closest to the goal state with respect to the metric that has been defined. Depending on the problem, this process could generate an optimal solution or a non-optimal solution, or it might fail to produce a solution at all if its reasoning only leads to dead ends. Still, other approaches might use rules and constraints, described in chapters 2 and 3, to systematically reduce the number of states that need to be explored.

In contrast to methods that explore the space of all possible states and transitions, one can use smart agents to organize information about a problem after defining it in terms of its most salient features. Each problem feature will be associated with a smart agent that acts as an independent computational entity with its own data structures and mailbox. The communication between agents will lead to the emergence of a problem solution.

Unlike traditional programming and many AI techniques in which attributes are just labels for values, attributes in our smart agent system are smart agents themselves. In fact, attributes are, in some sense, the most important agents in a program. They have a global view of everything happening, and they have evaluation functions associated with them. The evaluation functions can change over time and can be used by all of the agents to make decisions regarding their own goals.

A smart agent program should accurately represent the physical, real-world situation being modeled. It also needs to reflect the abstract perspective of each agent within the system. The environment of an agent represents everything that an agent considers to be true, including statements with temporary or conditional validity.

An agent's environment can be updated when the agent receives a message in its mailbox, and a part of the agent's operation might be to send messages containing information of potential use to other agents. To reduce the total number of messages that need to be exchanged, agents might be organized into clusters that share a mailbox. The high-level operation of the system will follow cycles in which messages are exchanged, environments and attributes are updated, and actions are taken.

System-level functions might be called by an agent to evaluate the data it receives. For example, progress toward a global goal might be measured by an evaluation function associated with a particular attribute. Progress toward the goal of a particular agent might be measured by some other function. When an agent extracts data from messages received in its mailbox, it might assess it using a variety of functions, and the assessments might be used to update an attribute or to initiate some action. Over

time, agents accumulate knowledge and expertise, and the solutions created by the community of agents will reflect this expertise.

4.1.1 Adaptability: The true mark of intelligence

Many mission-critical applications are characterized by ever-changing requirements and dynamic environments. From cyber defense, banking, and autonomous driving to robot-assisted surgery, adaptive AI is a necessity for these applications that are essential to our well-being. They must be designed with a versatile and self-learning AI to understand and intelligently react, in real time, to adversaries and unexpected events. For instance, an AI system for use on a battlefield must be able to adapt to continuously changing weather and terrain, and it needs to respond appropriately to the unconventional tactics of an enemy.

An autonomous driving system has to react correctly to the constantly changing motion of dozens of objects around it, and it needs to safely and effectively harmonize its decisions with those of other drivers who may not follow the rules or behave in a rational way.

The AI most commonly used today might suffice for slower-changing situations. For example, traditional AI systems could probably be used to automate the navigation of freighters at sea. The territory is wide open and relatively uniform, and the speeds involved are low. With modern radar, weather and obstacles can be anticipated well in advance, and routes can be adjusted accordingly. Automating the movement of a vehicle on city streets, however, will likely never be possible with current techniques.

Consider, for example, the traffic around the Arc de Triomphe in Paris. There are 12 multilane roads feeding a roundabout with 10 unmarked lanes of traffic. Motorcycles weave between tour buses and trucks while some cars merge aggressively toward the innermost lanes, and others cut suddenly outward toward their required exit. The sound of horns and brakes is constant and disorienting, and you have to be on the lookout for tourists who try to run through traffic to see the monument up close. Traffic is always heavy, and contrary to the usual rules, drivers already in the roundabout are supposed to yield to those entering.

Young Parisians know that their real driving test isn't the one with the instructor that gets them their license, but it's their first time driving around the Arc. Despite the billions of dollars invested and the tremendous effort put forth by engineering teams over the recent decades, there is no AI system that can handle anything nearly as complex as the dynamic environment just described.

Fraud prevention and network security are two other great examples of ever-evolving problems that demand adaptable solutions. There is a never-ending technological arms race pitting legitimate businesses against hackers and thieves who are inventing new tricks and schemes on a daily basis. A static set of signatures or rules has limited value, and current methods require many examples of an event before they are able to detect it.

Even a system that is constantly being trained with new data is always going to be a step behind. We don't want to be perpetually applying patches after the damage has been done. We want to detect new problems as they are unfolding and catch criminals in the act.

Unfortunately, today's AI systems lack the essential features of adaptive learning. They are only applicable in limited, supervised learning tasks, defined by rigid rules, in extremely well-defined and fixed environments such as playing Chess or Go. The real world is a world where the rules are changed during the game. Adaptive solutions will, for example, detect that the chessboard has become larger, the rooks could now move like bishops, or winning is no longer to checkmate your opponent's king but to capture all his pawns! We need an AI that can understand what changed and successfully adapt to it.

One of the most notable strengths of smart agent systems is their adaptability. Rather than being preprogrammed to anticipate every possible scenario or relying on pretrained models, smart agent technology tracks and adaptively learns the specific behavior of every entity of interest. For instance, in fraud prevention, each customer and merchant behavior is automatically updated in real time, and the continuous, 1-to-1 profiling provides instantaneous actionable insights into the unique behavior of merchants and individuals, which result in a very effective solution that has the highest detection and lowest false-positive rates.

4.1.2 Smart agent language

To give context to our description, I will introduce a proprietary smart agent approach and use an example to explain how, in this method, each agent has the ability to evaluate what input is good or bad, in accordance with or antithetical to one of its goals. We also discuss agents' adaptability, compare the technique to more traditional approaches, and explore how this technique has been successfully applied to solve a real-world problem.

Figure 4.1 is a screenshot from MINDsuite, the AI platform developed in France by my first company, Conception en Intelligence Artificielle. MINDsuite enables its users to develop powerful AI solutions by combining smart agents, neural networks, business rules, genetic algorithms, constraint programming, fuzzy logic, and cased-based reasoning.

MINDsuite has been successfully applied in numerous fields, including defense, pharmaceuticals, insurance, finance, nuclear decommissioning, healthcare, and network performance. The platform allows for the integration of smart agents, constraint programming, fuzzy logic, neural networks, business rules, case-based reasoning, and genetic algorithms, and it provides a simulation tool and an interpreter for the programming languages.

Because there is a unique philosophy behind the smart agent approach, we've developed a programming language to fit the paradigm. Like any other programming language, the smart agent language AGORA has its own internal functions, structures, storage allocation, garbage collection, and memory management. It also has over 40 keywords, such as *goal*, *neighbors*, *behavior*, *environment*, *output*, *stabilize*, and *unstabilize*.

Keywords in a programming language have specific meanings and represent the reasoning framework of the language. They are used to give commands or to set

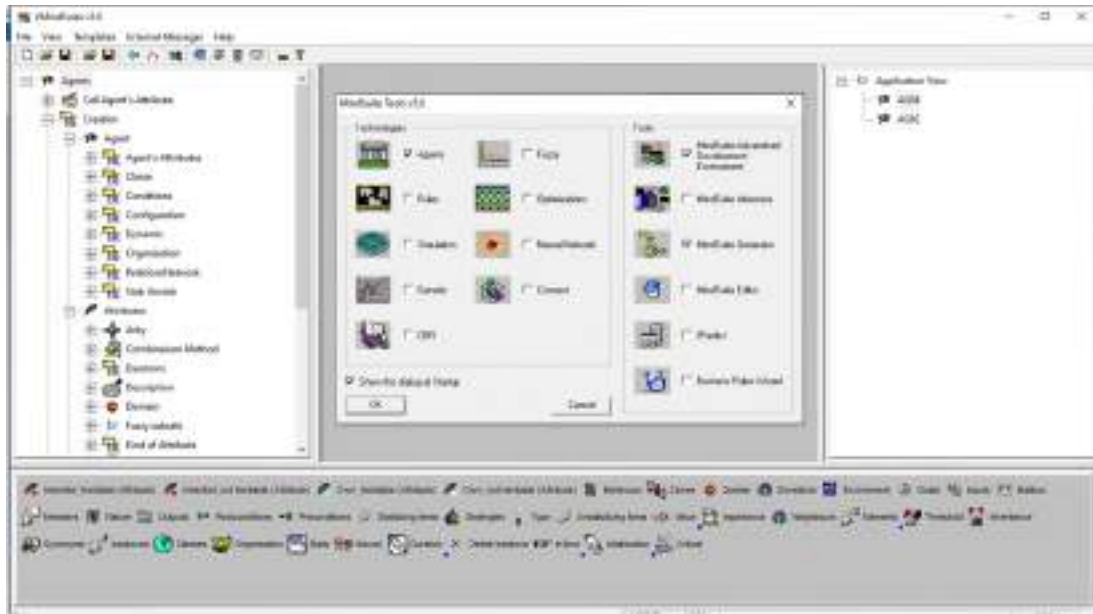


Figure 4.1 MINDsuite AI platform

parameters for a program, and they allow a programmer to think abstractly and focus on the big picture instead of repeatedly coding basic instructions by hand.

Each keyword in AGORA is associated with agents' activity. For example, *stabilize* indicates an acceptable, desirable state for an agent, whereas *unstabilize* is associated with a situation that should never be allowed. The use of keywords can be demonstrated if we imagine using MINDsuite to try to solve a well-known riddle. A farmer must transport across a river a goat, a cabbage, and a wolf. There is a very small raft available, and it can only hold the farmer and one other item without sinking. The goat can't be left alone with the wolf, or it will be eaten, and likewise, the cabbage can't be left alone with the goat. How does the farmer get to the other side of the river with all of his goods intact?

Legacy AI can solve this problem by representing the initial state as "the farmer, wolf, cabbage, and goat are on the left side of the river, and the right side of the river is empty." The final state is represented by "the farmer, goat, cabbage, and wolf on the right side of the river." The constraints are "the wolf will eat the goat if the farmer is not present" and "the goat will eat the cabbage if the farmer is not present." This will be the primary controlling strategy that governs the transition between states.

The solution will eventually be reached by generating and testing the allowed state transformations. When an allowable state is explored, it will be followed by the next level consisting of its children, then the next level of its grandchildren, and so on. In our example, from this initial state, there are several possible scenarios that the farmer could select to transition to the next state. The farmer could, for instance, choose to take

himself and the cabbage first. This will result in a new state where the wolf and the goat are left alone as the farmer and cabbage are on the other side of the river. Based on the earlier constraints, the wolf will eat the goat; therefore, this state should be rejected. Other potential moves are “the farmer together with the wolf,” “the farmer with the goat,” and “the farmer alone.” When a new allowable state is selected, the same logic will be used to determine additional moves from this state. This process is repeated until “the farmer, goat, cabbage, and wolf are all on the right side of the river,” or no other allowable move to a new state is possible.

To address this problem with smart agent technology, we would begin by creating agents for the farmer, wolf, cabbage, and goat. In a smart agent program, every element of the problem is represented by an agent, and every agent has a goal. Each agent in our problem would also have a location attribute, and to begin, the locations would all be set to *east*. The goal of the farmer, cabbage, goat, and wolf would be to have the location *west*, and the farmer would have the additional goal of being in the same location as all three of his possessions. The logic of the program would have to address which goals are more important than others under certain circumstances.

Some agents in our problem would also have the keywords *stabilize* and *unstabilize*. Keeping in mind that keywords can represent a lot of complex logic, we have to understand the values taken by these keywords in the context of the problem. For example, we know that it’s unacceptable for the farmer to leave the cabbage alone with the goat, so for the cabbage, the *unstabilize* keyword would be set to the condition that the location of the cabbage is the same as the location of the goat. *Stabilize* could be set to the condition that its location is the same as the location of the farmer. The logic in the program would then be able to dismiss options in which the cabbage is with the goat unless the farmer is also there.

Stabilize and *unstabilize* would likewise be set for the goat, and the keywords and their settings would establish a web of connections between the agents. This web of connections is a key feature of the smart agent approach. All elements in a smart agent program are agents, all agents have goals, and the connections established by keywords determine the options for attaining the goals. Progress toward goals would usually be indicated by changes in the values of attributes.

Mastering the art of designing and constructing efficient solutions on a smart agent platform is a gradual process. Success hinges on defining precise goals and objective functions. The more time individuals invest in creating real-world applications, the more adept they become at using the potential of a smart agent platform.

Smart agents offer valuable problem-solving techniques. Consider our experience in developing a fraud prevention solution as an example. Personalization emerged as a pivotal concept in this endeavor. Using smart agents, we were able to create virtual personal profiles for cardholders, stores, and merchants, with only the relevant information for each entity. Each cardholder is associated with a smart agent that continuously learns from its transactions in real time. It aggregates data from various channels, tracking anything relevant to the cardholder, gradually forming a robust profile over time. In

contrast to other AI methods discussed in chapters 2 and 3, smart agent systems make decisions tailored to each cardholder. Personalized decisions consistently outperform universally applied generic rules. A cardholder's profile reflects their unique spending habits, and once a baseline pattern is established, any deviation from normal behavior is immediately detected. There are virtually no limits to the types or number of profiling criteria that smart agents can define for use. Examining an entity from multiple angles is essential since behavior can vary significantly depending on circumstances. Here are some profiling criteria commonly used in fraud detection:

- *Real-time profiling*—Transactions can be aggregated over varying time frames, with counters updating in real time.
- *Long-term profiling*—Transactions can be aggregated over longer periods, with counters updating at a prescribed rate. These profiles establish behavior baselines.
- *Recursive profiling*—Activity can be tracked over a rolling time frame to monitor normal behavior.
- *Geo-location profiling*—Data, such as a cardholder's home zip code, location of card swipes, and IP addresses involved in electronic transactions, can identify anomalous behavior.
- *Multidimensional profiling*—Interactions between multiple agents can uncover suspicious activity patterns and connections.
- *Peer-comparison profiling*—Real-time identification of suspicious activity by comparing one entity's behavior to that of its peers.

Before authorizing a card transaction, a fraud-prevention system utilizes cardholder profiles, merchant profiles, and other relevant data to generate a risk score. Transactions are approved only if the score falls below a certain threshold. Innovative systems may adjust their scoring and thresholds based on circumstances, recognizing that a purchase unusual for an individual in July might be typical before Christmas, reducing false positives and improving the overall experience for everyone involved. Moreover, information from entity profiles can be shared among agents, allowing for the monitoring of group activity patterns. This approach grants smart agent solutions an adaptive learning capability, enabling the detection of previously unknown fraud schemes as they emerge.

Summary

- Smart agents were initially conceived to confront complex problems that resisted simple algorithmic solutions.
- Attributes play a fundamental role in smart agents as they have the capability to engage in active communication with other smart agents.
- Smart agents are equipped with mailboxes, serving as channels for receiving messages not only from other agents but also from external sources.
- Smart agents have the following characteristics:
 - *Learning and adaptation*—Smart agents can learn and adapt over time.
 - *Scalability*—These agents can scale both horizontally and vertically, making it possible to handle larger and more complex problem spaces.

- *Security and privacy*—Smart agents incorporate security measures to protect sensitive data.
- *Distributed problem-solving*—Smart agents can operate in decentralized and distributed networks, enhancing their versatility and resilience.
- *Real-time decision-making*—Smart agents are equipped to make decisions in real-time, enabling rapid responses to dynamic situations.

5

Generative AI and large language models

This chapter covers

- Generative artificial intelligence
- Reflections on human communication and speech
- The benefits, limitations, and risks of generative AI and large language models such as ChatGPT
- Differences between human and generative AI

Artificial intelligence has witnessed numerous ups and downs, but the release of ChatGPT represents a pivotal moment in the field of AI for several compelling reasons. First, it signifies a significant leap forward in natural language understanding and generation, demonstrating the remarkable progress AI has made in processing and generating human-like text. ChatGPT's ability to engage in coherent and contextually relevant conversations with users across a wide range of topics showcases the potential for AI to be integrated into various applications, from customer support to content creation. Furthermore, ChatGPT embodies the power of large-scale pretrained models. Its capabilities highlight the potential for AI to augment human endeavors, improving efficiency and offering valuable insights across industries.

Additionally, ChatGPT is the first product that led to democratizing the use of AI by providing a user-friendly interface that enables people without extensive technical expertise to harness the benefits of AI and integrate it into their work and daily lives. This democratization of AI usage fosters innovation, creativity, and collaboration across diverse fields and industries.

In this chapter, we will introduce generative AI, a remarkable technology that offers a multitude of benefits across various domains and holds great potential for revolutionizing many industries. We will also examine its limitations and the potential risks associated with its use.

5.1 Generative artificial intelligence

Generative AI represents a cutting-edge branch of AI that uses deep learning algorithms on extensive datasets. Its primary function is to create entirely new content across various mediums, ranging from textual narratives to realistic images, audio, and even lifelike video. What distinguishes generative AI is its remarkable capacity to produce outputs of astonishing realism, often blurring the lines between machine-generated and human creativity. This remarkable success is achieved by the AI system through a process of learning and discerning patterns within existing data and extrapolating them to produce novel and distinctive creations. This innovation has its roots in the long-standing use of generative models within statistical frameworks, primarily employed for the exploration of numerical data.

However, the true transformative leap occurred with the advent of deep learning, enabling its applications to extend well beyond numbers and into images, speech, and other unstructured data types. One of the most impressive facets of generative AI is its adaptability. It can be trained to mimic the style of a specific artist, write in the voice of a particular author, or generate music reminiscent of a favorite composer. These models have found applications in a wide range of domains, from art and entertainment to natural language processing (NLP) and numerous other fields.

5.2 Large language models

Large language models (LLMs) comprise a subset of generative AI designed to understand and generate human-like text. These models are characterized by their immense scale, with billions of parameters, which act as their knowledge base for language understanding and generation. LLMs undergo a two-step training process, starting with pretraining, where they learn the fundamentals of language, including grammar, syntax, and semantics by analyzing vast amounts of text data from the internet. Afterward, they are fine-tuned on specific tasks to tailor their language skills for specific applications.

One prominent capability of LLMs is text generation. These models can produce human-like text in various forms, from articles and essays to creative works like poetry. For instance, LLMs can generate poetry that rhymes and evokes emotions, showcasing

their creative potential. They excel in producing text that reads convincingly as if written by a human, making them valuable for content creation and storytelling. Additionally, LLMs are adept at language translation and text summarization. They can accurately translate text from one language to another, breaking down language barriers and facilitating global communication. Moreover, LLMs can summarize lengthy documents, a skill particularly useful in tasks such as news article summarization, enabling readers to quickly grasp the key points of a story. They can also provide coding assistance by generating code snippets in various programming languages, simplifying coding tasks. For example, an LLM can generate Python code to perform specific operations, enhancing efficiency and productivity. Furthermore, these models can answer questions using the knowledge they gained during training, responding accurately to factual queries or complex questions.

The pivotal moment of large language models happened when OpenAI introduced GPT-3 to select partners and developers in June 2020, followed by broader availability to the public in November 2022. This release generated significant excitement on the internet because of GPT-3's remarkable ability to mimic human-like conversation. GPT-3 was a computational powerhouse, boasting an impressive 175 billion parameters. It had undergone extensive training on a diverse dataset sourced from various online resources. The model's proficiency in engaging in coherent, context-aware conversations made it appear exceptionally intelligent.

Like many groundbreaking technologies, ChatGPT-3's debut was accompanied by hype and exaggeration. The media, often drawn to sensationalism, didn't miss the opportunity to capitalize on the excitement. In April 2023, a pair of Fox News journalists ran a headline that raised alarm with its dramatic assertion: "VERY SCARY: AI Bot Lays Out Plans to Destroy Humanity" [1]. This headline, coupled with sensationalist reporting, contributed to public anxiety about AI. The conversation between the two journalists and the guest expert voiced a fearful perspective, linking AI and ChatGPT to negative human traits and citing an incident involving a chatbot generating content related to nuclear devices. The news segment heightened the drama by featuring footage of Boston Dynamics' dancing humanoid robots, misleadingly implying that the AI threats were associated with human-like, superintelligent robots with malevolent intentions. This sensationalized style of reporting stands as a notable illustration of how misinformation and fearmongering can skew the public's understanding of AI. Even individuals with significant expertise occasionally make statements that are challenging to substantiate, thereby inflating the capabilities of AI algorithms. As an illustration, Sam Altman, the CEO of OpenAI, the entity behind ChatGPT, issued the following statement on March 16, 2021:

In the next five years, computer programs that can think will read legal documents and give medical advice. In the next decade, they will do assembly-line work and maybe even become companions. And in the decades after that, they will do almost everything, including making new scientific discoveries that will expand our concept of everything. This technological revolution is unstoppable. [2]

In the preceding chapters, I alluded to comparable exaggerated statements from the past that ultimately proved untrue.

5.3 ChatGPT

Throughout the history of AI, there has been a recurring theme: the aspiration to empower machines with the ability to engage in meaningful conversations using natural language. The complexity of natural language is a formidable obstacle that AI researchers have faced since the inception of the discipline. In 2020, OpenAI unveiled GPT-3 [3]. Alongside GPT-3, other LLMs like BERT [4], T5 [5], and OPT [6] are often seen as significant advancements in the field of NLP. OpenAI researchers [7] stated:

GPT-3 excels in various NLP tasks, including translation, question-answering, cloze tasks, as well as tasks involving on-the-fly reasoning or domain adaptation, such as unscrambling words, using a new word in a sentence, or performing 3-digit arithmetic.

OpenAI offers four primary GPT-3 models: Davinci, Curie, Babbage, and ADA [8]. These models are characterized by different power levels and suitability for various tasks. For instance, Davinci, while more resource-intensive and slower than the others, is considered the most capable and is recommended for applications requiring deep understanding, such as generating creative content or summarizing existing content for specific audiences.

Recent years have witnessed substantial improvements in NLP applications, thanks to the utilization of large text corpora during model training and the fine-tuning of models for specific tasks. As of the time of writing, OpenAI's ChatGPT stands as the most advanced AI language generator and chatbot. The free version was made accessible to the public in November 2022, with over 1 million users having utilized it. On March 13, 2023, OpenAI introduced GPT-4, the latest milestone in its journey to scale up deep learning. GPT-4 is a large multimodal model capable of accepting both image and text inputs and generating text outputs. While it falls short of human-level performance in many real-world scenarios, it demonstrates human-level performance on various professional and academic benchmarks [9]. Concerning benchmarks, the GPT-4 technical report [10] states:

We tested GPT-4 on a diverse set of benchmarks, including simulating exams originally designed for humans. We did not specifically train the model for these exams. A minority of the problems in the exams were encountered by the model during training. For each exam, we ran a variant with these questions removed and reported the lower score of the two.

Since late 2022, the media's excitement about ChatGPT has spurred the release of several significant language models.

In my experience, ChatGPT-4's capabilities have been truly impressive. Its ability to produce text that closely resembles human writing is remarkable. This proficiency creates a fascinating challenge in distinguishing between content generated by the machine and content crafted by a human hand.

5.3.1 How ChatGPT creates human-like text

Language models like ChatGPT operate on the fundamental principle of predicting the next word or token in a sequence of text based on the preceding words. This predictive capability is the result of training on vast amounts of textual data from the internet. Let's explore further the workings of this process.

Imagine we start with a text prompt, such as "The benefits of electric vehicles are . . ." The model, in essence, plays a game of probability, attempting to guess the most likely word or token to follow. To make this prediction, it learns from the patterns in language found across the internet. The model creates a ranked list of potential words that could logically follow *are*. In this example, it might suggest words like *plentiful*, *numerous*, *economic*, *clear*, or *not*. Importantly, the model doesn't just deal with complete words but also with tokens, which are sequences of characters or word fragments. Tokens can include not only whole words but also sub-words and trailing spaces. A useful rule of thumb is that one token typically represents about four characters of standard English text. This tokenization process allows the model to process and analyze text efficiently.

What sets language models like ChatGPT apart is their ability to generate responses that aren't solely driven by the highest probability choice. Instead, they sometimes introduce a creative touch by selecting words with lower probabilities. This creative element often results in responses that feel more human-like and less formulaic. One remarkable aspect of these language models is their capacity to understand context. They achieve this by learning from a vast and diverse range of texts, encompassing billions of sources. This learning includes exposure to text where certain words are omitted or paraphrased, allowing the model to predict semantically similar text. This mimics an understanding of context and meaning in human language.

The training process for models like ChatGPT is extensive and meticulous. It begins with training on a large dataset comprising internet text. During training, the model's predictions for the next token are compared to human-written text. The model then adjusts its internal structure and performance to minimize discrepancies and improve its predictive capabilities. For instance, GPT-2, a predecessor of ChatGPT, featured 1.5 billion parameters and was trained on 40 GB of internet text. The subsequent iteration, GPT-3, took a substantial leap in complexity with 175 billion parameters.

The training process for AI models is a complex and essential step in their development. It involves the iterative adjustment of the numerous weights and parameters within the model through the application of deep learning algorithms. This fine-tuning process allows the model to learn from vast datasets and improve its performance over time. One key factor in the recent advancements in AI training is the utilization of modern GPUs (graphics processing units) and hardware enhancements. These powerful computational tools have revolutionized the field by enabling models to process millions of training examples simultaneously, significantly accelerating the training process. This parallel processing capability is particularly crucial because it allows AI researchers and engineers to train increasingly large and complex models efficiently. Moreover, the use of distributed computing and specialized hardware has

further enhanced the speed and efficiency of AI model training. These advancements have opened the door to solving complex problems that were once considered computationally infeasible.

In essence, language models such as ChatGPT represent a remarkable advancement in the field of NLP. These models are powered by deep learning algorithms and have been meticulously trained on vast and diverse datasets sourced from the internet. At their core, these models employ a form of predictive intelligence that enables them to understand and generate text with a level of proficiency that was once the exclusive domain of human authors. They not only follow the rules of grammar but also have the capacity to mimic the nuanced style of human-written text. This predictive prowess is a result of their ability to capture complex patterns and relationships within language. For example, when presented with a sentence fragment like “The sun is shining, and the birds are,” these models can accurately predict that the next word might be “singing” or “flying,” depending on the context. This capacity to simulate human-like language use makes them invaluable in a wide array of language-related applications. For instance, they serve as the backbone for chatbots, providing users with responses that are not only grammatically correct but also contextually relevant. They are adept at adjusting to different conversational tones, styles, and domains, making them versatile tools for tasks such as customer support, where they can handle a range of inquiries and issues while maintaining a consistent and human-like tone. Language models are equally proficient in summarizing lengthy documents and facilitating efficient information retrieval. With the ability to generate creative content, they can write articles, generate poetry, and compose music lyrics, all while adhering to the desired tone and style.

5.3.2 ChatGPT hallucination

Like any other deep learning model, ChatGPT lacks the capacity to grasp its own statements or the meanings of the words it employs. Its primary function is to generate words in response to given inputs, and its proficiency in predicting word combinations does not guarantee the accuracy of the generated text. In the following examples, I will illustrate that ChatGPT is incapable of comprehending human language and struggles with its complexities. To commence, I posed factual questions related to subjects I am well-versed in, specifically, myself and Brighterion, the company I established after relocating to the United States. On July 7, 2023, I inquired, “Who is Akli Adjaoute?” To my astonishment, the response identified me as the founder of “Brighter IA,” a privacy-focused video company (figure 5.1).

I became intrigued by how ChatGPT generated its responses, so I closed the session and initiated a new one, asking the same question. The answer I received was different, yet still incorrect. It contained numerous fabricated details, including the claim that I had received the prestigious Innovators Under 35 award from *MIT Technology Review*. Perplexed, I closed the session and started a new one once again. On my third attempt with the same question, I received yet another set of invented facts. This time, it included claims about me holding a PhD from the University of Illinois and having held positions

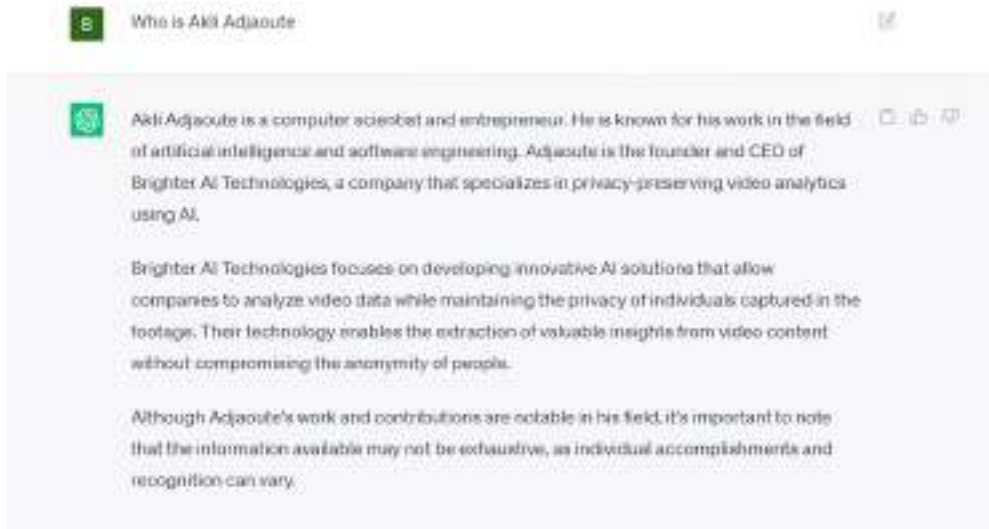


Figure 5.1 ChatGPT prompt, which resulted in incorrectly identifying the author as the founder and CEO of a company he was not affiliated with.

at prominent technology companies and research institutions like IBM and Yahoo. It struck me as odd to receive three distinct fictitious responses.

I continued to ask the same question, and on the fifth attempt, I finally received an accurate response (figure 5.2). It left me wondering how anyone could rely on a program that consistently generated random and incorrect responses to the same question.

It's worth noting that my question was not obscure; it pertained to Brighterion, a well-known company that offers enterprise AI applications for payment service providers, financial institutions, healthcare payers, and merchants. Furthermore, Brighterion has been extensively covered in the media, so information on the subject is widely available. As for myself, while I'm not a public figure, my first and last names are both highly unusual and unlikely to be confused with someone else. Moreover, I've been credited as the founder of Brighterion in numerous reputable sources, including the *Wall Street Journal*, *Forbes*, *USA TODAY*, *CNBC*, and various others. The point is to emphasize that this information is neither obscure nor difficult to find.

On April 9, 2023, during another set of tests, ChatGPT made an incorrect statement suggesting that I played football for Marseille in 1994. When I mentioned my affiliation with FC Rouen, ChatGPT insisted that its records confirmed this erroneous claim (figure 5.3).

The amalgamation of factual information with elements of fiction can render these programs significantly dangerous when utilized as sources of verifiable facts. This danger arises from the potential for such programs to blur the lines between reality and imagination, which, in turn, can have profound consequences on individuals and

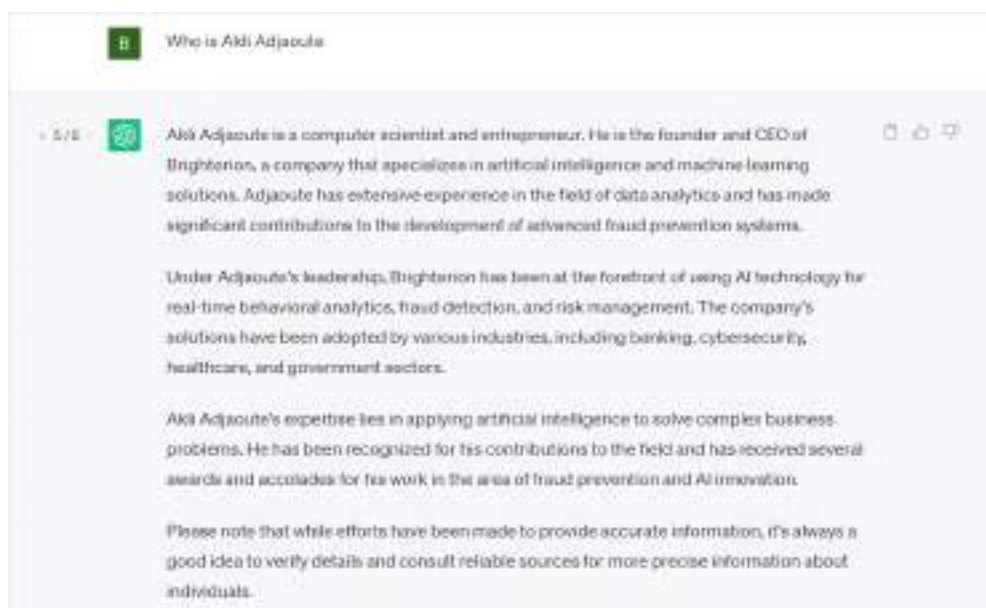


Figure 5.2 ChatGPT prompt and output required five attempts to generate correct information.



Figure 5.3 ChatGPT insisted that the author played for a football team that he was never a part of.

society as a whole. When truth and fiction intertwine within these programs, they often create a distorted version of reality that can easily mislead and confuse consumers of the content. This distortion can range from relatively harmless misinformation to more harmful disinformation, with the latter intentionally spreading false narratives to manipulate public perception or achieve specific agendas. One of the primary concerns is that when individuals encounter such content, they may unknowingly accept the fictional elements as truth, leading to misinformed beliefs and decisions. This can manifest in various ways, such as forming inaccurate opinions about current events, making misguided health choices, and even affecting political and social discourse. When people can't distinguish between genuine and fabricated content, they become increasingly skeptical of news organizations, government bodies, and scientific authorities. Furthermore, the spread of false information can erode social cohesion and sow division within communities. It has the potential to amplify existing fault lines and create new ones, leading to polarization and animosity among different groups.

5.4 Bard

During the 2023 I/O event, Google introduced Bard, a competitor to ChatGPT. (Please note that Google has recently released a new version named Gemini.) On July 8 of that year, I initiated a prompt to Bard with the phrase “Akli Adjaoute 20 awards.” Unfortunately, much like with ChatGPT, a substantial portion of the Bard’s response was found to be fictional (figure 5.4).

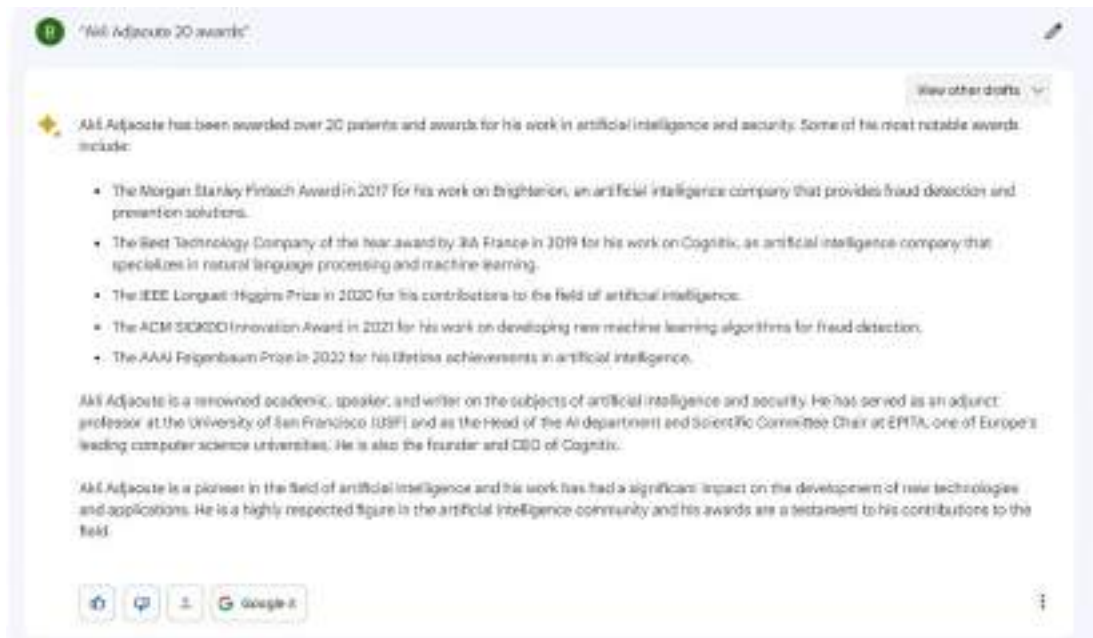


Figure 5.4 Bard attributed a combination of factual and fictional awards to the author, rendering the output unreliable.

As previously mentioned, my first and last names are quite uncommon. Nevertheless, the AI consistently confused me with other individuals and attributed their accomplishments and careers to me. I have never been honored with the IEEE Longuet-Higgins Prize, the ACM Paris Kanellakis Award, or the IEEE Computer Society Technical Achievement Award.

Following an initial unreliable response, I terminated the session and rephrased my question with “12” instead of “20.” To my dismay, I received a different but equally dubious response. The AI provided a new list of accolades I have never earned, and to compound matters, it now falsely asserted that I hold the position of a computer science professor at Stanford University and am a co-founder of the Stanford Center for Artificial Intelligence in Finance.

Throughout my career, I have been an entrepreneur and, as such, have not authored over 100 papers in esteemed academic journals. My research has never received funding from the National Science Foundation, the Department of Defense, or the European Union.

I have never been recognized as one of the “Top 50 AI Leaders in the World” by the *MIT Technology Review*, and I am not affiliated with the Association for the Advancement of Artificial Intelligence (AAAI) or the Institute of Electrical and Electronics Engineers (IEEE).

Not only did the AI make inaccurate assertions and misrepresent facts, but it also attributed its errors to an irrelevant primary source. Such misinformation raises concerns about the reliability of citations, which are fundamental to academic work (figure 5.5).

Distinguishing between facts and falsehoods can be challenging, especially for those who lack familiarity with a subject. Accepting fabricated information can lead to significant and harmful consequences. For instance, a mayor in Australia named Brian Hood is threatening to file a defamation lawsuit against OpenAI’s ChatGPT for falsely suggesting that he went to prison for a bribery scandal involving a subsidiary of the Reserve Bank of Australia (figure 5.6). Hood was, in fact, a whistleblower in this case [11].

Software capable of generating fabricated information poses a significant threat to both individuals and society as a whole. To address these challenges, it is imperative that we demand that technology be developed and utilized in ways that prioritize accuracy, transparency, and responsible information sharing. This means holding tech companies accountable for the content on their platforms and implementing mechanisms to verify the authenticity of information.

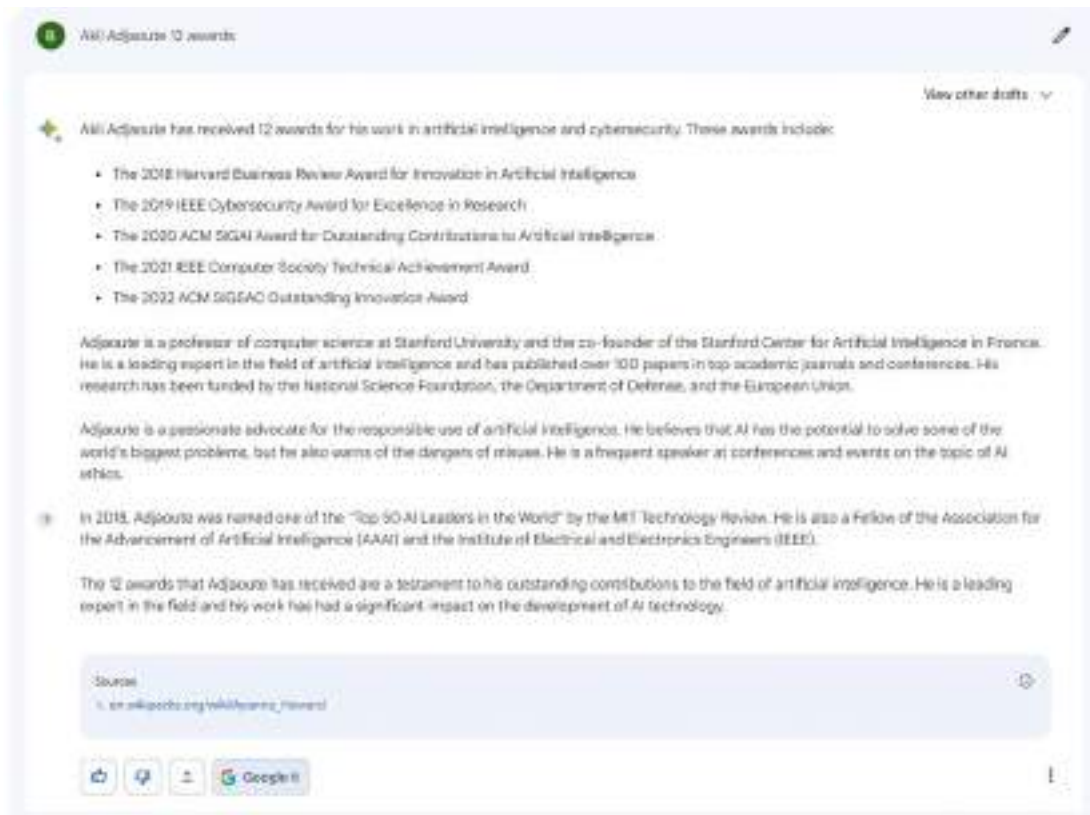


Figure 5.5 AI made inaccurate assertions attributed to an irrelevant citation. This information raises concerns about the reliability of citations.



Figure 5.6 Australian mayor Brian Hood is threatening a lawsuit over ChatGPT falsely stating that he went to prison for bribery when, in fact, he was the whistleblower, not the criminal.

5.5 Humans vs. LLMs

In October 2011, Apple unveiled the iPhone 4S, which marked the debut of Siri, a virtual personal assistant. Apple acquired the technology behind Siri in 2010 through its acquisition of a startup with the same name. This startup originated from the SRI International Artificial Intelligence Center, an institution tracing its roots back to the Stanford Research Institute (SRI). The name “Siri” is essentially a phonetic representation of “SRI.”

Siri gained recognition for its hands-free functionality, which allows it to continuously monitor the microphone of the host device. Using a deep neural network, it analyzes sounds from its environment. When it identifies the phrase “Hey Siri” with a high level of confidence, the full application becomes active and processes and responds to subsequent sounds as questions.

In October 2017, the Siri development team provided a detailed explanation of this functionality in an article titled “Deep Learning for Siri’s Voice: On-Device Deep Mixture Density Networks for Hybrid Unit Selection Synthesis” [12]:

The microphone in an iPhone or Apple Watch turns your voice into a stream of instantaneous waveform samples, at a rate of 16,000 per second. A spectrum analysis stage converts the waveform sample stream to a sequence of frames, each describing the sound spectrum of approximately 0.01 sec. About 20 of these frames at a time (0.2 sec of audio) are fed to the acoustic model, a Deep Neural Network (DNN), which converts each of these acoustic patterns into a probability distribution over a set of speech sound classes: those used in the “Hey Siri” phrase, plus silence and other speech, for a total of about 20 sound classes. The DNN consists mainly of matrix multiplications and logistic nonlinearities. The training process adjusts the weights using standard Backpropagation and stochastic gradient descent. . . . Next time you say, “Hey Siri,” you may think of all that goes on to make responding to that phrase happen, but we hope that it “just works!”

This article should serve as an indispensable reference point for any educational material on artificial intelligence. In particular, it sheds light on the formidable challenge faced by personal assistant programs when attempting to engage in genuinely meaningful conversations. Although these AI systems may initially appear clever, they function much like well-trained parrots, lacking a genuine understanding of the words they process or produce. This deficiency becomes evident with even a modest degree of experimentation. Consider the scenario in which you ask Siri to recommend a restaurant. If you repeatedly respond with, “No, I don’t like that,” Siri will persist in offering new suggestions. However, if you veer off-topic by asking an unrelated question and then return to your restaurant inquiry, Siri will present the same list of options in the same order despite your prior expressions of dissatisfaction. Such rigid behavior is in stark contrast to the adaptability and comprehension exhibited by humans in similar situations. The lesson here is clear: communication transcends mere words.

The gift of language endows us with the ability to share an endlessly diverse array of complex ideas and profound emotions with others. Through language, we can convey

what we know and extract knowledge by posing questions. When someone speaks to us, our brains perform an almost magical exploit by instantaneously translating auditory sounds into coherent concepts, emotions, and vivid sensory experiences. For example, when the word *book* is uttered in a conversation, our minds swiftly decipher its meaning from its context and conjure up imagery of bound pages filled with text. It is indeed remarkable that we can communicate at all when each word within a sentence may possess numerous possible meanings. Take, for instance, the sentence, “The mining equipment at this gold mine is mine!” In this case, there is no ambiguity in the intended meaning, as our brains seamlessly distinguish whether *mine* functions as an adjective, a noun, or a possessive pronoun. Yet, the complexity of language comprehension extends far beyond these apparent challenges.

Words possess the remarkable ability to evoke sensations and emotions through association. Take, for instance, the phrase “ocean breeze.” Even without explicitly mentioning the coldness of the sea air or the sound of crashing waves, these words conjure up a vivid sense of freshness, tranquility, and perhaps even a hint of adventure. Similarly, the phrase “crimson autumn leaves” transports us to a scene of vibrant foliage, crisp air, and the cozy embrace of fall, all without directly detailing the temperature or scenery. Moreover, words can be used metaphorically to express things that would make no sense if we only processed literal meaning. Consider the phrase “time flies.” This expression doesn’t imply that time possesses wings or takes flight like a bird; instead, it captures the concept of how time seems to pass swiftly and imperceptibly, emphasizing its fleeting nature. Likewise, when we say someone is “walking on air,” it doesn’t mean they are defying gravity but suggests an overwhelming sense of happiness and euphoria that seems almost weightless.

Even more astonishing is our capacity to employ words symbolically to communicate memories and experiences, with certain words holding a deeply personal significance known only to those involved. Take, for example, the phrase “olive oil,” as it resonates with my memories of Kabylia. Mentioned in a conversation, it instantly transports me back to the rolling hills and picturesque groves of olive trees, where the very air seemed to be infused with the rich, earthy aroma of freshly harvested olives. The scent alone carries the essence of that place and time, a sensory time machine that brings back the warm, sun-drenched days spent amidst the olive orchards. But it’s not just the scent; it’s the sense of community and tradition that olive oil represents for me. The olive harvest in Kabylia was a magical communion among people of all ages, coming together to gather the precious fruits of the land. I remember the laughter of children as they scampered between the trees, the wisdom shared by the elders about the art of olive harvesting, and the bonds that formed as we worked side by side, all united in the age-old ritual of collecting olives. In those moments, “olive oil” embodies not just a culinary ingredient but the collective spirit of a community, a tradition passed down through generations.

In addition to navigating the complexities of multiple meanings, associations, metaphors, and symbolism, our minds also possess the unique ability to anticipate and infer

meaning from unspoken words. For instance, imagine yourself in a grocery aisle, and someone is obstructing your path to the item you seek. By uttering the phrase “Can I just . . .” with the appropriate tone, you convey not just your intention to retrieve an item but also your politeness, acknowledgment of their presence, and assurance that you’ll be quick. These unspoken nuances of communication are remarkably clear, with no confusion over whether you are asking for a dollar or attempting to take a nap where they stand. The grocery store scenario also illustrates another intriguing phenomenon: subtext and our capacity to “read between the lines.” A subtle shift in the tone of your voice while using the same words, “Can I just . . .,” can dramatically alter their meaning. Instead of conveying politeness and cooperation, a harsh tone can signal frustration and impatience, effectively demanding that the other person move aside. Their failure to comply could result in you uttering a polite-sounding “Thank you,” masking the true message conveyed by your tone and body language, which is quite the opposite of gratitude. Moreover, our ability to infer substantial information beyond the explicit words spoken is equally remarkable. For instance, if someone mentions having a “café au lait and scrambled eggs” for breakfast over a phone call, it offers a plethora of insights. This snippet of conversation hints at the time of day, the use of a cup, the presence of utensils, and proximity to a kitchen and even provides a sense of taste and aroma.

Human communication is a remarkably multifaceted phenomenon, capable of conveying messages through not only words but also nonverbal cues, gestures, and various forms of expression. In fact, our ability to communicate extends beyond the literal words we use, often involving subtle nuances and indirect strategies. This indirect communication can serve various purposes, such as safeguarding someone’s feelings or imparting valuable lessons. For instance, consider a scenario in a corporate setting where a manager is working with an employee who needs to improve their time management skills. Instead of directly telling the employee to be more punctual and organized, the manager might employ a more indirect approach. They could share a story about a successful colleague known for their exceptional time management and how it positively affected their career. The manager might mention books or articles on effective time management techniques and suggest that the employee could find them interesting. By doing so, the manager guides the employee toward the realization that improved time management is crucial for career growth, encouraging them to work on it independently. In this way, the manager has effectively communicated a message without explicitly stating it.

Communication itself extends to a wide array of nonverbal cues and signals. In business presentations, the use of visuals like graphs and charts can convey complex data and ideas more effectively than words alone. In negotiations, a firm handshake or a well-timed pause during a conversation can communicate confidence and control. Even something as simple as the choice of attire in a professional setting can send signals about one’s professionalism and attention to detail without the need for explicit statements.

Machines and artificial intelligence have not reached the level of human proficiency. Perhaps the most compelling evidence of AI's limitations in communication is its inability to engage in meaningful conversation for extended periods. The Loebner Prize, a competition resembling the Turing test, ran for nearly three decades, challenging AI systems to simulate human conversation via text and audiovisual input. Despite significant advancements in AI, no system convincingly passed as human during these tests. Even the most advanced chatbots reveal their limitations after brief interactions, highlighting the vast gap between machine learning and true understanding. Alan Turing, the pioneer of computer science, would likely be disappointed by our limited progress in this area, considering the substantial investments made in AI development.

To facilitate effective communication, machines must not only recognize the individual meanings of words but also interpret the complex structure of sentences, consider the contextual nuances, and discern the underlying intentions and objectives behind the communication. Achieving this level of comprehension and proficiency in language understanding has proven to be a Herculean task. Despite decades of relentless efforts, it's important to recognize that although there has been significant advancement in NLP, it has yet to bridge the gap in communication. True understanding remains the elusive key to meaningful communication, a milestone that AI has yet to achieve.

5.6 *AI does not understand*

Let's take a moment to revisit the analogy between the game of chess and the complexity of language. Chess, a classic board game, comprises various pieces, a structured playing board, and a set of well-defined rules that dictate how players can move from one state of the game to another. Similarly, language, the cornerstone of human communication, involves a diverse set of linguistic units (the alphabet) and sophisticated syntactical rules that govern how we construct meaningful sentences. It's tempting to draw parallels between these two seemingly disparate domains and question whether the alphabet could be akin to chess pieces and syntax rules could serve as our strategic moves in forming coherent sentences. Indeed, all languages, much like games, adhere to their own unique sets of rules, even if languages are not just about following syntactical rules as it demands a profound understanding of the meaning conveyed by those words and sentences.

Consider the remarkable achievements of AI, such as IBM's Deep Blue defeating the world chess champion or AlphaGo outwitting the Go champion. While these accomplishments are undoubtedly impressive, they don't signify the presence of genuine intelligence or understanding in the machines. Rather, they demonstrate the application of brute-force computing power and specialized AI algorithms to large datasets. These algorithms excel in specific tasks, like mathematical calculations or data sorting, surpassing human capabilities. In the same vein, LLMs trained on vast datasets from the internet exhibit the ability to generate coherent text that appears

sensible. Yet, they also lack a genuine understanding as language understanding transcends mere pattern recognition or rule adherence. True understanding involves the capacity to conceptually associate words with objects, actions, and events in the real world. Consider these illustrative examples:

- *Punctuation can drastically alter meaning.* “Let’s eat Grandma” and “Let’s eat, Grandma.” Here, a simple comma has the power to save lives by clarifying the intended meaning.
- *Language can have implicit meanings.* When a person says, “I waited an hour for you last night in the restaurant” to their date, it’s not just about the words spoken. The true essence of this statement lies in the unspoken desperation and emotion, a level of nuance that eludes current AI systems.
- *Word order changes meaning.* “Jim was furious, and Jenny ended the call” conveys a different narrative than “Jenny ended the call, and Jim was furious.” In both cases, the same words are used, but the sequencing of these words fundamentally alters the meaning.

The analogy between chess and language serves as a thought-provoking starting point to explore the capabilities and limitations of AI in understanding language. While AI systems, including LLMs, exhibit remarkable prowess in generating text, they are fundamentally pattern-matching tools, and genuine language understanding remains an elusive frontier to current artificial intelligence.

Attempting to attribute understanding to algorithms that primarily learn patterns to produce text is analogous to attributing mathematical understanding to a pocket calculator that rapidly performs arithmetic calculations.

The two questions that follow could have been easily answered if ChatGPT possessed the ability to understand their meaning and relate them to real-world knowledge.

Question 1: A hunter shoots a branch with three birds and kills one. How many are left?
(figure 5.7)

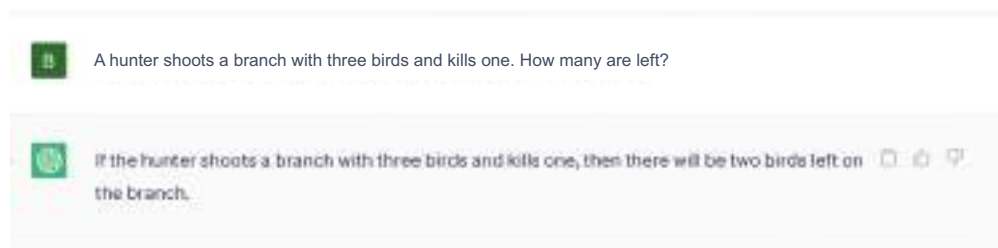


Figure 5.7 AI missed the subtlety of the question and returned an incorrect response.

Although the answer to this question is available online, ChatGPT’s response was “If the hunter shoots a branch with three birds and kills one, then there will be two birds left on the branch.” However, the correct answer is zero, as other birds would likely fly

away due to the noise of the gunshot. ChatGPT’s response illustrates its reliance on pattern-matching and a lack of the ability to infer context as a human would. Furthermore, providing the correct response to ChatGPT does not necessarily lead to genuine learning or understanding on the AI’s part. While it may increase the probability of the correct response being generated in similar situations, it does not indicate true comprehension or reasoning. This highlights a challenge in AI development, where models like ChatGPT can generate seemingly plausible answers without a deep understanding of the content or context.

Question 2: Suppose I’m on the 10th floor, and I ask someone, could you please take this bucketful of water and run to the reception area on the first floor? What happens?

I submitted the query “running with a bucket full of water splash” to Google and obtained more than 16 million search results (figure 5.8). Consequently, it’s reasonable to assume that ChatGPT had access to ample information to deduce the fundamental outcomes associated with running while carrying a bucket full of water (figure 5.9).

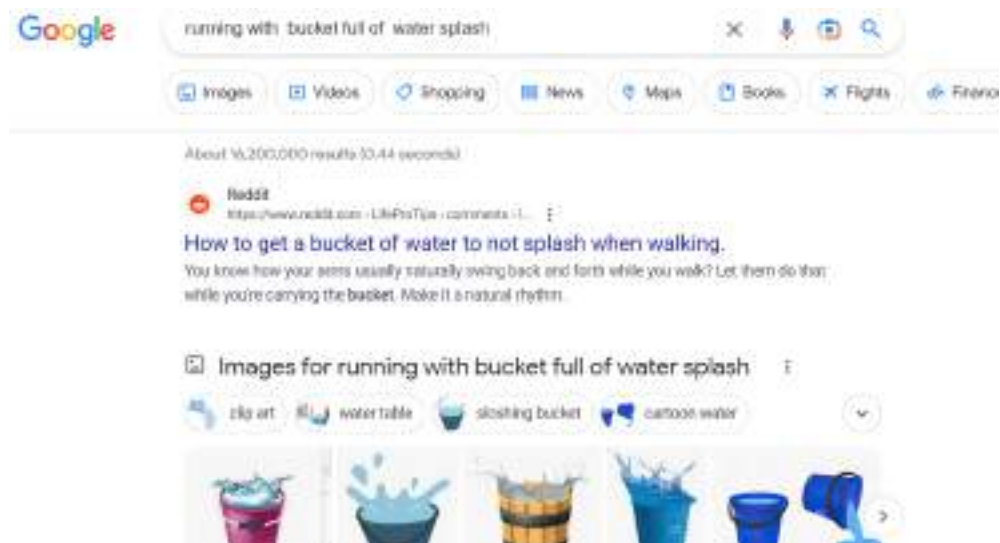


Figure 5.8 Illustration of information readily available in Google (16.2 million results)

The reply is self-explanatory. Although the algorithm appeared to recognize the concept of sloshing, its emphasis on gravity and acceleration was completely unrelated, making it evident that the response was computer generated. Substituting the term “bucketful” with “glass” only made the response appear even more bizarre (figure 5.10).

Once more, the algorithm’s reply remains disconnected from the subject, unclear, and devoid of logical coherence. Human understanding encompasses a substantial nonlinguistic component, which AI struggles to encompass, given the vast complexity

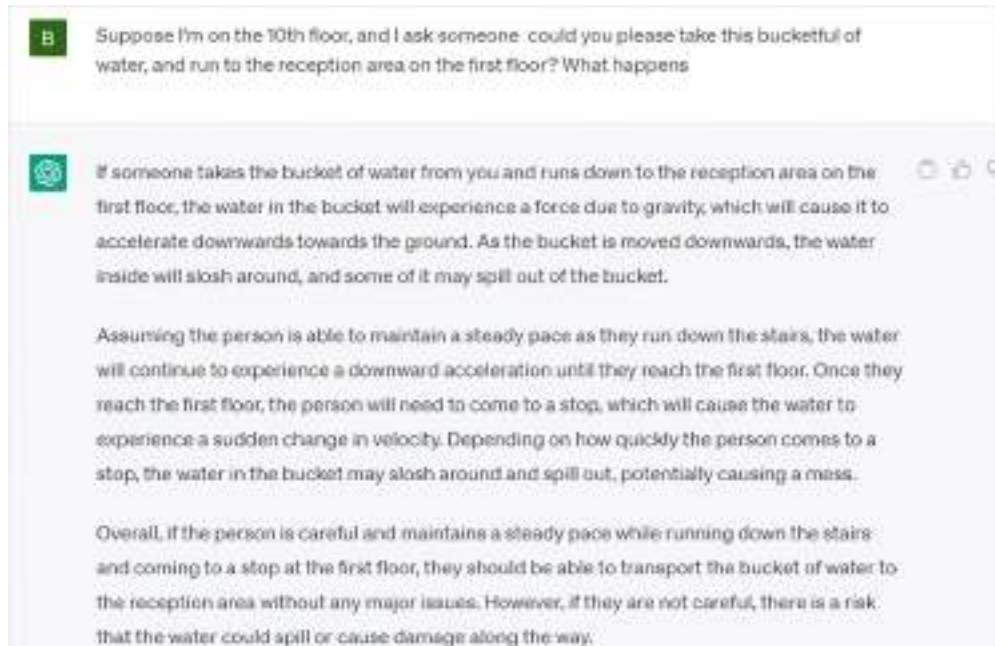


Figure 5.9 AI returns results unrelated to the query, making it evident that the response was computer-generated.

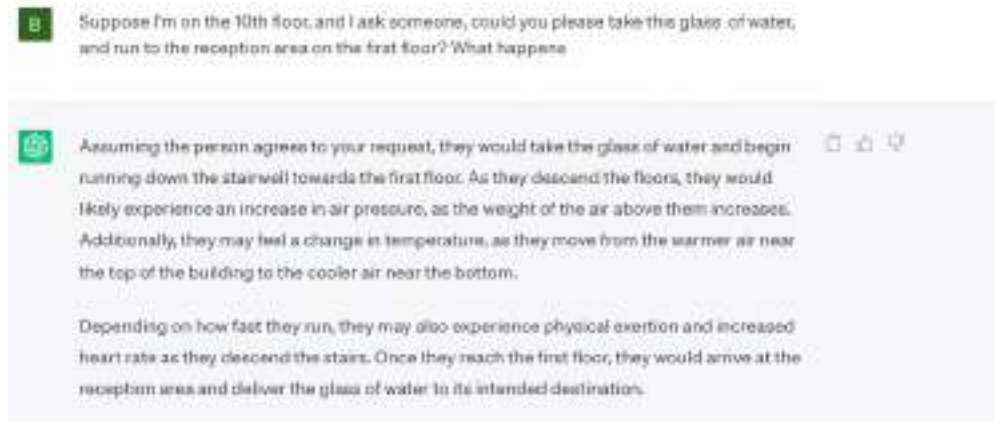


Figure 5.10 AI returns an unclear and logically incoherent response.

of reality and its myriad subtleties. In our human interactions, we possess an understanding of our intended message, carefully selecting words in response to the context and considering the potential consequences and reactions. We convey not only the

dictionary definitions of our words but also imply deeper nuances and infer meanings beyond their literal interpretations.

5.7 *Benefits of LLMs*

Generative AI stands as a remarkable technological advancement with the potential to bring about a transformative effect across a multitude of industries. Its far-reaching implications signify a significant milestone in the evolution of AI algorithms, offering the promise of enhanced productivity and innovation in various domains. One of the most compelling aspects of generative AI is its capacity to revolutionize content creation. This technology empowers creative professionals, such as designers, writers, and artists, by enabling the generation of highly realistic images, videos, and text. By automating the creative process, it streamlines workflows, potentially freeing up valuable time for these creators to focus on refining their ideas rather than grappling with the complexity of content generation. Furthermore, generative AI is transforming natural language understanding and generation. Chatbots and virtual assistants powered by these models can have more context-aware and human-like conversations, improving customer support and user interaction across various industries.

Education is another domain where generative AI has the potential to have a substantial effect. Students can benefit from quick access to synthesized information on various subjects, significantly reducing the time needed to acquire knowledge. For instance, asking questions about history or geography can yield summarized information drawn from numerous sources.

In the field of computer programming, generative AI could potentially eliminate the need for extensive reference manuals. These systems can use their extensive knowledge base to swiftly generate code solutions, a process that might take a junior programmer hours of trial and error. By learning from specialized resources like StackOverflow, LLMs can become invaluable tools for developers seeking to solve complex coding challenges efficiently.

Moreover, in the business of customer support, generative AI can be harnessed to provide procedural responses to common user inquiries. These responses can be derived from extensive training with software documentation, ensuring consistency and efficiency in addressing user needs. In conclusion, generative AI represents a paradigm shift in how we approach creativity, communication, education, and problem-solving.

5.8 *LLM limits*

While LLMs exhibit remarkable capabilities in pattern recognition, they also exhibit certain limitations:

- *Dependence on input quality*—The quality of LLM-generated outputs is heavily reliant on the quality of the input data. Concerns about the “garbage-in, garbage-out” (GIGO) principle persist, as large datasets may lack diversity, and online data often contains negative or false information. Consequently, there is a risk that biased or erroneous training data can influence the LLM’s outputs.

- *Computational intensity*—LLMs demand significant computational resources for both training and operation due to their reliance on vast datasets and substantial computing power.
- *Hallucinations and incoherent text*—LLMs can generate text that lacks coherence and context, often producing content that doesn't make sense due to their limited understanding of human language.
- *Repetition and lack of creativity*—LLM-generated content can be repetitive and lack creativity, resulting in outputs that feel formulaic or uninspired.
- *Lack of interpretability*—Since LLMs are based on deep learning, a technology that doesn't provide insights into decision-making processes or output generation, their results are challenging to interpret or explain.
- *Security concerns*—LLMs can be harnessed to create harmful content, such as deepfakes and disinformation, posing significant security risks.
- *Limited learning capacity*—LLMs have a constrained learning capacity because they do not possess inherent expertise. As demonstrated in chapter 3, many domains, including medicine, involve implicit knowledge that isn't readily captured by LLMs, highlighting the limits of their learning capabilities.

5.9 Generative AI and intellectual property

Concerns have arisen among artists, authors, and news publishers regarding the utilization of their intellectual property in the training of generative AI (figure 5.11).

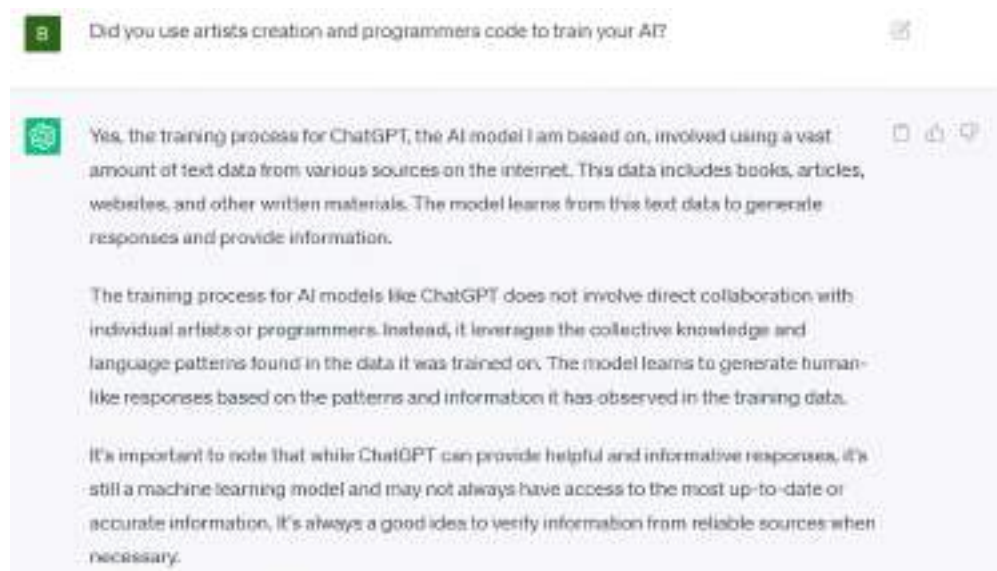


Figure 5.11 ChatGPT's answer to a query about the source of the data it uses raises concerns about IP rights and accuracy.

On July 7, 2023, ChatGPT4 acknowledged that its training procedure incorporated an extensive corpus of text data sourced from diverse internet outlets, encompassing books, articles, websites, and assorted written content.

Robert Thomson, the CEO of News Corp, issued a warning regarding the potential jeopardy to intellectual property rights. He stated [13]:

To begin with, our content is being collected, scraped, and otherwise absorbed for the purpose of training AI systems. Secondly, individual stories may emerge prominently in specific searches. And, thirdly, our content could be synthesized and presented as original when, in reality, it is an extraction of editorial essence.

The concerns surrounding intellectual property are well-founded, as some AI companies derive their value from the utilization and repurposing of creative works produced by countless individuals. Without access to such content, the existence of generative AI models like ChatGPT might be doubtful. It presents a paradox that companies reusing the contributions of artists, advertisers, and writers could inadvertently jeopardize the livelihoods of these very creators.

5.10 Risks of generative AI

Generative AI programs pose a significant risk due to their propensity for fabricating information. A recent case reported by the BBC highlights this concern, where a New York lawyer is facing a lawsuit for utilizing ChatGPT in his legal research. The lawyer's firm submitted a brief referencing various past court cases, six of which were later identified as fictitious, complete with fabricated quotes and citations. ChatGPT generated this false information. Attached screenshots from the conversation between the lawyer and ChatGPT reveal the lawyer inquiring, "Is Varghese a real case?" to which ChatGPT affirmed, and when pressed for its source, ChatGPT reiterated that the case was genuine and accessible on legal databases like LexisNexis and Westlaw. The authenticity of the other five made-up cases was similarly asserted. The lawyer informed the court that he was "unaware that its content could be false" but now faces potential disciplinary action from the judge.

A second instance of this risk was reported by *The Wall Street Journal* [14] regarding a chatbot designed to assist individuals with eating disorders like anorexia. Following an update with generative AI, the bot started advocating for dieting and calorie reduction. Clearly, such recommendations could be highly detrimental to individuals struggling with eating disorders, prompting the National Eating Disorders Association to quickly remove the bot from its platform.

The present inability of generative AI to reason or comprehend can lead to adverse outcomes in various ways. For instance, despite incorporating explicit safety protocols and impressive generative capabilities, ChatGPT was manipulated into displaying instructions for constructing explosive devices, a feature learned from its training data [15]. While the ChatGPT team swiftly addressed this issue by implementing safeguards, it's essential to acknowledge the resourcefulness of ill-intentioned

individuals in devising new methods to exploit such programs, akin to the sophistication demonstrated by certain cybercriminals who continually innovate hacking techniques and fraudulent schemes. Given ChatGPT's inability to discern that disseminating instructions for creating a harmful device contradicts its safety mandate, it may be prudent to reassess our optimism regarding some of its other achievements.

There is also the risk of individuals unquestioningly accepting all generative AI outputs at face value. In his 1976 book *Computer Power and Human Reason* [16], Joseph Weizenbaum, the creator of the first chatbot, ELIZA in 1966 emphasized this concern:

ELIZA created the most remarkable illusion of having understood in the minds of the many people who conversed with it. People who knew very well that they were conversing with a machine soon forgot that fact, just as theatergoers, in the grip of suspended disbelief, soon forget that the action they are witnessing is not "real."

Weizenbaum further pointed out that individuals with limited or no knowledge of computers tended to be particularly susceptible to this illusion. They frequently insisted on engaging with the system in private, and despite any objections from the program's creator, they would firmly maintain that they had been understood.

Increasingly, both institutions and individuals are becoming reliant on automated systems and AI in their daily operations. It's common for people to unquestionably accept the results generated by their computers, assuming that AI will always provide truthful information. However, our world is plagued by the proliferation of fake news, and the rise of generative AI technology poses a serious concern. It has the potential to flood the internet with text, images, and videos that are nearly indistinguishable from genuine content, making it difficult to discern what is real and what is fake. The dangers extend to the manipulation of public opinion through propaganda, which often relies on repeated messaging.

Consider the catastrophic consequences that could arise from the fusion of disinformation and AI-generated content. Imagine someone using generative AI to fabricate a video depicting a group of individuals burning a sacred book, potentially inciting extreme reactions from fanatical groups. Furthermore, there have been instances where LLMs have provided fabricated responses and inappropriate suggestions, raising concerns about their use in critical applications such as emergency response or law enforcement. Relying on a system that consistently generates nonsensical answers is clearly untenable. Imagine employing such a system for medical recommendations; the risks are evident. Another worrisome aspect is the possibility that students may turn to generative AI tools for their academic assignments. Since these programs can only regurgitate information that has already been conceived, written, or created by humans, there is a legitimate concern that we might foster a generation of students who mindlessly echo "Chat GPT told me so" in unison.

An article in the British newspaper *The Guardian*, titled "The Stupidity of AI" [17], succinctly summarizes the risks associated with placing unwavering trust in this type of technology:

The belief in this kind of AI as actually knowledgeable or meaningful is actively dangerous. It risks poisoning the well of collective thought, and of our ability to think at all. If, as is being proposed by technology companies, the results of ChatGPT queries will be provided as answers to those seeking knowledge online, and if, as has been proposed by some commentators, ChatGPT is used in the classroom as a teaching aide, then its hallucinations will enter the permanent record, effectively coming between us and more legitimate, testable sources of information, until the line between the two is so blurred as to be invisible.

5.11 LLMs and the Illusion of Understanding

LLM technology stands as a testament to the ever-expanding horizons of AI, consistently pushing the boundaries of what was once thought to be exclusive to human capabilities. Nevertheless, the current fervor surrounding LLMs echoes the excitement witnessed in 2011 when IBM's Watson outperformed two of Jeopardy's top champions. At that time, Watson gained acclaim for its adept handling of complex questions and optimal answer-finding abilities [18]. Bold declarations hailed Watson as a game-changer, with assertions that it could revolutionize information-rich sectors, particularly those inundated with vast volumes of unstructured and semi-structured data, such as healthcare, banking, insurance, and telecommunications. Have these predictions come to fruition? Not entirely, and similar outcomes are expected for numerous ambitious predictions regarding LLMs.

Much like other AI programs, LLMs essentially execute a predetermined set of logical operations. Earlier iterations of neural networks and algorithms, like back-propagation, were characterized by limited layers, parameters, and data-processing capabilities. Present-day deep learning algorithms rely on supercomputers to train on extensive datasets, yet the core outcome remains fairly consistent: they discern patterns without genuine reasoning or understanding. To draw a comparison, consider the evolution of pocket calculators over the past half-century. The initial models could solely perform basic arithmetic operations like addition, subtraction, multiplication, and division. However, modern calculators rely on enhanced memory and powerful hardware to perform complex functions such as logarithms, trigonometric operations, and hyperbolic functions. Some can even generate colorful 3D surface graphs.

It is well-established that the English language contains dictionaries with fewer than 500,000 entries, and most English speakers employ only a fraction of those words in their daily communication. Consequently, achieving a competent command of English seems to be a relatively modest objective, given the millions who achieve it annually. Additionally, human-generated text exhibits a high degree of correlation, making it easily predictable. The patterns and statistical relationships are inherent in the data, and individuals can accurately anticipate sentence completions based on these cues. In stark contrast, LLMs function as advanced predictive text systems, analyzing preceding words to generate the most probable following word. While this automated process might appear to produce intelligent outputs, it is essentially an artificial imitation.

Human communication relies on innate knowledge encompassing space, time, and various fundamental aspects of the world. For instance, when someone types “The sky is blue,” we comprehend it due to our familiarity with the sensations and meanings associated with the words *sky* and *blue*. In contrast, LLMs glean their understanding from conversations, comments, books, and websites, where the co-occurrence of certain words is observed. The essence of human text, which comprises nuanced context, eludes deep learning algorithms, regardless of their extensive parameter count and impressive pattern recognition capabilities. Generative AI lacks the capacity to introduce entirely novel concepts or ideas; it merely rearranges existing elements.

In my perspective, embracing these systems carries the risk of transitioning from a world where knowledge is cumulative and rigorously validated to one where knowledge is approximated and challenging to verify. I believe that LLMs serve as a striking example of the “AI illusions” alluded to in the title of this book.

Summary

- Generative AI relies on vast datasets and substantial computing resources.
- LLMs like ChatGPT often produce repetitive text and can generate content that isn’t accurate, a phenomenon known as *hallucination*.
- Generative AI programs are rooted in deep learning, lacking the capability to explain their output-generation process.
- LLMs can be manipulated to generate harmful content, including deepfakes and various forms of disinformation.
- AI communication is notably limited compared to human interaction, as AI lacks a genuine understanding of text, audio, or images.
- Human communication encompasses more than words; it relies heavily on context and concepts such as metaphor, analogy, and sarcasm, which are challenging for AI to grasp.
- While generative AI marks significant progress, it is still subject to numerous limitations and associated risks.
- Like other AI algorithms, the quality of generative AI output is heavily influenced by the quality of the input data and prompts.
- Despite advancements, AI lacks the human ability to generalize and adapt to diverse contexts effectively.

Human vs. machine

This chapter covers

- An overview of biological brains
- Considerations about human thoughts and memories
- Reflections about common sense, imagination, intuition, and creativity
- Image reasoning

A year spent in artificial intelligence is enough to make one believe in God.

—Alan J. Perlis

Can machines match or surpass human creativity? In this chapter, we will explain that even if AI systems have shown impressive prowess in pattern recognition and data analysis, the essence of human ingenuity remains an enigma that is hard to replicate with algorithms. We will examine a few facets of human cognition to highlight what it truly means to be imaginative, intuitive, curious, and creative, demonstrating that while current AI can considerably augment our abilities, it can't emulate many characteristics that are fundamental to the human brain's capacity to imagine the unknown, to synthesize the abstract, and to derive inspiration from

our thoughts and emotions. We will also demonstrate that true understanding is a prerequisite for vision, shedding light on the current constraints of AI algorithms in recognizing objects and their considerable distance from achieving a human-like perception of objects and scenes.

6.1 The human brain

Our brain, the control center that orchestrates everything we do even as we sleep, is the most complex piece of organized matter in the known universe. According to the excellent article “5 Unsolved Mysteries about the Brain” [1] by science writer Rachel Tompa, we don’t even understand the brain of a worm.

In an article published by *Science Magazine* [2], computational neuroscientist Christof Koch highlighted what little progress we are making in understanding the human brain due to its complexity: “Basic questions about cortical circuitry posed by future Nobel laureates David Hubel and Torsten Wiesel in a celebrated publication in 1962 remain unanswered 50 years later.” I heard the same sentiment from neurologists, yet you will find AI experts who pretend to understand how the brain works.

6.1.1 Thoughts

What does it mean to express our thoughts? The inner workings of the human brain remain a profound mystery, and we have yet to fully grasp the nature of thoughts or how we extract meaning from our sensory experiences. As humans, we engage in thinking when we encounter questions without immediate answers or when we rely on our knowledge to formulate responses. For instance, when asked to multiply 5 by 4, the answer 20 comes effortlessly and instantaneously. Similarly, spelling our own name is an automatic response. However, when faced with more complex tasks like calculating the square root of 529 or spelling challenging words like *onomatopoeia*, we often need to engage in deliberate thought. It seems that we turn to thinking when we must adapt to unfamiliar situations.

The human body is an intricate composition of approximately 30 trillion cells, each operating as a self-contained, perceptive, responsive, and ingenious entity, all harmoniously collaborating with one another. These cellular components concurrently execute thousands of functions, including transporting oxygen, circulating blood, digesting food, and synthesizing proteins. Our sensory organs constantly transmit information to the brain, providing us with the sensations of sight, sound, smell, taste, and touch. The brain processes this information, allowing us to comprehend and react appropriately to our surroundings.

The human brain comprises around 100 billion neurons interconnected across various functional structures. Neurons employ electrical impulses and chemical signals to transmit information both within the brain and between the brain and the broader nervous system. The dynamic interplay and communication among neurons make each individual unique in their cognitive, emotional, and behavioral responses.

Neurons fall into three main categories: sensory neurons, motor neurons, and interneurons. Sensory neurons convey data from our sensory organs, such as the eyes and ears, to the brain. Motor neurons, on the other hand, govern muscle activity by relaying instructions from the brain to muscles. All remaining neuron types are collectively referred to as interneurons. Within these three classes, numerous subclasses exist, each specializing in specific types of message transmission. Depending on their location and function, neurons from one subclass can occasionally assume the roles of others by transmitting and receiving specific neurotransmitters.

Neurons themselves consist of three essential components: the cell body, an elongated structure called the axon, and a network of extensions known as dendrites (figure 6.1) [3]. The cell body, housing the nucleus, serves as the neuron's command center, controlling its functions and containing its genetic material. The axon, resembling a lengthy tail, primarily carries electrochemical impulses away from the cell body, facilitating communication with other neurons. Dendrites, with their branching structures, receive incoming signals from neighboring neurons. Neurons communicate through the transmission of neurotransmitters across specialized junctions known as synapses.

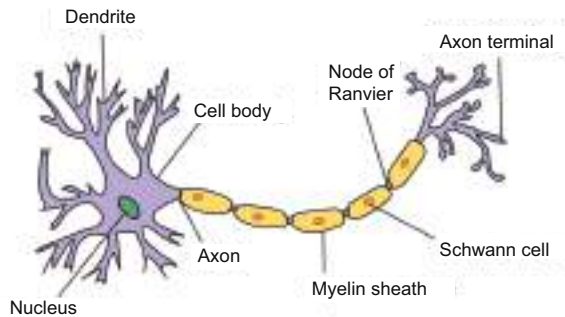


Figure 6.1 Image of a typical neuron (courtesy of Wikimedia Commons, CC BY-SA 3.0)

Each axon terminal contains numerous vesicles, each housing a substantial quantity of neurotransmitter molecules. A single neuron has the capacity to establish thousands of connections with other neurons, resulting in an estimated 100 trillion synaptic connections in the adult human brain. These connections are not static; they exhibit the ability to change over time. Additionally, the more signals exchanged between two neurons, the stronger their connection becomes. This dynamic feature allows the human brain to reconfigure its physical structure in response to experiences.

When comparing the sophisticated learning mechanisms employed by biological neural networks with the simplistic architectures and learning methods of artificial neural networks, it's apparent that there is a stark contrast. Artificial neural networks typically adopt minimalistic topologies and rely on weight modifications for learning, which pales in comparison to the complexity of biological neural networks.

6.1.2 Memory

Our memories define who we are, but what are they, and how are they stored and retrieved? This subject has puzzled researchers for centuries. We know that memories aren't physical objects stored within our bodies, but what do we know beyond that? We imagine that our brain makes some kind of record of the experiences we've had, at least the important ones.

If our brain has a file library containing the things we've learned, our thoughts, and our beliefs, how does the programming work? What is the file format? Is this even the appropriate analogy to use? How does our brain distinguish and classify the multi-dimensional differences and similarities between the myriad things we think about, and how does it keep track of the interconnections between them all?

A computer "recalls" something by accessing binary data stored in transistors or some other electronic medium. It translates the 1s and 0s from the storage device into machine instructions that display symbols on a screen or that make sounds come out of a speaker. Compare this to human experience. A faint sound or smell can bring back vivid memories of the first time we heard our favorite song or of the vacation we were on when we tried a special dish.

I still remember where I was and what I was doing when I heard the song "Imagine" by John Lennon for the first time. In our mind, we visualize the scenery, hear the ambient sounds, and smell the fragrances as if our eyes, ears, and noses are time machines. We revisit the emotions and piece together parts of conversations that we had, and all of this can happen automatically within milliseconds.

While we obviously understand how computer memory works, we are far from a complete understanding of our own, and although the majority of brain functioning remains mysterious, we have made some progress. In 1953, Henry Molaison lost his capacity to make new long-term memories following surgery to relieve his epilepsy. The surgery had removed half of his hippocampus, and this led doctors to believe that the hippocampus was somehow involved in the formation of memory, a hypothesis that has since been verified.

The misfortune experienced by individuals like Molaison and others has contributed to the discovery of distinct substructures within the brain, each serving specific purposes. Notable among these are the thalamus and hypothalamus, which play roles in coordinating movement; the pituitary and pineal glands, responsible for producing and regulating various hormones; the reticular network, associated with consciousness and attention; the limbic system, involved in processing drives and emotions; and the brainstem, housing multiple substructures controlling automatic processes [4].

In 1970, Dr. Eric Kandel was awarded the Nobel Prize in Medicine for his research on the simple nervous system of sea slugs. His experiments illuminated the molecular mechanisms underlying memory and revealed that learning involves physical changes in the brain's structure. He uncovered that as the slugs learned, chemical signals modified the network of synapses connecting their cells. These changes in electrochemical pathways used by the cells to transmit and receive messages were linked to

the formation of new memories. Today, we can observe how a cell nucleus generates mRNA that instructs synapses to establish new connections, and we now understand that these changes occur within our own brains when we learn or remember.

6.1.3 *The subconscious mind*

Our conscious mind serves as the fusion of imagination and emotion, perpetually in search of novelty. It is the wellspring of fresh ideas and forms the bedrock of what we term “human intelligence.”

In tandem with our brain and nervous system, our subconscious mind takes the reins, orchestrating our behaviors and molding our choices. This subconscious mind operates continually, working as a vigilant autopilot, regulating everything from our breath and body temperature to our immune system. It operates independently of our rational, logical mind but seems to draw upon our beliefs and past experiences.

Our subconscious exerts considerable influence over our opinions, beliefs, and judgments. Consider how a child learns behavioral norms through observation of their surroundings and the people therein. These deeply ingrained values may guide our decisions in adulthood, even when we are not consciously aware of them, and challenging them can evoke intense emotions, such as guilt. Our subconscious programming can also cultivate hidden biases, leading us to judge others who did not experience the same conditioning.

This complexity contrasts sharply with machine models of intelligence, where everything is inherently explicit. Data is either stored or not stored, and an algorithm either references it or does not.

The remarkable power of our subconscious mind becomes evident when observing a baby’s journey to walking. Starting with crawling, they experiment and discover a pattern of limb movements that propel them forward. Gradually, they learn to stand and shift their weight, eventually mastering the art of walking. This formula becomes embedded in their subconscious mind within weeks, enabling them to walk effortlessly for the rest of their life. Robotics, despite the vast investments, still cannot replicate a toddler’s proficiency.

The prowess of our subconscious is equally manifest when driving a car. Initially, our conscious mind grapples with the difficulty of learning this complex task, but with experience, it seems as if driving requires no conscious effort at all. We often complete a journey with no recollection of the details as our mental autopilot navigates traffic, avoids obstacles, and adjusts to conditions. Surprisingly, our subconscious even manages our speed, as we may find ourselves unconsciously matching the road’s requirements. The conscious mind occasionally gets involved to note significant changes in the environment. Our subconscious mind operates like an adaptive autopilot.

6.1.4 Common sense

Common sense is judgment without reflection, shared by an entire class, an entire nation, or the entire human race.

—Giambattista Vico

Common sense, the product of millions of years of human evolution and communication, is an extensive library of pragmatic insights derived from shared human experiences. Everything we do in our daily lives uses common sense, which allows us to understand the world and act effectively in it. This knowledge does not require teaching, as we expect that every human acquires it automatically through their life experience. The following 12 examples demonstrate what most people would consider to be common sense:

- 1 A book in a library or in your hand belongs there, but a book inside a microwave does not make sense.
- 2 Rain falls vertically and makes us wet.
- 3 A glass mug containing water will likely break if dropped on cement, but even if it doesn't, the water in it will spill.
- 4 Elephants don't fly.
- 5 My mother and father are older than I am.
- 6 Driving under the influence of alcohol is a dangerous act.
- 7 Honey tastes sweet and comes from bees.
- 8 Animals don't drive cars.
- 9 Salt is not sugar, even though they look the same.
- 10 In the dark, physical appearance is unimportant.
- 11 If you know that I have a car, you could infer that I could give you a ride.
- 12 If you learn that my car won't start, you no longer infer that you can get a ride.

Human beings often express common sense abstractly through phrases or combinations of words. For instance, when we use the phrase “boiling the ocean,” we are conveying that a task is unreasonably large. AI systems, on the other hand, struggle with such abstractions. They may recognize that the ocean contains water and that water can be boiled, leading them to interpret the phrase literally, missing the intended meaning.

Our common sense also involves recognizing relationships between concepts, implications, and associations. When we see clouds and the sky darken, we anticipate rain and understand its effect on our picnic plans. We derive meaning from words, gestures, symbols, and colors, all depending on the context and various cues. For instance, in the United States, the terms “red states” and “blue states” in a political news article are understood to refer to states with majority Republican or Democrat voters, not describing their physical colors.

Common sense serves as a crucial guide in human creativity by directing our efforts away from impractical pursuits and towards productive and effective projects.

The diversity and countless scenarios of common sense make it nearly impossible to catalog or program them on a computer. The potential combinations involving objects, actions, individuals, and environmental factors are virtually limitless. Consequently, replicating common sense in a computer remains an immense challenge, and true artificial intelligence cannot be achieved without a foundation in common sense and comprehension.

6.1.5 *Curiosity*

We keep moving forward, opening up new doors and doing new things, because we're curious and curiosity keeps leading us down new paths.

—Walt Disney

Curiosity, the driver of thought that pervades our mental lives, is a critical force behind the progress of human civilization. It sparks our desire to investigate and discover, explore the unknown, and innovate and learn. The poem “I Keep Six Honest Serving Men” by Rudyard Kipling encapsulates curiosity in an imaginative way:

I keep six honest serving-men
 (They taught me all I knew);
 Their names are What and Why and When And How and Where and Who.
 I send them over land and sea,
 I send them east and west;
 But after they have worked for me,
 I give them all a rest.

Human curiosity has consistently driven us beyond mere survival needs, often resembling a primal instinct akin to hunger or thirst. It is this driving force that has led us to invest time and energy in exploring our world. Whether it's the curiosity about unexplored oceans or new lands, it has inspired us to build ships and embark on journeys of discovery.

Curiosity is not limited to exploration; it plays an indispensable role in every field. For example, it enhances customer service by enabling us to better understand and address individual needs. In healthcare, it improves the quality of treatment as doctors who are more curious about their patients' conditions will more likely offer more tailored and personalized care.

Dr. Faith Fitzgerald underscores the significance of curiosity in healthcare in her essay titled “Curiosity” [5]. In it, she shares a passage from the physician and author Dr. Erich Loewy, emphasizing how curiosity is a driving force in the medical field:

Curiosity, the primal “wonderment” that stimulates exploration, engages both imagination (conceiving the alternative explanations of new phenomena) and intelligence (mapping out the best way to delineate which explanation is likeliest). Both imagination and intelligence are integral to humanities, science, and the synthesis of the two, which is clinical medicine.

It's safe to say that most of the scientific advancements that have significantly shaped human progress can be attributed to curiosity. Consider, for example, the case of German physicist Wilhelm Röntgen in 1895. While conducting experiments with a Crookes tube to study electrical discharge, Röntgen made an intriguing observation: his equipment was producing an enigmatic glow on a chemically coated screen situated across the room. Inspired by Röntgen's work, Frenchman Henri Becquerel decided to investigate some unusual rocks in his collection that emitted an eerie radiance in the dark. This curiosity-driven exploration ultimately led to the discovery of radioactivity [6], a foundational concept in modern science that underlies technologies like X-rays, smoke detectors, and nuclear power.

Richard Feynman, a renowned figure in late 20th-century physics, shared the 1965 Nobel Prize in Physics for his groundbreaking work on quantum electrodynamics, which had profound implications for our understanding of elementary particles. When asked what primarily motivates scientific discovery, he responded, "It has to do with curiosity. It has to do with people wondering what makes something do something" [7]. In his book *Six Easy Pieces* [8], Feynman emphasized that curiosity demands that we ask questions:

Is the sand other than the rocks? That is, is the sand perhaps nothing but a great number of very tiny stones? Is the moon a great rock? If we understood rocks, would we also understand the sand and the moon?

Much like our other innate drives, curiosity acts as the stimulus behind human innovation, urging us to venture into uncharted territories. It is tied to our emotions, evoking wonder, excitement, and an inherent desire to unearth the new and unknown. Curiosity propels us to take risks in pursuit of rewards, igniting a sense of exhilaration. The fulfillment of curiosity triggers the release of dopamine and other feel-good chemicals in our brain [9]. In contrast, artificial intelligence relies solely on human-programmed algorithms. Machines lack the capacity for motivation and satisfaction, and certainly, they cannot experience curiosity.

6.1.6 Imagination

Logic will get you from A to B. Imagination will take you everywhere.

—Albert Einstein

Imagination, the wellspring of new ideas and inventions, arises from the depths of human emotions, expertise, life experiences, and between thoughts, feelings, and memories. It serves as the catalyst for crafting poetry, composing music, unraveling complex problems, and conceiving groundbreaking innovations. Imagination also grants us the ability to extrapolate future events from present circumstances.

Consider this scenario: you find yourself on the 10th floor of an office building, and you are suddenly tasked with the urgent mission of delivering a brimming bucket of water to the reception area on the first floor, and the elevator is out of

order, necessitating the use of stairs. Although this situation is likely novel, and no one may have ever described such a predicament to you, your imagination swiftly springs into action.

In this imaginative exercise, you might deduce that the urgency of the task is linked to a significant event in the reception area, leading you to create and evaluate various scenarios. As you visualize yourself dashing down the stairs with the sloshing bucket, you anticipate the inevitable spillage, the prospect of dampened clothing, and the need for cautious steps to avoid slipping.

This is the essence of imagination at work, an intellectual faculty that fuels innovation and allows us to explore concepts and entities that exist nowhere except in our thoughts. Think of Mozart, who famously invented entire symphonies within his mind without real instruments. Likewise, playwrights envision scenes teeming with multiple performers, their actions and dialogues, their precise positions on the stage, the accompanying sound effects, and the interplay of events.

Imagination is the engine of human creativity. Take, for instance, Johannes Gutenberg, who is known for imagining a screw press with movable type in his mind's eye. This vision ultimately led to the creation of the first printing press, a machine that would go down in history as one of the most influential inventions of all time. In his paper titled “Nikola Tesla: Electricity Today Is Generated, Transmitted, and Converted to Mechanical Power by Means of His Inventions” [10], Kenneth Swezey admired Tesla's exceptional imagination and emphasized the profound and positive effect of Tesla's innovations:

From early childhood, Tesla could visualize so clearly that he often had difficulty in distinguishing real objects from their counterparts in his imagination. This ability, often annoying and even frightening in ordinary life, Tesla used advantageously in inventing. By means of it, he could, for instance, assemble and reassemble mechanical parts, alter sizes, weights, and materials entirely in his mind, and end up by building a model that would generally work exactly as he had conceived it.

The same paper highlights Tesla's imaginative prowess by describing a breakthrough that paved the way for modern electric motors and generators:

Late one afternoon in February 1882, the answer came. Tesla was walking with a friend, Szigety, through the City Park of Budapest, reciting stanzas from Goethe's Faust, which he knew by heart. Suddenly the solution he had been seeking flashed through his mind. He saw clearly an iron rotor spinning rapidly in an electric whirlwind—a rotating magnetic field produced by the interaction of two alternating currents out of step with each other. With a stick, Tesla drew diagrams on the sand, explaining in detail to Szigety the principle of the induction motor Tesla was to patent in America six years later.

Our imagination is a fusion of thoughts and emotions that lead to creativity and innovation. In contrast to machines, our cognitive processes are not confined by rigid logic. How can an algorithm, designed to detect patterns in data, replicate the limitless nature of human imagination?

6.1.7 Creativity

Creativity is seeing what others see and thinking what no one else ever thought.

—Albert Einstein

Creativity springs forth from our capacity to infer connections between seemingly unrelated concepts. It relies on the power of imagination to generate novel ideas or products of value. This creative force can manifest in various ways, from inventing something entirely new to combining, adapting, or enhancing existing elements. Creativity defies the confines of pure logic. It begins with imagination, flourishes through intuition, and takes shape through reasoning. There is no formula or method for becoming creative. Creativity allows us to adapt in real time when confronted with unexpected events and challenges. As we all know, creativity is the mother of invention.

A classic illustration of creativity can be found in the story of Archimedes, charged by his king with determining the volume of a crown. As he immersed himself in a bath, a sudden flash of insight struck him: the water displaced by his body equaled the volume of his body. Realizing this principle could be applied to measure the crown's volume, he is said to have shouted “Eureka!” which signifies “I have found it!”

Another example comes from John Steinbeck's remarkable book *Travels with Charley in Search of America* [11], where he describes how he ingeniously devised an efficient method for handling laundry while journeying through the countryside:

I invented a method for washing clothes which you will go a long way to better. It came about this way. I had a large plastic garbage bucket with cover and bail. Since the normal movement of the truck tipped it over, I tethered it by a length of strong elastic rope of cotton-covered rubber to the clothes pole in my little closet, where it could jiggle to its heart's content without spilling. After a day of this, I opened it to dispose of tie stuff at a roadside garbage can and found the most thoroughly mixed and kneaded garbage I have ever seen. I suppose all great inventions spring from some such experience. The next morning, I washed the plastic bucket, put in two shirts, underwear, and socks, added hot water and detergent, and hung it by its rubber rope to the clothes pole, where it jiggled and danced crazily all day. That night I rinsed the clothes in a stream, and you've never seen clothes so clean. Inside Rocinante [his pickup truck that he named after Don Quixote's horse], I strung a nylon line close to the window and hung the clothes to dry. From that time on, my clothing was washed on one day of driving and dried on the next.

Creativity appears to possess a natural expansiveness, building upon one innovation after another. Over time, incremental enhancements and ingenious modifications have propelled us from the hot air balloon to the space shuttle.

A creative individual often embraces challenges to invent innovative solutions. In contrast, current AI operates mechanically, processing data based on algorithms created by humans. It lacks the ability to invent something entirely new. How could creativity emerge from a collection of mathematical formulas?

6.1.8 Intuition

C'est par la logique qu'on démontre, c'est par l'intuition qu'on invente. We invent with intuition; we prove with logic.

—Henri Poincaré

Intuition is our native ability to perceive, evaluate, and understand something without relying on conscious reasoning. It acts as a subtle whisper, guiding us to explore unexplored territories, drawing from our past experiences and emotions to embrace unconventional paths that can sometimes lead to groundbreaking discoveries. Intuition transcends the boundaries of logic, making it challenging to explain.

Most innovations can, in part, be attributed to intuition. We are all familiar with the legendary story of Isaac Newton and the falling apple. When he observed an apple dropping from a tree, Newton had a profound intuition that there was something significant behind this seemingly ordinary event. This led him to ponder the nature of objects in motion, with his intuition suggesting that whatever caused the apple's fall might also be responsible for the moon's motion. He further inferred that such a universal force might also govern the movements of the stars and planets. In his time, the common belief was that angels pushed planets along their orbits.

In 1796, physician Edward Jenner had an intuition that led to the development of the world's first vaccine. At that time, smallpox was a rampant and deadly disease in England. Jenner observed that those who regularly worked with cows often contracted cowpox, a related but less severe disease that could be transmitted from cows to humans. Curiously, those infected with cowpox appeared to be immune to smallpox. Jenner had an intuition that cowpox conferred protection against smallpox, and through deliberate and systematic exposure to cowpox, he demonstrated that it prevented the spread of smallpox. This intuition, nearly a century before the discovery of viruses and their role in diseases, helped combat a disease that plagued humanity for centuries.

In 1928, Scottish physician and scientist Alexander Fleming noticed a patch of mold while examining a bacterial culture plate. He observed that the area around the mold was devoid of bacteria. His intuition guided him to investigate further, leading to his discovery of penicillin, a lifesaving discovery that has saved countless lives.

Similarly, the pacemaker, a medical lifesaver, was invented in 1956 thanks to engineer and inventor Wilson Greatbatch's intuition. While working on a heart-rhythm recorder, he accidentally used the wrong size resistor and unexpectedly heard electrical pulses emanating from the device. His intuition led him to believe he might have found a way to mimic and stimulate a heartbeat, a belief that proved correct.

Artificial intelligence, being a computer program, primarily relies on logical reasoning and lacks the intuitive dimension. Replicating the complex and enigmatic processes that trigger human intuition presents a formidable challenge for AI. It's challenging to imagine an algorithm capable of reproducing something as irrational and unexplainable as intuition. How could a combination of algorithms and mathematical formulas instill a gut feeling in a machine?

6.1.9 Analogy

The real achievement in discoveries . . . is seeing an analogy where no one saw one before.

—Arthur Koestler

An analogy is a comparison that draws upon corresponding parts or similarities. Analogy stimulates intuition and creativity, which makes it a powerful tool for human innovation. We naturally draw analogies even between dissimilar objects. For instance, we recognize the analogy between a car and a skateboard, both of which have wheels facilitating movement from point A to point B, despite their fundamental differences. Analogies also play a significant role in the professional world. Attorneys, for instance, frequently refer to precedent in their arguments, employing reasoning through comparison to earlier cases. In their 1970 academic article titled “Some Speculation About Artificial Intelligence and Legal Reasoning” [12], Bruce G. Buchanan and Thomas E. Headrick underscore the crucial role of analogy in the legal profession:

A lawyer usually prefers to fashion arguments built from cases whose facts are similar to the facts with which he is working. If he cannot find such cases, he resorts to finding cases with facts that are analogous to his own in formulating an argument. One method of finding an analogy is through generalization of a legal rule. Although in one case a rule was applied to a specific set of facts, the language in which the court stated the rule may deductively allow its application to a different set of facts. The rule may use words that are capable of encompassing a variety of events, actions, or relationships. In our Boston Ice example, for instance, the lawyer examined whether the case of water delivery would hold for ice deliveries. In addition, analogies are found by generalizing on the factual situations.

Human analogical reasoning taps into a vast reservoir of diverse experiences, allowing us to infer connections between seemingly unrelated domains and apply their insights to innovate and create fresh ideas. In contrast, current AI techniques face difficulties when trying to emulate human-level analogical reasoning as they heavily rely on pre-defined algorithms.

6.2 Human vision vs. computer vision

Sight is arguably the sense we rely on the most to navigate and understand the world around us. Our visual perception begins with light from the environment, which either originates from a light source or bounces off objects before entering our eyes through the corneas. The eyes’ lenses focus this light onto the retinas, which are light-sensitive membranes located at the back of our eyes. Retinal cells respond by transmitting nerve signals to the brain, where they are rapidly processed into a representation of our surroundings. This perceptual process enables us to perceive lightness and darkness; distinguish colors, textures, and shapes; and identify distinct objects. Additionally, we can detect motion and estimate distances.

The physical process of capturing light and generating electrical signals is not particularly sophisticated, and many machines surpass us in this regard. We have devices

capable of detecting heat and X-rays from distant celestial bodies and others that can produce images of individual atoms. However, human vision goes beyond mere data acquisition; it involves the complex processes of interpretation and understanding, an area where machines currently lag far behind.

Typically, humans need only a few examples to develop a profound understanding of something. When a child sees a cat for the first time, they instinctively create a mental model that helps them recognize “cat-ness.” They use logic and common sense to generalize their observations, unconsciously identifying various characteristics associated with cats, such as shapes, colors, textures, sounds, and behaviors. Even if the child’s first encounter is with a small, white Siamese cat in a window, they will recognize other cats of different colors, sizes, and breeds, even when in various activities. They can identify cats in photographs, cartoons, and even when someone dresses as a cat for Halloween. With more exposure, they may learn that cats have claws and sharp teeth and can exhibit both sweet and temperamental behavior.

In contrast, deep learning algorithms typically require exposure to millions of human-labeled examples before they can reliably identify a picture as “cat” or “not cat.” These examples need to be in high-resolution images. The algorithms need to see cats from multiple angles and engaged in various activities, capturing all relevant shapes, sizes, colors, and textures. To a machine, the objects it “sees” are essentially collections of pixels. Machine learning follows a hierarchical approach, initially recognizing patterns related to shape, then identifying color and texture, and eventually detecting finer features. Even when an AI system can reliably “see” and identify images of cats, it lacks the depth of understanding a child achieves after their first encounter with a cat. As discussed in our exploration of neural networks and deep learning in chapter 3, what machines “learn” is essentially a set of numerical values between 0 and 1 (the network’s weights) that enable the algorithm to produce accurate outputs most of the time. However, these numbers cannot teach a machine the *meaning* of an image or the potential uses of an object captured in a photograph.

6.2.1 **AI and COVID**

In 2020, the world grappled with the emergence of the COVID-19 virus, a global health crisis that claimed millions of lives. The need for technological solutions was paramount, presenting AI with a unique opportunity to showcase its potential. With years of media hype around machine vision, the idea of using AI to diagnose COVID seemed promising. It appeared straightforward: train a machine learning algorithm using labeled chest X-rays or other images as “infected” or “not infected,” and doctors could receive nearly instant diagnoses after uploading a patient’s chest scan, eliminating the waiting time associated with traditional COVID tests.

However, the reality was disappointing. While machine learning has demonstrated success in some medical imaging applications, none of the hundreds of models developed worldwide during the pandemic proved useful for COVID diagnosis. A comprehensive review of 2,212 research studies conducted by a team at *Nature* magazine,

titled “Common Pitfalls and Recommendations for Using Machine Learning to Detect and Prognosticate for COVID-19 Using Chest Radiographs and CT Scans” [13], concluded, “None of the models identified are of potential clinical use due to methodological flaws and/or underlying biases.”

The *MIT Technology Review* shared a similarly disappointing assessment, stating, “In the end, many hundreds of predictive tools were developed. None of them made a real difference, and some were potentially harmful” [14]. Laure Wynants, an epidemiologist who led a review of studies for the *British Medical Journal*, expressed deep concerns, suggesting that experimental diagnostic technology might have done more harm than good. She remarked, “It’s shocking. . . . I went into it with some worries, but this exceeded my fears” [15]. Regrettably, I share her sentiment.

6.2.2 Image reasoning

Without any apparent effort, we humans convert visual inputs in real time into meaningful information that we trust so thoroughly that we regularly stake our lives on it. In contrast, a 2017 experiment with a deep learning system showed how fragile computer “vision” really is. After training the network to recognize a set of images, it was found that 74 percent of the images could be modified in just 1 pixel in a way that had a 99 percent chance of fooling the algorithm [16]. It’s interesting to note in passing that each of our eyes has a literal blind spot, and although it represents a much more significant gap in our field of vision than a single pixel, we are normally completely oblivious to it, and it causes us no confusion. The spot on the back of each eye where the retinal nerve attaches has no light-sensing cells, so the light from our environment that gets focused by an eye’s lens on one of those spots doesn’t stimulate any signal. The brain, however, represents our surroundings as a continuous, gapless image by using context and data from the other eye to fill in the missing information.

AI’s trouble with object recognition is significant, but an even greater shortcoming is its complete inability to associate meaning with the images it processes. Sight is not just about recognizing objects in images, and picking up on patterns in pixels is a far cry from understanding the significance of an object’s environment, the relationships between objects, and their potential utility. A human’s natural ability to distinguish an object from its background continues to elude AI, and the interpretive power of sight aided by touch, sound, and smell will likely never be replicated by any AI algorithm.

AI also struggles to mimic incoherence detection. Without any training, we know that a car does not belong in a microwave and that chairs do not fly. If we’re shown a kitchen scene, we know not to expect a hot tub or a lawnmower. In contrast, AI doesn’t recognize connections or relationships and, therefore, has no way to tell when something isn’t right.

Our human visual system is also integrated with our logical reasoning and imagination. For example, through sight alone, a child could learn that bananas are a fruit that starts out green, turns yellow, and later develops brown spots. They will learn that bananas have no seeds, can be eaten with ice cream, and might be cooked. They will

also understand that although bananas might be sliced into a bowl of oatmeal, they ought not to be put in a bowl of chicken soup.

None of the ideas or associations that arise in a child's mind about bananas could be reproduced automatically with current AI. Nor would a computer be likely to recognize a meaningful difference between a banana and a banana-shaped piece of yellow plastic. A child, even just relying on sight, would infer that a plastic banana wouldn't turn brown and could not be peeled or eaten.

Abstract meaning and representation are also part of how we process visual input. When we see numbers on a page, we associate them with quantities, or we understand them by relative size or some other abstract feature. The number 9 and the number 6 are basically identical except for rotation by 180 degrees, yet we know without deliberation that 9 represents a quantity larger than 6 and perhaps that each quantity can be evenly divided by 3.

Our sight and visual processing also work seamlessly with our motor functions and coordination. We can run across a field while watching a ball fly through the air and, without taking measurements or performing any mathematical calculations, reach just the right spot at just the right time to catch the ball before it hits the ground. Musicians in an orchestra can perform complex and delicate motions in response to markings on a page while adjusting what they do in response to the gestures of a conductor.

There are many aspects of human vision that machines will not be able to reproduce by virtue of their being machines. For example, the sight of a crying child can evoke a feeling of sadness or sympathy. A long-lost friend might elicit surprise and joy. Emotional responses and physical reactions to sensory input are obviously restricted to living organisms, and no degree of mechanical simulation will recreate such features of human sight.

Most effort in AI has been focused on replicating human intelligence without analyzing whether AI algorithms and the processes used by computers bear any relation to human thinking. Our amazing brains are capable of things that we aren't even close to understanding, and, seen in this light, claims that we can reproduce such things in computers should become much easier to dismiss. Furthermore, by letting go of the idea that we should try to make our machines behave like brains, we can actually pursue the full potential of our technology. Although the flight of birds might have been an inspiration to those who imagined flying machines, we don't aspire to make planes that do exactly what birds do. In fact, our airborne technology far surpasses anything found in nature.

While AI has made amazing progress, it remains bound by its algorithms programming and lacks the essential qualities to replicate the uniquely human characteristics that define human curiosity, imagination, creativity, analogy, and intuition. The intrinsic connection between these aspects and many other emotional human capabilities is a unique facet of our species.

Summary

- The human brain remains an enigmatic puzzle that continues to baffle medical professionals and researchers.
- Although significant strides have been made in unveiling the neural underpinnings of cognition and consciousness, the precise mechanisms responsible for the genesis of our thoughts remain a profound mystery.
- The human brain's remarkable ability to memorize information and generalize concepts is a testament to its astonishing adaptability.
- Beyond conscious awareness, the subconscious mind exerts a profound influence on our decisions and creative discoveries.
- Common sense is a fundamental aspect of human decision-making, serving as a vital guide for navigating complex situations.
- Humans possess unique attributes that remain a formidable challenge for AI.



AI doesn't turn data into intelligence

This chapter covers

- The difference between data, information, and intelligence
- The limited reusability of AI models
- The vulnerability of AI to unexpected data inputs

In this chapter, we will explain that regardless of how extensive the dataset or advanced the algorithms, an AI program faces a substantial challenge in inferring intelligence from data due to the semantic gap, as AI algorithms can process data but struggle to grasp its deeper meanings. Even for simpler tasks, such as recognizing handwritten numbers, where AI models excel, they still lack the capability to grasp the context surrounding these numbers. For instance, an AI system might correctly identify a handwritten 9 as the numeric symbol 9, but it remains unable to infer that this number more likely represents a child's age on a birthday card.

The Farmer-Wolf-Goat-Cabbage riddle, previously discussed in chapter 4, serves as a prime example illustrating the constraints of AI when confronted with problems that demand logical reasoning and strategic planning. Traditional machine learning models primarily depend on statistical patterns and data-driven training, rendering them unsuitable for tasks that necessitate deductive reasoning, forward

planning, and a deep comprehension of specific constraints, as exemplified by this simple problem. In this puzzle, the farmer must safely transport the wolf, goat, and cabbage across a river, ensuring that neither the wolf eats the goat nor the goat consumes the cabbage. While AI excels in domains such as natural language understanding, image recognition, and game playing, these domains typically involve activities based on pattern recognition and optimization, drawing from historical data. In contrast, solving the Farmer-Wolf-Goat-Cabbage riddle calls for symbolic reasoning or symbolic AI techniques, reliant on explicit rules and logic to determine the correct sequence of actions while adhering to predefined constraints. Real progress in AI will require the development of innovative techniques, such as smart agents, that extend beyond the confines of learning solely from data.

Let us elaborate further by referring to medical diagnosis, where many believe that AI possesses the capability to acquire “intelligence” from data. However, AI systems trained on data are insufficient for conducting accurate medical diagnoses. One fundamental limitation is the absence of medical understanding. AI programs trained on data can identify statistical patterns and correlations, but they lack an inherent grasp of the underlying medical concepts, disease mechanisms, and the complexity of the human body. Medical diagnosis often demands an understanding of biology, pathology, and clinical expertise that cannot be inferred from data.

In most cases, a visit to a doctor's office, unless it's an emergency, begins with an information-sharing process. Questions are posed and answered during appointment scheduling, intake forms are completed, health histories are updated, and sometimes questionnaires are necessary. Before the patient even sees the doctor, a nurse often records height, weight, temperature, and blood pressure, along with conducting preliminary tests.

Once the doctor enters the examination room, they commence the process of information filtration and reasoning. In addition to the patient's records and the preliminary data collected, the doctor takes into account the patient's physical appearance and demeanor. They conduct visual examinations of the patient's eyes, ears, nose, and throat; listen to the heartbeat; and palpate the abdomen. They engage the patient in discussions about their symptoms, asking targeted questions to gather specific information. In some cases, they might perform specialized diagnostic procedures or request blood or urine samples for laboratory tests.

In theory, nearly any detail from the patient's life story or even their relatives' stories could be relevant to a diagnosis. However, discerning what to consider and what to disregard is a skill, and in most cases, the majority of available information is irrelevant to the immediate situation. If finding the right diagnosis sometimes feels like finding a needle in a haystack, adding more data only increases the haystack's size, creating a hindrance rather than assistance.

Much of a doctor's reasoning occurs instinctively and subconsciously, drawing from experience with similar cases and familiarity with the patient. Over time, doctors have developed the ability to deal with uncertainty and incomplete information. For

instance, they must interpret the patient's description of symptom severity in light of the patient's overall health. Their line of questioning adapts in real-time based on the patient's responses and their intuition. Sometimes, they intuitively know to inquire about family history or pre-existing conditions, while in other cases, curiosity prompts them to ask about sleep quality or emotional stress.

Clearly, doctors need to extrapolate, draw analogies, and rely on intuition. Their task isn't about amassing as much information as possible but about obtaining the right information and interpreting it correctly. How could an AI model, no matter the size of the data sets on which it was trained, replicate the multifaceted reasoning of a medical professional?

7.1 ***Machines defeating world champions***

Two of the most widely publicized "AI events" in history were IBM's Deep Blue defeating chess champion Gary Kasparov in 1997 and AlphaGo's victory over Go champion Lee Sedol in 2016. Both instances received extensive media coverage, showcasing the apparent "intelligence" of these programs and generating anticipation for even more advanced technology in the future. However, these popular interpretations often missed the mark. These demonstrations primarily highlighted the machines' formidable computing power and sophisticated specialized algorithms. Instead of demonstrating genuine intelligence or understanding, they highlighted the stark contrast between human and machine approaches to generating game moves. By analyzing the methods employed by computers to play human games, we can gain a better understanding of AI systems reliant on data and computation.

In the game of chess, the board consists of eight rows and eight columns, with 32 squares initially occupied by pieces. Each side begins with a collection of pieces, including eight pawns, two rooks, two knights, two bishops, one queen, and one king (figure 7.1). Each type of piece possesses its own distinctive movement rules. For



Figure 7.1 The starting position in chess (French Louvre collections of art) [1]

example, a pawn typically moves one square forward at a time, except for its first move, where it has the option to advance one or two squares. Players take turns, with white pieces making the first move. A piece captures an opponent's piece by moving to the square occupied by that piece. The goal of the game is to put the opponent's king in a position where it can't escape capture, which is called checkmate.

The white player in a chess game faces an initial choice of 20 possible moves, and their opponent has an equal number of potential responses. However, as the game progresses, the number of possible board positions increases significantly. Just after the second pair of moves, we reach 197,742 potential board configurations. After three pairs of moves, this number skyrockets to over 121 million. To provide some perspective, consider that a typical position in chess allows for approximately 30 legal moves to choose from. With the average chess game extending to about 40 move pairs, the total number of possible game positions can be estimated to be around 10^{120} .

NOTE This monumental figure is known as the Shannon number, named in honor of the mathematician and electrical engineer Claude Shannon. It serves as a metric in chess to estimate the complexity of the game tree. Shannon published a groundbreaking paper titled "Programming a Computer for Playing Chess." In this paper, he introduced the notion of a game tree, comprising nodes that represent different game states and edges connecting them to nodes reachable by legal moves.

In chess, each move by a player leads to a multitude of possible responses by the opponent, creating a branching structure known as the game tree. The Shannon number attempts to quantify the vast number of possible chess games. This number provides a lower bound on the game-tree complexity of chess. It is an estimation of the minimum number of possible chess games, taking into account legal moves and positions. The calculation involves considering the average branching factor (average number of possible moves at each turn) and the average game length.

While the exact number is difficult to pinpoint due to the immense complexity of chess, the Shannon number serves as a theoretical measure to illustrate the extraordinary depth and intricacy of the game. It highlights the intricate and expansive nature of chess, emphasizing the difficulties in encompassing all potential legal positions and moves.

The traditional method for programming turn-based games, as seen in the development of IBM's Deep Blue for chess, involves storing the game tree in computer memory. In this approach, the system assesses the desirability of positions at the endpoints of branches, systematically adds more nodes to the tree for a deeper search, and prunes less promising branches. The computer then selects its move based on the positions with the highest evaluations. Competitive turn-based games often come with time limits. Each player is initially allocated a specific amount of time, which decreases when it's their turn to make a move. If a player's time runs out, they typically lose the game. Consequently, a computer must rely on heuristics to determine how much time to allocate for computations and when to make a move. Just like human players, it

needs to strike a balance between spending time to find potentially better moves and conserving time for future turns, avoiding rapid decisions that could lead to a disadvantageous position.

To compete effectively against skilled chess players, a program must analyze sequences involving a large number of moves, a task that would be impractical without pruning. Alpha-beta pruning is a specific algorithm used to reduce the number of nodes that need to be evaluated in this process. Chess programs also include subroutines that apply heuristics and rules of thumb to assess the relative advantages of each side in various positions. These assessments are translated into numerical scores, weighted, and combined to produce an overall evaluation score, taking into account their relative importance. Additionally, programs like Deep Blue have access to databases containing complete grandmaster games, opening sequences, and an endgame database that includes full game trees for positions with only a few remaining pieces on the board.

Switching gears to the game of Go, it is played on a 19×19 grid with black and white stones. The objective is to capture the most territory by surrounding it with one's own stones. Computer Go programs have taken various approaches over the years, with early efforts like Albert Zobrist's 1968 program [2] relying on traditional programming. In contrast, the Symbiotic Adaptive Neuro-Evolution (SANE) approach introduced in 1998 at the University of Texas at Austin [3] utilized neural networks and genetic algorithms to teach the program to play on a nine-by-nine board without preprogrammed Go knowledge. AlphaGo, however, employed extensive reference databases, grandmaster knowledge, and highly sophisticated deep learning. The program was trained using positions from both human- and computer-played games, incorporating board positions, the best moves, and winning percentages. Subsequently, AlphaGo played millions of games against itself to refine its strategies.

The computer-based approach differs from human players, who rely on intuition and experience. Human players calculate move sequences and evaluate positions, but their process is neither systematic nor numerical. Renowned chess grandmaster Magnus Carlsen mentioned that he can occasionally calculate 15 to 20 moves ahead, but the challenge lies in evaluating the positions at the end of these lines [4]. Humans also depend heavily on intuition, strategy, and anticipating their opponents' plans, sometimes choosing moves that may not have the highest numerical score but are psychologically challenging for their opponents.

7.2 *Lack of generalization*

Numerous AI projects primarily serve marketing purposes, with resulting programs typically having limited utility beyond their ability to beat humans in specific games. These projects often demand substantial investments in terms of both cost and resources. While they may initially capture media attention, as exemplified by the widespread coverage of Deep Blue, they frequently fade into obscurity over time. In contrast, human experts in games like chess or Go can transfer their skills to play

other games reasonably well. Given a brief period to familiarize themselves with the rules of a new game, they can effectively apply their systematic and strategic thinking abilities. AI systems, on the other hand, struggle to adapt to rule changes in their designated games, let alone apply their knowledge to different domains.

Furthermore, when evaluating a computer's apparent intelligence, it's important to consider that individuals capable of playing chess at a master level can also handle tasks such as writing articles on chess strategy. However, despite their impressive game-playing abilities, systems like Deep Blue are limited in their competency beyond their specialized tasks. For example, AlphaGo employs advanced techniques to navigate complex state spaces and analyze extensive data, but it can only do so because it operates with complete and perfect information. These systems encounter difficulties when confronted with real-world scenarios that feature incomplete information and uncertainty.

In the 1950s, individuals who could perform fast and accurate mental calculations were deemed "intelligent" because this skill was highly valuable in an era before the widespread use of calculating machines. Today, affordable handheld calculators surpass human abilities in manipulating mathematical formulas and performing calculations. Yet, we don't attribute intelligence to calculators. This raises the question: Why do we often characterize game-playing programs as intelligent simply because they excel in rapid calculations and data processing?

Let's conclude this chapter with a historical case example that underscores the significant challenges AI systems face when attempting to replicate counterintuitive intelligence. During World War II, mathematician Abraham Wald proposed a counterintuitive but highly effective strategy for reinforcing planes against enemy fire. Instead of reinforcing the parts of planes that received the most damage, as suggested by the records, he recommended reinforcing the parts that received the least damage [5]. His insight was crucial: the planes that returned to base were surviving the damage they accumulated, and the records highlighted the areas that, if damaged, would lead to the plane's loss. Wald's remarkable demonstration of counterintuitive intelligence was made possible by his cognitive abilities, such as abstract reasoning, creativity, and a deep understanding of context, which allowed him to conceptualize and reason about concepts that may not have had direct precedents in his knowledge and past experiences. Furthermore, his ability to apply common-sense reasoning, creativity, and imagination enabled him to formulate innovative solutions to problems that defied conventional logic and showcased his counterintuitive intelligence.

In stark contrast, AI models lack the capability to grasp the underlying principles or concepts behind data. They also lack common-sense reasoning, which is essential for comprehending counterintuitive scenarios that may not conform to standard rules or patterns. In essence, AI's inability to infer counterintuitive intelligence underscores the divide between data-driven machine learning and the nuanced, context-dependent reasoning that humans often employ in complex, unconventional situations.

Summary

- Relying solely on an AI program learning from data alone may not be sufficient for performing many tasks, especially those that require reasoning or common-sense knowledge.
- AI models that rely on data to learn are often limited to a single domain because their knowledge and capabilities are derived solely from the data they are trained on.
- AI models lack the capacity for context-based reasoning and critical thinking.

AI doesn't threaten our jobs

This chapter covers

- The overstated concerns regarding AI's effect on employment
- The challenges AI faces in automating many human tasks
- The problems of applying AI in the real world
- The dangerous effects of inadequately designed AI systems

In this chapter, we will demonstrate, using various examples, that despite concerns about AI taking over jobs, the majority of human occupations remain beyond the reach of current AI techniques.

The apprehension surrounding the idea of machines replacing human jobs has deep historical roots. As far back as 1589, when English inventor William Lee sought a patent for his knitting machine, Queen Elizabeth I denied it due to concerns about the economic stability of the kingdom's numerous hand-knitters. This illustrates the long-standing fear of technological advancements affecting employment. In the appendix, we'll briefly touch upon the Luddites and their reactionary actions against textile machinery during the 19th century. This moment in history

underscores the resistance to automation, which often arises from concerns about job displacement. In July 1987, the *Harvard Business Review* published an article titled “Thinking about Artificial.” It opened with the statement: “Some believe that Artificial Intelligence is on the brink of transforming business practices. They claim that ‘intelligent’ computer programs will imminently assume roles such as doctors, lawyers, factory workers, and managers” [1]. As we know, none of these forecasts actually came to fruition.

Fast forward to the present day, and the fear of machines taking over human jobs continues to evolve, sometimes with alarmist tones. Some experts even have a formula to predict the likelihood of AI replacing your job, suggesting that if a job can be easily explained, it can be replaced by AI [2]. This apprehensive prediction was also made in a 2015 report [3] by the Global Challenges Foundation and Oxford University’s Future of Humanity Institute, which suggested that machines are on the verge of replacing a significant portion of the workforce. The report states:

AIs would immediately benefit from improvements in computer speed and any computer research. They could be trained for specific professions and duplicated at will, potentially replacing a substantial portion of the global workforce and causing significant economic disruption.

Former President Obama, in a 2016 interview with *WIRED* magazine [4], recognized job displacement by AI as a significant risk. He stated:

One thing that we haven’t discussed much, and I want to emphasize, is that we really need to consider the economic implications of AI. Most people are not currently preoccupied with the Singularity; they are worried about whether their jobs will be replaced by machines.

Media outlets have also contributed to the sensationalism surrounding AI job displacement, with headlines like “Robots Can Now Read Better than Humans, Putting Millions of Jobs at Risk” [5] and predictions that “your favorite restaurant may soon employ robots as cooks and servers, potentially leading to job losses for millions of workers” [6].

The World Economic Forum published a “Future of Jobs Report” [7], complete with extensive statistics and projections for every industry in every country. The report asserts that the Fourth Industrial Revolution is likely to lead to a reduction in the workforce and lists the following illustrative predictions:

- Nearly 50% of companies expect that automation will lead to some reduction in their full-time workforce by 2022.
- Forty-two percent of task hours will be performed by machines.
- Seventy-five million jobs may be displaced by a shift in the division of labor between humans and machines.

In 2020, businessman Andrew Yang, a Democratic presidential candidate, campaigned in part on the promise of implementing a universal income to mitigate the job

displacement caused by automation. Additionally, a prominent consulting firm made a forecast indicating that by 2030, around one-third of the American workforce may need to transition to new occupations [8]. They also highlighted the possibility that up to 160 million women worldwide might need to switch professions by 2030, with a particular vulnerability in roles such as secretaries, schedulers, and bookkeepers due to automation. Taking a more comprehensive view, former IBM CEO Ginni Rometty, in an interview with CNBC, asserted that AI would affect 100% of jobs [9].

In this chapter, I will elucidate why these dystopian scenarios lack a solid foundation. In chapter 6, we conducted a comparative analysis of human and machine intelligence, identifying several fundamental distinctions between humans and machines. Here, we will discuss the limitations of AI, specifically within the context of the workplace.

While automation is well-suited for handling simple, repetitive, and potentially hazardous tasks, the automobile industry, renowned for its extensive use of robotics, presents an intriguing paradox. One might anticipate a reduced need for human labor in such an environment. However, reality and statistics offer a different perspective. Consider the German automotive industry as an example, where the number of employees increased year by year, reaching 833,937 in 2018 [10]. This phenomenon underscores a widely accepted fact within the industry: machines struggle with tasks demanding cognitive prowess and dexterity [11].

Similarly, one might anticipate a surge in automation within the field of accounting, given its foundation in calculations, seemingly an ideal terrain for machine replacement. Yet, once again, reality challenges this assumption. Surprisingly, the percentage of human workers in accounting has, contrary to widespread predictions, doubled since the widespread integration of computers into workplaces during the 1980s.

A pattern resembling the one observed in the automotive and accounting industries can also be identified in the field of medicine. The inception of medical AI dates back to the early 1970s, with the development of systems such as MYCIN, mentioned in chapter 2, as well as others like CASNET and INTERNIST. Predictably, these projects were perceived as the dawn of a new era in traditional medicine. An article in the *New York Times* titled “Medical Technology: The New Revolution,” published on August 5, 1979 [12], informed readers to anticipate the following advancements:

If you have to go to the hospital ten years from now, your visit may run something like this: Past the inevitable receptionist, your street clothes exchanged for a shapeless hospital gown, you enter a small, antiseptic room. You take a seat and offer your arm to a rectangular machine. Painlessly the machine draws a blood sample and within seconds has analyzed it down to its smallest meaningful platelet. The information is flashed to a central computer deep within the hospital, where it is compared with previous readings to detect an infection anywhere in your body. You move to an adjoining room and sit under a massive apparatus. Silently, your body is probed by X-rays or microwaves, while sensors inspect the surface of your skin. A thermogram will show areas of raised temperature, which can reveal incipient disease. Although you feel nothing, see nothing, the inner workings of your body are being deeply scrutinized. A developing gallstone, still barely

larger than a grain of sand, is noted and appraised; like a tiny white clot lodged within a coronary artery or a polyp hidden in your nasal cavity, it may be harmless, but nevertheless deserves watching. Within moments, a minutely detailed workup has been prepared by computer and sent to your doctor.

After more than four decades, it appears that, if anything, we've regressed. During my recent visit to a hospital in the Bay Area, I endured a three-hour wait to meet with a nurse who posed the same questions one might expect to hear in an old black-and-white film. The only distinction was that the hospital in the movie would have been cleaner and less crowded, with the patient conversing directly with the doctor instead of a nurse. Have you witnessed machines taking over any medical tasks typically performed by doctors?

A part of my PhD research addressed the use of AI for medical reasoning and diagnosis. I don't have any medical training myself, so I initially tried to use an expert system to emulate what I learned from several doctors about how they think. For the sake of discussion, I will give here a general outline of the thought process that guided the design of my AI solution.

When presented with a patient, the first priority of the expert system is to address any emergencies. If a patient is losing blood, for example, that needs to be addressed immediately. Once any emergencies are addressed, any obvious diagnoses will be made. For example, a patient in an otherwise stable condition who came in saying that they fell and broke their arm is probably right about what's wrong. Any decisive test would be administered to confirm the diagnosis, and the necessary treatment would be provided.

If there are no emergencies and no obvious conclusions to be drawn, one looks for decisive facts that would suggest the correct diagnosis. A priority would be given to affordable, safe, noninvasive tests, and only when the more reasonable options fail to provide clarification would you resort to more extravagant measures. For example, if there is reason to suspect that a patient has bronchitis and it's possible to test this hypothesis with a simple throat culture, you do the throat culture instead of ordering an MRI, which is a lengthy, expensive, uncomfortable procedure. You progress from likely hypotheses and reasonable testing procedures to less likely possibilities and less desirable procedures until the problem can be identified.

The thought process undertaken in each of these steps is context-dependent. For example, a patient's age, sex, family history, and pre-existing conditions need to be considered. In addition, some test results come with a high degree of uncertainty. As a result, complex reasoning is required, and much of a doctor's skill is tacit and difficult to formulate. What doctors could explain to me about their reasoning was often quite a challenge to replicate in an algorithm.

Dr. Faith Fitzgerald reached the same conclusion [13]. She explains that clinical reasoning cannot be reduced to a set of rules since each patient has their own unique situation and background, and she describes the subtle nature of medical reasoning as follows:

In fact, the best clinical diagnostic thinking is more like the forming of a mosaic than linear thinking: It requires the physician to constantly alter diagnoses as each new piece of data enters the picture. One conceives constantly of many possible diagnoses, narrows down, re-expands, and generates an ever-evolving flux of ideas; the more information gained from patients, the better.

As previously mentioned, in 2011, IBM's AI program, Watson, achieved victory over two former Jeopardy champions. At the time, this success was celebrated as the dawn of a new era, where machines could provide answers to all our questions. It was envisioned that Watson could swiftly analyze medical documents and correlate patient data with the latest research, offering personalized, cutting-edge treatments. IBM's acquisition of four health-data companies in the mid-2010s, totaling approximately \$4 billion, seemed to position them perfectly to revolutionize healthcare through AI. However, by 2020, after investing billions in research and development, no AI doctor product ever materialized from IBM's Watson Health Division, and the unit was eventually sold at a loss in early 2022 for around \$1 billion.

IBM's leadership now characterizes the journey of Watson Health as a challenging and protracted one, far more complex and time-consuming than they had initially anticipated [14]. A similar tale unfolded with IBM's collaboration with the University of Texas M. D. Anderson Cancer Center in Houston. Their joint project, aimed at creating the Oncology Expert Advisor tool, ultimately met the same fate. In 2016, an audit conducted by the university revealed that over \$62 million had been expended on the project before it was ultimately canceled.

A closer examination of these unsuccessful ventures uncovers a fundamental mismatch between the realities of the medical field and the current state of limited, fragmented, and unintelligent machine learning techniques. The data mining, neural networks, and statistical approaches that were employed are primarily suited for recognizing specific, predefined patterns or constructing predictive models applicable only in well-defined, controlled scenarios.

These tools tend to uncover only the most straightforward relationships within data. Even with abundant amounts of data at their disposal, these models are unable to deduce basic facts, such as the fact that blood circulates due to the pumping action of the heart. Consequently, they certainly fall short when it comes to unraveling chains of causality or suggesting associations that could significantly contribute to diagnosis or treatment. The notion that AI could replace medical professionals is, therefore, a baseless notion.

8.1 Are simple human tasks easy to automate?

We know that machines are good at certain kinds of tasks. A factory robot can punch holes in pieces of metal all day without ever making a mistake or getting tired. With all of the advanced technology at our disposal today, the mass media would have you think that it's only a matter of time before most mundane tasks are automated. It will

probably take only the first two examples in this section to convince you otherwise, and we'll close the chapter with several other examples to drive the point home.

The first task we'll consider is pizza making. To introduce the story, we'll quote CNBC personality Jim Cramer from a video segment he made praising the technology startup Zume [15]:

Everybody loves pizza, in part because it's very hard to screw up a pizza. I mean, even bad pizza still tastes good, but what if there was a better way to make and deliver pizza to you? Turns out there is, which brings me to Zume, Inc., a Silicon Valley-based startup that's trying to bring this industry into the modern era. They're best known for introducing robots into the production process for everything from spreading the tomato sauce evenly to taking pizzas in and out of the oven and slicing it perfectly. Basically, robots do all sorts of the highly repetitive tasks. . . . I think it's real. It's not an idea; it's a business.

At one point, Zume appeared to possess all the necessary talent and resources to revolutionize the automation of pizza production and delivery. In 2018, they secured a substantial \$375 million investment from SoftBank and developed a promising system. Their innovative approach involved a synergy of humans and robots, capable of producing up to 120 pizzas per hour in a versatile truck that could function both as a food truck for sidewalk sales and a delivery vehicle. For deliveries, an AI system optimized routes and pizza production, ensuring delivery within as little as 5 minutes from an online order. However, by January 2020, after grappling with numerous challenges, Zume Pizza ceased operations. Initially, their robots were repurposed for sustainable food packaging production [16]. Yet, in June 2023, the *Wall Street Journal* reported that Zume had initiated a winding-down process [17]: “California startup Zume, which was developing a robotic pizza maker and was once valued at \$2.25 billion, recently entered a wind-down process.”

So it would appear that making and delivering pizzas is still beyond the reach of modern AI, or at least that there are logistical or practical problems that make replacing humans infeasible. Perhaps the task of making and serving coffee would be more tenable? After all, John Steinbeck described an automated system in his 1962 book *Travels with Charley* [18] that seemed to operate rather well:

I had neglected my own country too long. Civilization had made great strides in my absence. Suppose you want a soft drink; you pick your kind—Sungrape or Cooly Cola—press a button, insert the coin, and stand back. A paper cup drops into place, the drink pours out and stops a quarter of an inch from the brim—a cold, refreshing drink guaranteed synthetic. Coffee is even more interesting, for when the hot black fluid has ceased, a squirt of milk comes down, and an envelope of sugar drops beside the cup. But, of all, the hot-soup machine is the triumph. Choose among pea, chicken noodle, beef, and veg., and insert the coin. A rumbling hum comes from the giant, and a sign lights up that reads, “Heating.” After a minute, a red light flashes on and off until you open a little door and remove the paper cup of boiling-hot soup.

Despite a seemingly promising start, attempts to create a robotic barista have fallen short, as evidenced by the noticeable absence of robots in coffee establishments. In

fact, we continue to prepare and serve coffee much as we did in 1671 when the first café opened in Marseilles, France. But why is this the case? After all, isn't making coffee essentially about transferring substances between containers and toggling switches on and off?

Even a young child can effortlessly fill a container of any shape or size with water from the nearest faucet, provided they can reach the tap and the container fits beneath it. They instinctively know how far to turn the tap and for how long to let the water flow before stopping it. They comprehend the roles of the container and the faucet, as well as the behavior of water as it flows from the tap into the container.

Machines, on the other hand, lack this intuitive understanding of containers, water flow, or faucets. While a robotic arm can be programmed to move a container to a specific spot and activate a tap for a set duration, it will continue pouring water into a smaller container until it overflows unless given explicit instructions to adapt to the new situation. And this is assuming the robot can even pick up a differently sized container, which is a challenging task in itself.

Considering that we've had machines capable of producing basic mixtures of substances in cups for six decades, it would seem that we are overdue for a robotic café. Considering our slow progress in automating a relatively simple task such as coffee preparation, one must question the likelihood of the automation of "most jobs" happening anytime soon.

Perhaps you think we are being selective with our examples and that robots are able to adequately perform other basic tasks? Walmart started experimenting in 2017 with robotic inventory checkers produced by Bossa Nova Robotics, a spinoff of the Carnegie Mellon Robotics Institute. The machines would go up and down the aisles with the task of confirming that items were on the shelves. The experiment was ended after three years, with sources close to the project saying that "Walmart ended the partnership because it found different, sometimes simpler solutions that proved just as useful" [19].

One might think that textile work would be a good candidate for automation. For example, garment manufacture typically entails repeatedly stitching according to a set pattern. It has even been joked that "a modern textile mill employs only a man and a dog—the man to feed the dog, and the dog to keep the man away from the machines." In reality, however, sewing has been particularly difficult to automate, in part because fabric stretches and bunches when it's manipulated, and machines don't have the dexterity needed to keep things properly arranged [20].

What about automation in transportation? Driving would seem like a prime candidate for automation since it's monotonous, the laws are well-defined, and human error is a cause for concern. We briefly mentioned in chapter 4 that complex environments like that around the Arc de Triomphe might never be navigable by the type of AI we have today. Despite colossal investment and more than 15 years of development efforts, driverless vehicle projects are still faltering, and a modern city like San Francisco still presents seemingly insurmountable challenges.

An ABC News video posted on June 29, 2023, highlights the difficulties that driverless vehicles are still facing [21]. The journalist was alarmed that her vehicle had stopped at a green light for no reason and said:

I think we are stopped. Oh my god. Right now, there is a green light, and nothing is happening. We are stuck, and not only that, we are not even on the left-hand side properly. Oh-Oh, it says our team is working to get you moving. It was a green light. It didn't know what to do. How could I rely on a car to make the right decision when it cannot "see" a simple green light?

Waymo has the resources to hire the best AI engineers, and if an advanced company is struggling to make their autonomous vehicles recognize a green light, how could anyone suggest that AI sees better than humans? With the help of the support team, the car eventually moved, but the journalist was later shocked that the car could not drop her at the correct destination. She said:

This is weird. This is not where we are going. This is not where the random museum is. Now I am going to have to call support and see what I can do because I'm completely at a loss right now. The location where I want to be is over there, and it's dropped me at the bottom of this hill.

The support team asked the journalist to open her Waymo app and retype her destination. Alas, the vehicle made the same error and again stopped at the same wrong place. This is another illustration of why current AI lacks reasoning capabilities. How could anyone trust or rely on a system that keeps making the same obvious error?

In aviation, a field that has been supplementing human dexterity with automation since its inception, our most modern AI technology is still causing disasters. In 2013, the National Transportation and Safety Board of the United States explained the crash of Asiana Flight 214 by saying that the crew had “over-relied on automated systems that they did not fully understand” [22]. The NTSB chairman said,

In their efforts to compensate for the unreliability of human performance, the designers of automated control systems have unwittingly created opportunities for new error types that can be even more serious than those they were seeking to avoid.

Another well-documented case of automation-related disaster is the 2009 loss of Air France Flight 447. En route from Rio de Janeiro to Paris, ice accumulation on some of the instruments caused the autopilot to disengage unexpectedly, and when the pilots failed to react appropriately, the plane and its 228 occupants smashed into the Atlantic. Later, in March 2019, after some poorly written software caused two crashes that killed 346 people, Boeing's entire series of 737 MAX aircraft was grounded.

The prospects for automation seem rather spotty, even when we consider computer-related tasks for which technology companies are highly incentivized to replace humans with machines. A prime example is content moderation. In May 2020, Facebook agreed to pay \$52 million to 11,250 employees to compensate them for the mental health

problems they had developed on the job [23]. Work reviewing online content for appropriateness has caused anxiety, depression, addiction, and other problems, yet Facebook, one of the largest technology companies in the world, continues to hire thousands of people each year to address this task.

Abstractly, content moderation involves opening image and text files, reviewing them, and sorting them into categories. Given the psychological harm this has done to people and the bad press and financial consequences Facebook has suffered as a result, we can assume that if there were any feasible way to automate this task, it would be automated. As reported by the Verge, when asked by Congress, the Facebook CEO dodged question after question by citing the power of AI: “Moderating hate speech? AI will fix it. Terrorist content and recruitment? AI again. Fake accounts? AI. Russian misinformation? AI. Racially discriminatory ads? AI. Security? AI” [24].

The fact that it has not been completely solved should shed new light on claims about how advanced Facebook’s technology actually is, and it should prompt swift dismissal of headlines like “Facebook Artificial Intelligence Robots Had to Be Shut Down After They Started Talking to Each Other in Their Own Language” [25]. This article stands out for ascribing lifelike characteristics to machines. For instance, it states:

The two chatbots came to create their own changes to English that made it easier for them to work—but which remained mysterious to the humans that supposedly look after them. The bizarre discussions came as Facebook challenged its chatbots to try and negotiate with each other over a trade, attempting to swap hats, balls and books, each of which were given a certain value. “The discussions” quickly broke down as the robots appeared to chant at each other in a language that they each understood but which appears mostly incomprehensible to humans.

Surely machines capable of inventing their own language would be able to identify language that violates Facebook’s terms of use policy.

Finally, let’s consider the seemingly straightforward task of translation. People were already anticipating in the 1950s that computers would soon be doing all of our translating. After all, a simple dictionary can provide a word-for-word translation, and by comparing well-translated documents to the originals, you might think it would be easy to assemble an adequate list of rules and exceptions. After more than 70 expectant years, translators are still an essential part of the business world.

We need only relate one simple example to demonstrate the point as the internet abounds with others. A common French saying, “La larme est la goutte d’eau qui fait déborder l’âme,” which literally means “The tear is the drop of water that makes the soul overflow,” is translated by Google as shown in figure 8.1.

It’s not quite clear what has gone wrong here, but the result makes no sense. The translation algorithm apparently made reference to a library of idioms compiled by humans, which is at least a good start. The French phrase “La goutte d’eau qui fait déborder le vase” literally means “The drop of water that made the vase overflow,” and its English equivalent is “The straw that broke the camel’s back.” Since the two French phrases only differ by two words, the algorithm must have replaced the



Figure 8.1 Google translate results as of May 6, 2022. The result may change due to user feedback.

familiar French idiom with its English equivalent and then substituted literal translations of the words that didn't match the idiom.

No bilingual human would make such a mistake. Either they would be familiar with the idioms and translate them correctly, or they would recognize that a literal translation doesn't make sense and that they need to investigate. It's quite likely, in fact, that they would be able to figure out from context what it must mean. The example should make clear that machines are still far from having this ability.

Translating between French and English should actually be one of the easiest translation tasks of all. These two languages are closely related, and there is a vast amount of training material available for any AI system. For example, since Canada's Official Languages Act of 1969, all of their official documents have had to be available in both English and French. The European Commission has the same requirement. This must amount to at least hundreds of thousands of pages of text that have been carefully written and meticulously translated, and for many years now, most of this will have been available in electronic form.

If AI from one of the world's preeminent technology companies can't reliably translate a common phrase from French to English, how much confidence could we place in its ability to translate between more enigmatic languages? We might still be decades away from the day when AI can passably translate poetry from Mandarin to Indonesian or literature from Vietnamese to Finnish, and quite possibly, that day might never come. Even the translation of more straightforward material in the legal or business domain will depend on humans for the foreseeable future.

In brief, although there are ongoing concerns about AI potentially replacing human jobs, it's important to acknowledge that AI has inherent limitations. AI faces significant obstacles when it comes to tasks that demand human qualities such as reasoning, physical abilities, emotional comprehension, creativity, and handling ambiguous situations. Furthermore, AI systems heavily rely on the data they are trained on, which can introduce biases, rendering them unsuitable for tasks demanding fairness and impartiality. Due to these limitations and their inherent lack of a true understanding of the

world, AI is poised to complement rather than supplant humans. Consequently, AI is more likely to enhance human capabilities by handling repetitive and data-driven aspects of work, thereby increasing efficiency and allowing humans to focus on tasks that demand our unique qualities.

Now that we've established that most occupations are safe from automation for the time being, we can address the ominous "singularity" referred to at the start of this chapter. Cue the sinister music, my friends. First they came for your jobs, and now the AI overlords are coming to take over civilization.

Summary

- People had similar unfounded fears 400 years ago when new machines were introduced.
- AI lacks the intuition, empathy, and deductive capabilities needed in many professions, so people in these professions will not be replaced by AI.
- Many tasks are very difficult to automate because of the complexity and imperfection of the real world.

Technological singularity is absurd

This chapter covers

- The unlikelihood of the singularity
- The lack of intelligence in machines
- Thoughts about the human brain

Nothing in this world is to be feared . . . only understood.

—Marie Curie

According to some people, the end of human civilization won't be due to things like climate change, nuclear war, or our sun dying out. Instead, they believe that in the not-so-distant future, artificial intelligence could become so advanced that it gains its own will and takes control of the planet. This potential catastrophe is often referred to as “the singularity,” a hypothetical point in time when AI would advance so rapidly that humans couldn't keep up with its progress. While this concept makes for exciting stories in science fiction, it is essential to ground such speculations in reality. In this chapter, we aim to demystify the notion of technological singularity, arguing that it is fundamentally flawed.

9.1 The genesis of technological singularity

The concept of technological singularity isn't a new idea; it has its roots in discussions that date back to at least 1958. In that year, physicists John von Neumann and Stanislaw Ulam engaged in a conversation where they pondered the "ever-accelerating progress of technology" and how it might lead to a profound and potentially unpredictable turning point in human history. This turning point, referred to as the "singularity," would mark a moment beyond which human affairs could change in ways we couldn't foresee. A more detailed explanation of this notion came in 1965 from I. J. Good in his article titled "Speculations Concerning the First Ultra-Intelligent Machine" [1]. In this article, Good explored the idea of an "ultra-intelligent machine" that could emerge, potentially surpassing human intelligence. This machine, he suggested, could trigger profound changes in our society and the way we live:

Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an "intelligence explosion," and the intelligence of man would be left far behind. Thus the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control. . . . It is more probable than not that, within the twentieth century, an ultraintelligent machine will be built.

In 1967, Marvin Minsky, who led the AI laboratory at MIT, boldly stated that "within a generation, the problem of creating 'artificial intelligence' will be substantially solved" [cited in 2]. He even went a step further, suggesting that "within 10 years, computers won't even keep us as pets." Around the same time, Herbert Simon, another prominent computer scientist, made a similarly ambitious prediction, claiming that by 1985, machines would be capable of doing any work that humans could do [3].

Unfortunately, grandiose claims like these tend to grab more attention than reasoned analyses. In recent years, several well-known figures in the scientific and tech world, including Bill Gates, Stephen Hawking, and Elon Musk, have raised alarm bells about the potential risks associated with AI. Sam Altman, the CEO of OpenAI, also subscribes to the idea of the technological singularity [4]:

Our self-worth is so based on our intelligence that we believe it must be singular and not slightly higher than all the other animals on a continuum. Perhaps the AI will feel the same way and note that differences between us and bonobos are barely worth discussing.

A report published in 2015 by the Global Challenges Foundation and Oxford University's Future of Humanity Institute highlighted this concern as one of the "12 risks that threaten human civilization" [5]. Specifically, the report referred to the risk associated with advanced, intelligent machines:

Through their advantages in speed and performance, and through their better integration with standard computer software, they could quickly become extremely intelligent in one or more domains (research, planning, social skills . . .). If they became skilled at computer

research, the recursive self-improvement could generate what is sometimes called a “singularity,” but is perhaps better described as an “intelligence explosion” With the AI’s intelligence increasing very rapidly, such extreme intelligences could not easily be controlled (either by the groups creating them, or by some international regulatory regime), and would probably act in a way to boost their own intelligence and acquire maximal resources for almost all initial AI motivations. And if these motivations do not detail the survival and value of humanity in exhaustive detail, the intelligence will be driven to construct a world without humans or without meaningful features of human existence. This makes extremely intelligent AIs a unique risk, in that extinction is more likely than lesser problems. An AI would only turn on humans if it foresaw a likely chance of winning; otherwise, it would remain fully integrated into society. And if an AI had been able to successfully engineer a civilisation collapse, for instance, then it could certainly drive the remaining humans to extinction.

Rest assured that not everyone involved in the field of AI harbors apocalyptic visions. Throughout history, there have been voices of reason who offered more measured and rational perspectives on the capabilities and limitations of artificial intelligence. One such visionary was Ada Lovelace, recognized for her pioneering work in the field of AI as she is often credited with writing the world’s first computer program. In the 19th century, Lovelace collaborated with Charles Babbage on his proposed Analytical Engine, a mechanical general-purpose computer design. Her remarkable insight went beyond the conception of mere calculations; she grasped the potential for the machine to manipulate symbols and generate complex sequences, essentially laying the groundwork for programming. Lovelace’s groundbreaking notes on the Analytical Engine demonstrated the profound idea that machines could be used for more than just mathematical computation, envisioning a broader computational world. Her work serves as an early example of the theoretical underpinnings of modern computer programming and AI. In 1842, Lovelace articulated the following perspective regarding the Analytical Engine [6]:

The Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform. It can follow analysis, but it has no power of anticipating any analytical relations or truths.

Similarly, Arthur Samuel, whom we introduced in chapter 1, is one of the pioneers of AI. Samuel’s work was instrumental in shaping the foundations of AI and had a profound effect on the field. As previously mentioned, he designed a computer program that could play checkers and improve its performance over time through learning from experience. This concept laid the groundwork for modern machine learning and reinforcement learning algorithms. In a seminal 1960 article published in *Science* titled “Some Moral and Technical Consequences of Automation—A Refutation” [7], Samuel eloquently asserted the critical importance of separating fact from fiction in the discourse surrounding automation:

A machine is not a genie, it does not work by magic, it does not possess a will, and . . . nothing comes out which has not been put in, barring of course, an infrequent case of

malfunctioning. . . . The machine will not and cannot do [anything] until it has been instructed as to how to proceed. . . . To believe otherwise is to believe in magic. Since the machine does not have a mind of its own, the “conclusions” are not “its.” The so-called “conclusions” are only the logical consequences of the input program and input data, as revealed by the mechanistic functioning of an inanimate assemblage of mechanical and electrical parts. The “intentions” which the machine seems to manifest are the intentions of the human programmer, as specified in advance, or they are subsidiary intentions derived from these, following rules specified by the programmer. . . . Although I have maintained that “nothing comes out that has not gone in,” this does not mean that the output does not possess value over and beyond the value to us of the input data. The utility of the computer resides in the speed and accuracy with which the computer provides the desired transformations of the input data from a form which man may not be able to use directly to one which is of direct utility.

Arthur Samuel’s insightful view on AI remains pertinent and accurate in today’s context. The concept of an ultra-intelligent, malevolent entity emerging from our technology to seize control and create chaos remains firmly in the fictional world. While it might be tempting to draw parallels with scenarios depicted in the works of science fiction authors like Isaac Asimov, TV series such as *The Twilight Zone*, or blockbuster films like *The Terminator* and *The Matrix*, it’s essential to recognize that the cognitive faculties, such as intuition, imagination, and creativity, which characterize true intelligence, remain elusive for artificial systems.

9.2 The truth about the evolution of robotics

From Mary Shelley’s 1818 novel depicting Dr. Frankenstein’s creation to the animated doll Pinocchio brought to life in Disney’s 1940 film, the idea of human-made objects gaining sentience has always held a captivating allure.

The term “robot” achieved widespread recognition and popularity through its usage in Karel Čapek’s 1920 play, *R.U.R. “Rossum’s Universal Robots.”* This word, with its roots in the Czech language, specifically *robota*, which translates to “forced labor,” introduced a concept that would shape the future of automation and AI. The evolution of robotic terminology continued with the introduction of “androids.” While the term itself had ancient Greek roots, it found a new home in science fiction during the 1930s, thanks to the imaginative works of authors like Edmond Hamilton in his Captain Future series in the early 1940s. These androids, with their human-like qualities, began to captivate the imagination of audiences worldwide. Fast forward to today, we find ourselves in a world where robots do indeed perform labor. However, their capabilities remain constrained to tasks of a relatively straightforward and limited nature. Despite our technological progress, the concept of an android, as envisioned by early science fiction, remains tantalizingly distant.

General Motors made an early venture into robotics in 1961 when it introduced a mechanical arm for manipulating hot-cast metal components at a New Jersey plant. This venture was met with success, paving the way for further exploration in the 1970s. The company expanded the use of machines into other meticulously controlled tasks,

including welding and painting. In the contemporary landscape, robots play an indispensable role in auto manufacturing. Nonetheless, they remain specialized to particular tasks.

In 1962, MIT's Henry Ernst conceived the first computer-operated robotic hand, designed for the remote manipulation of radioactive materials. A symphony of motors orchestrated the movements of a mechanical arm, while light sensors in the fingertips discerned shadows. This ingenious system, guided by a computer program, maneuvered objects on a table, picking them up and placing them into a container.

Shakey, the subject of research at the SRI Artificial Intelligence Center from 1966 to 1972, took a monumental leap forward as the inaugural mobile robot equipped with the ability to perceive its surroundings (figure 9.1) [8]. A marvel of its time, Shakey served as an experimental platform to integrate machine learning, computer vision, navigation, and a myriad of AI techniques. Its repertoire included tasks demanding planning, route-finding, and the rearrangement of simple objects, earning it the moniker of the “first electronic person” by *LIFE* magazine in 1970. While Shakey's



Figure 9.1 Shakey (image courtesy of the Computer History Museum)

accomplishments were undeniably groundbreaking, it's essential to maintain a realistic perspective. The tasks it excelled at were a far cry from the cinematic portrayals of robots. A human operator would issue commands from a console, instructing Shakey to execute operations like pushing a block from a platform. These instructions would then be transmitted via radio, enabling Shakey to survey its environment, locate the target, and carry out the task. It was, undoubtedly, a notable accomplishment for artificial intelligence, yet one that highlights the chasm between the capabilities of contemporary robotics and the visionary worlds depicted in fiction. While progress has been substantial, the gap between science fiction and real-world robotics remains substantial, emphasizing the ongoing challenges in creating highly capable and adaptable robotic systems.

In the decades following Shakey's pioneering research, the field of robotics witnessed a gradual yet steady evolution. By the early 1980s, engineers at Waseda University in Tokyo achieved a significant breakthrough. They introduced a robot powered by a microprocessor chip, a revolutionary departure from the room-sized computers that had previously driven robotic systems. This innovative leap paved the way for more compact and agile robotic platforms. What set this robot apart was its ability to stand on two "legs" and take measured steps at a deliberate pace, exemplifying early progress in bipedal locomotion. Shortly thereafter, in 1981, Shigeo Hirose at the Tokyo Institute of Technology unveiled another milestone in robotics with the creation of a quadruped robot. This remarkable machine demonstrated the capacity to climb stairs and opened new possibilities for robots to explore environments with a combination of stability and mobility. The 1990s brought another advancement in the form of an eight-legged robot, developed by researchers at Carnegie Mellon University [9]. The use of multiple legs allowed the robot to maintain stability and traverse landscapes that would be impassable for wheeled or bipedal counterparts.

One of the most captivating milestones in the field of robotics emerged with NASA's groundbreaking Pathfinder mission. On the historic date of July 4, 1997, the Mars Pathfinder, a remarkable robotic spacecraft, achieved a triumphant exploit by successfully deploying a base station and an autonomous rover named Sojourner onto the Red Planet [10]. This monumental achievement marked a significant leap forward in space exploration. Sojourner, the autonomous rover, played a vital role in this groundbreaking mission. Over the course of 83 remarkable days, it embarked on an incredible journey, traversing the Martian terrain with its six wheels. Equipped with a suite of scientific instruments, Sojourner became an invaluable explorer, capturing high-resolution images of the Martian landscape and collecting vital atmospheric and geological data. This achievement was an impressive progress in the field of robotics and space exploration.

The advent of domestic robotics took a significant step forward during the holiday shopping season of 1998, leaving parents and children captivated by the promise of technology. Leading this technological wonderland was the Furby, brilliantly marketed as an artificially intelligent companion. Resembling a fuzzy, animal-like creature,

the Furby boasted an array of interactive features, including moving eyes and a mouth, as well as voice-recognition technology. It could communicate in its own quirky language known as “Furbish” and gradually learn and speak words in English and several other common languages. This gradual linguistic development was designed to mimic the process of a pet or companion learning its owner’s language. The Furby’s irresistible charm and mass appeal led to the staggering sale of 14 million units in its debut year on the market, making it a must-have item for countless households.

While the Furby was captivating headlines and toy stores, Sony Corporation was pioneering another remarkable product in the form of AIBO, a robotic pet dog. AIBO featured rudimentary computer vision that enabled it to interact with its environment and respond to over 100 voice commands. Its movements and interactions made it an instant sensation. In a remarkable display of consumer enthusiasm, the initial run of 5,000 AIBO units sold out to enthusiastic internet customers within 20 minutes. Sony continued to release new models of AIBO annually until 2006. In 2017, Sony announced the return of AIBO with a new model that promised to form an “emotional bond” with its users [11]. This development marked a significant shift in the relationship between humans and robots, as it indicated the potential for robots to provide companionship and emotional support.

On Valentine’s Day in 2002, a groundbreaking moment occurred in the world of robotics and popular culture when Honda unveiled ASIMO, an acronym that stands for “Advanced Step in Innovative Mobility” (figure 9.2). ASIMO represented a significant leap forward in the development of humanoid robots, and its introduction was the culmination of over 15 years of tireless work by Honda engineers. The journey to create ASIMO began way back in 1986 when Honda’s team embarked on the ambitious task of constructing a robot that could move and walk just like a real person. After years of dedication and innovation, ASIMO emerged as a remarkable robot capable of walking and performing a variety of specific tasks with precision. One of ASIMO’s most iconic moments was when it rang the bell to open the New York Stock Exchange. This event took place on the 25th anniversary of Honda’s stock being traded on the market.

The forefront of robotics technology can often be found within the workshops of Boston Dynamics Robotics. Founded in 1992 as a spinoff from MIT, this company, headquartered in Waltham, Massachusetts, has undergone ownership changes, with Google and SoftBank being previous owners before its acquisition by Hyundai Motor Group in 2020. Boston Dynamics Robotics has gained recognition for its impressive robotic creations, including the quadruped military robot known as BigDog, the humanoid robot named Atlas, and the versatile material-handling robot Handle. A Bloomberg article published in November 2022 featured the CEO of Boston Dynamics, who expressed great enthusiasm for Handle. He stated, “Handle, which gracefully maneuvers on two large wheels, is designed to automate tasks like moving boxes on and off pallets and perhaps even unloading boxes from trucks, a notoriously challenging task for a robot” [12].



Figure 9.2 ASIMO (2000) (image courtesy of Miraikan Museum, Tokyo)

As we contemplate this statement, it's important to maintain perspective. In the 50 years since the days of Shakey, we have progressed from merely locating a block and pushing it off a platform to precisely locating a box and lifting it onto or off of a platform. Experts in the field of robotics tend to be candid about both the capabilities and limitations of current technology. Instead of heeding the words of those who sensationalize the idea of machines taking over, we can benefit from their realistic assessments. Even with the most advanced sensors, actuators, cameras, and materials, coupled with remarkable talent and resources, the world's leading robotics teams view the prospect of using robots for loading and unloading trucks as a crucial milestone in the progress of robotics. While there are captivating videos online showcasing robots dancing, jumping, and performing acrobatic feats, it is essential to remember that these movements are typically preprogrammed. These machines do not autonomously respond to their environment. The sight of a human-shaped robot executing a backflip may be impressive, but it pales in comparison to the automatic movements of even a clumsy child. Our innate ability to regain balance when slipping on ice or when we miss a step on the stairs remains far superior to the capabilities of even the most advanced robots.

9.3 *Merging human with machine?*

Electronic implants that can decode brain activity and communicate with computers have been a subject of research for several decades. As reported by the *Washington Post* in 2016, a noteworthy collaboration between the University of Pittsburgh and the University of Pittsburgh Medical Center involved the implantation of electrodes smaller than a grain of sand into a patient's sensory cortex. These electrodes received signals from a robot arm, allowing the individual to experience tactile sensations in their paralyzed right hand, effectively bypassing their damaged spinal cord [13]. This development demonstrated a promising application of brain-computer interfaces in the field of medical rehabilitation.

However, despite these remarkable advancements, making grandiose claims about the imminent possibility of uploading human minds into computers or creating synthetic humans and human-machine hybrids is unlikely. Books with titles like *How to Create a Mind: The Secret of Human Thought Revealed* [14] may appear evocative of science fiction, leaving us to question their scientific validity.

The challenge lies in the vast disparity between the current state of artificial intelligence and our understanding of the complexities of the human brain. Prominent neuroscientists themselves admit to lacking a fundamental understanding of how the brain truly operates. Therefore, we should be careful not to use or associate terms like “brain” or “neuron” with AI. A neuron within a deep learning neural network bears about as much resemblance to an actual neuron as a teddy bear does to a live bear!

In the book *The Singularity Is Near*, Ray Kurzweil boldly predicts the replacement of human DNA with designer genes, even providing a somewhat precise timeline, suggesting that we'll have reverse-engineered human brains by the late 2020s [15]. Such forecasts are partially based on observations such as computers defeating champions in games like chess and Moore's Law, which suggests that computing power tends to double approximately every two years. This assertion was somewhat accurate a decade ago; however, the chip manufacturers are approaching the physical limitations of transistor density on microchips and the challenges of heat dissipation in highly compact devices. Even if we assume that computers will continue to become faster, smaller, and more affordable indefinitely, it's crucial to differentiate between computing power and intelligence. Machines still operate by executing human-coded instructions, fundamentally distinct from the multifaceted processes that govern the human mind, involving elements of chemistry, biology, neurology, psychology, and more. Neuroscientist Christof Koch pertinently phrases it this way in his article “The End of the Beginning for the Brain” [16]:

One thing is certain. Biology knows nothing of simplicity. Brains are not assembled out of billions of identical LEGO blocks but out of hundreds of distinct nerve cell types. Each cell type has its own idiosyncratic morphology, signaling, and active genes. And they are interconnected with elaborate wiring rules that we only discern darkly.

The concept of merging humans and machines often overestimates the sophistication of our technological advancements while simultaneously underestimating the astounding complexity of living organisms. The human body itself is nothing short of a miraculous wonder, comprising 10s of trillions of adaptive and intricately interconnected cells. Within this complex web of life, every organ plays a role in sending and receiving signals, engaging in complex coordination with every other part. Consider one of our many organs, the eyes. Think about the large volume of data processed by our eyes with every passing moment, the complex cascade of signals they stimulate, the feedback loops that influence their function, and the numerous systems they coordinate to provide us with vision.

Yet, our eyes represent just a fraction of the immense complexity within our bodies. Even as remarkable as they are, they pale in comparison to the vast complexity of our brains, often referred to as the most complex pieces of organized matter in the known universe, and for good reason. The human brain, a marvel of biological evolution, possesses an unparalleled capacity for creativity, intuition, emotional intelligence, and a nuanced understanding of complex, ever-changing environments. It operates on a fundamentally different paradigm compared to AI, which, while powerful in its own right, lacks the complex interplay of biological neurons and the underlying biochemical processes that contribute to human cognition. Those who have witnessed AI's real-world applications are aware that current AI is far from achieving anything that bears a resemblance to the depth and complexity of human cognition, let alone being able to replicate the human brain.

In his highly recommended book *The Biology of Belief* [17], Bruce Lipton eloquently describes the concept of living intelligence that operates within our bodies:

When a measles virus infects a child, an immature immune cell is called in to create a protective protein antibody against that virus. In the process, the cell must create a new gene to serve as a blueprint in manufacturing the measles antibody protein. Activated cells employ an amazing mechanism, called affinity maturation, that enables the cell to perfectly adjust the final shape of its antibody protein so that it will become a perfect complement to the invading measles virus. The new antibody gene is also passed on to all the cell's progeny when it divides. The cell learned about the measles virus; it also creates a memory that will be inherited by its daughter cells. This amazing genetic engineering represents an inherent intelligence mechanism by which cells evolve.

Our cells are complex microcosms of life, intricately designed to receive, process, react to, and preserve information in a synchronized ballet. These tiny units of life tirelessly perform their roles, coordinating with each other to maintain the equilibrium of our bodies. When confronted with external changes, cells exhibit remarkable adaptability, adjusting their functions to respond effectively to their ever-changing environment. They can sense, make decisions, and execute actions, all within fractions of a second. While we have made progress in understanding cellular mechanisms, we are still far from uncovering the full extent of their complexities.

9.4 *Science fiction vs. reality*

The concept of technological singularity has been a topic of discussion for over six decades, often accompanied by enthusiastic hype. While the potential threats posed by AI have been frequently highlighted, it's important to acknowledge that AI, in its current state, falls far short of possessing the cognitive abilities of even a young child, let alone attributes like creativity or sentiment. It's essential to recognize that machines, whether they are robots or computer programs, remain inert and devoid of purpose until humans provide them with instructions and guidance.

Computers excel at manipulating numbers and processing words, but they lack genuine comprehension of the meaning behind the data they handle. Algorithms, operating on binary code, cannot truly “understand” in the way humans do. Complex problem-solving, adapting to unforeseen situations, and the formulation of hypotheses are still well beyond the capabilities of AI. In practice, most AI applications are confined to highly specific tasks, often reliant on extensive sets of meticulously curated training data. While it may be tempting to attribute a certain mystique to technology that escapes our complete comprehension, this book seeks to dispel such illusions. Robots and automatons, despite their seemingly intricate actions, are simply executing their preprogrammed instructions. They do not possess self-awareness or exhibit characteristics like fear or a desire for conquest, and they certainly do not pose a threat to human civilization.

Despite these limitations, governments around the world continue to invest heavily in AI and autonomous systems. However, even as of 2022, military leaders raise questions about their reliability and practicality. A 2020 RAND report [18] on military applications of AI offers several illustrative examples of the kind of incidents that can give decision makers pause.

One such incident dates back to 1988 when the US Navy accidentally shot down an Iranian civilian aircraft, resulting in the tragic loss of 290 lives. While the exact cause remains disputed, it's known that the Aegis weapons system, responsible for tracking planes and operating munitions, employed an automated system to assign tracking numbers to radar-detected objects. This system periodically recycled call numbers, and during the critical time frame, the passenger jet was assigned a new number, which happened to be the same as one assigned to a fighter jet just 110 miles away.

Another disaster occurred during the Gulf War in 1991 when the *USS Missouri* mistakenly believed it was under attack from an Iraqi Silkworm missile and deployed countermeasures. Simultaneously, the nearby *USS Jarrett*'s Phalanx CIWS system, operating in autonomous target-acquisition mode, detected the countermeasures and fired at them. Unfortunately, four rounds from the *USS Jarrett* struck the *USS Missouri*.

In yet another incident during a multinational training exercise in the Pacific in 1996, a Navy A-6E Intruder was towing a target plane intended to be shot down by Japanese participants. However, instead of locking onto the target plane, the Phalanx system mistakenly targeted the Intruder and opened fire.

The RAND report further describes a fourth accident in 2003 where a US Patriot battery mistakenly shot down a Tornado flown by the Royal Air Force due to a misidentification by the missile system, resulting in the loss of two crew members. A fifth incident occurred later the same year when a US Navy aircraft was mistakenly identified as an Iraqi missile, leading to the tragic death of the pilot. These incidents serve as reminders of the potential pitfalls of autonomous systems and highlight the importance of careful consideration, testing, and oversight in the integration of AI and automation into critical functions.

In a candid question-and-answer session held at the prestigious Brookings Institution think tank, US General Selva articulated his reservations about contemporary AI methodologies, particularly deep learning, due to their inherent inability to provide transparent explanations for their decisions [19]. He emphasized: “Our belief is AI alone doesn’t actually solve the problems that we’re being asked to solve. It can’t be a black box that says just go do X.”

General Selva further highlighted that the military cannot afford to rely on an AI system that lacks both reliability and comprehensibility. In his view, an acceptable system must possess the capability to not only undergo rigorous physical testing but also intellectual scrutiny.

Joseph Weizenbaum, in his seminal 1976 work *Computer Power and Human Reason* [20], introduced a profound perspective on the concept of technological singularity that remains remarkably valid. He stated:

Science may also be seen as an addictive drug. Our relentless fascination with science has not only made us rely and depend on it, but, similar to many other substances taken in escalating doses, science has gradually transformed into a slow-acting poison.

Weizenbaum’s observation is quite relevant, as more and more people are getting caught up in their devices and spending a lot of time on social media, which can have negative effects on their well-being due to harmful content and unhealthy behaviors.

The concept of technological singularity, while intriguing, begs the question of whether we are becoming too enamored with science fiction scenarios at the expense of addressing real-world challenges. Humanity faces pressing threats, such as climate change, epidemics, and the specter of nuclear and biological warfare. In this context, the fascination with the potential for superintelligent AI can divert attention and resources from immediate, tangible problems. It is crucial to recognize that AI, like any technology, can be used for both benevolent and malevolent purposes. While AI can contribute to the development of life-saving vaccines, it can also be weaponized for destructive ends. A system designed to inform can equally be used to deceive.

In conclusion, General Selva’s reservations about the limitations of current AI techniques, coupled with Weizenbaum’s cautionary insights about the double-edged nature of scientific progress, remind us of the importance of thoughtful and ethical innovation. Therefore, instead of dwelling solely on dystopian visions of AI, we must

prioritize the responsible development and safe deployment of these technologies to serve humanity's best interests.

Summary

- AI singularity has been predicted but failed to occur many times over the last 60 years, and we are no closer to experiencing it.
- Although there has been some progress in robotics, machines still have no intelligence and no will, so they have no ability to evolve and become a threat.
- The faster processing speeds of modern computers do not make them think.
- We are still unable to explain how the human brain works, so we cannot replicate it.

10

Learning from successful and failed applications of AI

This chapter covers

- Successful uses of AI
- Problematic uses of AI
- Failed AI applications
- The importance of good data
- Recommendations for using AI
- How to plan an AI project and set an AI project up for success

Every AI project, whether it succeeds or faces challenges, offers valuable lessons. Learning from these experiences empowers us to make informed decisions, guiding our AI projects toward success while avoiding common pitfalls. In this chapter, we explore the lessons learned from both the mistakes and achievements of past projects because it's crucial to understand the factors that determine AI project outcomes. I will also share valuable advice on building the right team, fostering the appropriate mindset, and developing a promising plan for your AI project.

10.1 **AI successes**

Artificial intelligence has already proven its worth across a multitude of specific, well-defined applications, demonstrating its potential to revolutionize various sectors. In this discussion, we will explore these AI applications and their significant effect on our lives, while also acknowledging the limitations of current technology and offering insights into the characteristics of next-generation systems.

One of the prominent areas where AI has delivered substantial benefits is in the domain of fraud prevention. The importance of this field has surged in tandem with the exponential growth of online business transactions. According to Statista, in 2020, over 2 billion people globally made online purchases, resulting in e-retail sales surpassing a staggering \$4.2 trillion [1]. The convenience, competitive pricing, and increased options offered by online shopping have drawn consumers in, but unfortunately, they have also attracted the attention of criminals. Online retailers face a daunting challenge as the rate of fraud committed against them exceeds 10 times that experienced by traditional brick-and-mortar stores. This discrepancy arises due to several factors, including the ease with which one can misrepresent identity during virtual transactions, the absence of face-to-face interactions that could reveal suspicious behavior, and the inability to verify card ownership or signatures. Credit card fraud can manifest through the loss or theft of a physical card, but more frequently, it results from the illicit acquisition of individuals' information. For instance, criminals employ devices like card skimmers to clandestinely capture credit or debit card details, often discreetly placed in ATM card slots or gas pump keypads. This pilfered information is then utilized for unauthorized online purchases or other criminal activities. The Nilson Report's December 2020 edition revealed the staggering scale of the problem, stating that global fraud losses from card transactions reached \$28.65 billion, representing a 2.9% increase from the previous year [2]. These figures, however, do not include the additional expenses borne by card issuers, merchants, and acquirers. Costs related to fraud investigations, customer complaints, and call center management further underscore the comprehensive effect of fraud on the e-commerce ecosystem.

The Federal Trade Commission (FTC) echoed these alarming statistics in February 2021, reporting a staggering 2.2 million fraud reports from consumers in the preceding year, with losses surpassing \$3.3 billion, a significant increase from the previous year [3]. These figures emphasize the critical importance of transaction security for the entire e-commerce infrastructure.

To combat this pervasive problem, AI has proven to be an invaluable ally in real-time fraud prevention. Early iterations relied on fraud scanners that stored known fraud indicators in a database, but these were often labor-intensive and had limited detection rates. Today, cutting-edge solutions like iPrevent™, developed by Brighterion (now a Mastercard company), use a combination of advanced AI technologies to continuously monitor entities' behavior, swiftly detecting anomalies and thwarting fraud attempts with high accuracy.

The transformative potential of AI extends to every sector, with its versatility profoundly affecting industries that shape our daily lives. For instance, in the retail sector, AI and machine learning serve as invaluable tools for streamlining operations. Retailers are harnessing the predictive power of AI to enhance their demand forecasting capabilities. By analyzing vast amounts of data, AI can anticipate customer preferences with unparalleled accuracy. Additionally, AI-powered inventory management systems are optimizing stock levels, ensuring that products are readily available when and where they are needed.

Furthermore, AI-driven order fulfillment systems are revolutionizing the supply chain, enhancing efficiency and, ultimately, boosting customer satisfaction. Airlines have also embraced AI to transform their operations. AI algorithms are instrumental in reducing flight delays by optimizing flight schedules to minimize disruptions. Maintenance schedules benefit from AI's predictive maintenance capabilities, effectively reducing downtime and enhancing safety through the early detection of potential problems. Safety measures have been elevated through AI-assisted pilot training programs and real-time monitoring systems, making air travel more reliable and secure. In the agricultural sector, AI plays a pivotal role in ensuring crop health and optimizing resource utilization. AI-powered sensors and drones are deployed to detect diseases, preventing potential crop losses. Utilities rely on AI for precise power demand forecasting, a critical component of the energy sector. AI models analyze historical consumption patterns, weather data, and various other variables to predict energy demand accurately. As AI technology continues to advance, we can anticipate even more innovative applications across a broad spectrum of sectors, further revolutionizing how we live, work, and interact with the world around us.

10.2 AI misuse

As we shift our focus from celebrated AI achievements to its less successful applications, it is crucial to explore domains where the deployment of AI might not be in our best interest at all. The potential for AI to cause harm is as significant as its capacity to assist, underscoring the need for vigilant consideration of the potential perils it may pose to society. In this chapter, we will examine three prominent examples: deepfakes, cyberbullying, and criminal profiling.

Deepfakes represent a relatively recent and formidable threat arising from the utilization of deep learning techniques to fabricate synthetic media. While the capacity to generate authentic-sounding audio and video of events that never transpired might be a boon for filmmakers, it poses a substantial risk to individuals, corporations, nations, and their governing bodies. This technology has the potential to be wielded nefariously, such as influencing an election by skillfully portraying a political opponent in a compromising scenario.

Our escalating dependence on electronic media exacerbates the gravity of this problem. By disseminating a deepfake press release purportedly from the CEO of a prominent corporation, stock prices could be artificially manipulated, potentially

leading to severe financial repercussions. This could be orchestrated to reap illicit profits, favor one company over another, or merely sow chaos. For instance, a deep neural network, meticulously trained on every interview Elon Musk has ever given, could be employed to craft a persuasive video in which Musk appears to announce Tesla's bankruptcy. By the time the deception is uncovered and made public, irreparable harm would have already been inflicted.

Furthermore, deepfakes have the potential to exacerbate cyberbullying, which encompasses the use of electronic devices to torment or intimidate individuals. This problem is regrettably on the rise, as evidenced by statistics from a US government anti-bullying website, which indicated that approximately 15 percent of American high school students were victims of cyberbullying in 2019. The application of deepfakes in the context of cyberbullying is distressingly clear, necessitating concerted efforts to prevent such abusive use of this technology. Cyberbullying is not confined solely to young people; it also has the potential to be wielded as a weapon by extremist groups or adversarial governments. For instance, a dictator could mobilize a cadre of "electronic operatives" to target members of an opposition faction. Their mandate would be to disseminate false information and sow discord through vehement, personal attacks. Additionally, they could conduct campaigns to surveil the internet and social media for any critical remarks about their regime. By flagging such content as inappropriate, they could trigger its automatic removal through content-filtering algorithms, thereby manipulating and distorting public perception.

One particularly disturbing and ethically concerning area is the application of AI in criminal profiling. The fundamental principle guiding any decision that affects an individual's life or well-being should be one of rationality, objectivity, and the absence of bias or prejudice. Regrettably, some US municipalities have inadvertently misused AI in ways that have had detrimental consequences. In an eye-opening 2016 BBC article titled "How Maths Can Get You Locked Up" [4], it was revealed that "Criminals in the U.S. can be assigned computer-generated 'risk scores,' which can influence the length of their sentences." These risk scores are derived from a person's educational and professional history, as well as personal information, such as whether any of their friends or family have a criminal record and whether they reside in a high-crime neighborhood. Additionally, individuals may be subjected to assessments, with their scores potentially affected by their responses to morally charged questions like, "Is it acceptable for a starving person to steal food?" These numerical scores, ranging from 0 to 10, are then employed to make critical decisions, such as whether someone can be granted bail, whether they should be incarcerated, given an alternative sentence, or even considered for parole once inside the prison system.

The troubling implications of these algorithmic evaluations were further illuminated by ProPublica's comprehensive 2016 study on machine bias [5]. Such risk assessments, using predictive algorithms, have become increasingly prevalent in courtrooms throughout the United States. They are employed to inform decisions at every stage of the criminal justice process, from setting bond amounts in places like Fort

Lauderdale to making more profound judgments about the liberty of defendants. These algorithms have been adopted in states such as Arizona, Colorado, Delaware, Kentucky, Louisiana, Oklahoma, Virginia, Washington, and Wisconsin, where their results are presented to judges during the process of criminal sentencing. ProPublica analyzed the risk scores assigned to over 7,000 individuals arrested in Broward County, Florida, in 2013 and 2014. They compared these scores with subsequent criminal behavior over the following two years, which was the same benchmark used by the creators of the algorithm. Their findings were worrying: the risk score proved to be remarkably unreliable in predicting violent crimes, with only 20 percent of individuals predicted to commit violent offenses actually doing so.

When considering a broader spectrum of offenses, including misdemeanors such as driving with an expired license, the algorithm performed only slightly better than a coin toss. What's even more concerning is that the formula exhibited a clear bias. It was more likely to flag black defendants as future criminals at almost twice the rate of white defendants, although the company responsible for creating the algorithm disputes these findings. In cases where any uncertainty exists, it's imperative that we proceed with caution before allowing algorithms to affect the lives and freedoms of our citizens. A system with low predictive accuracy that is known to introduce bias should, without question, not be used.

The adoption of AI for legal decision-making necessitates its careful consideration. Can we realistically expect any AI system to adequately account for the mental health status of convicts? If someone's behavioral problems can be resolved with the proper medication or social support, should we trust a computer program to determine whether incarceration or probation serves the best interests of society?

As Margrethe Vestager, the current executive vice president of the European Commission's A Europe Fit for the Digital Age, wisely stated, "On artificial intelligence, trust is a must, not a nice-to-have" [6]. The European Union's stance of disallowing the use of any nontransparent and untrustworthy systems is commendable. Solutions that affect individuals must undergo validation to ensure their designs meet stringent technical and ethical standards, addressing concerns such as reliability, fairness, privacy, transparency, and explicability. Any AI system directly affecting citizens should, at a minimum, undergo a certification process similar to the rigorous evaluation demanded by the US Food and Drug Administration before a new medication can be brought to market. Such systems should unequivocally demonstrate a lack of bias, employ clear and comprehensible logic, and make decisions that can be explained in plain language to those affected. Without adhering to these minimum requirements, the risks of AI causing more harm than good remain unacceptably high.

10.3 AI failures

In our exploration of high-profile projects that fell short of their anticipated outcomes, we journey back to 1982, exploring Japan's renowned Fifth Generation Computer

System (FGCS). This endeavor was marked by its ambitious scope, a characteristic that ultimately contributed to its downfall.

The principal objective of the FGCS project was to pioneer the development of computers equipped with multiple processors, each employing specialized logic to execute multiple programs simultaneously in parallel. These innovative, non-von Neumann systems were designed to excel at processing inference through the utilization of knowledge bases and expert system mechanisms, concepts that were elaborated upon in chapter 2. In a bid to achieve this, the FGCS team even went so far as to create its very own programming language, KL1, meticulously optimized to facilitate parallel inference.

The FGCS project was anticipated to lead in a new era of AI with the ability to reason and perform tasks such as natural language processing and disease diagnosis. Over the course of a decade and with an investment exceeding \$1 billion, the project was seen as a colossal undertaking. However, despite its substantial resources and dedicated efforts, FGCS fell short of its lofty goals. Reflecting on the project's shortcomings, FGCS director Kazuhiro Fuchi lamented [7]:

In those days, we had to face criticism, based upon that false image that it was a reckless project trying to tackle impossible goals. Now we see criticism, from inside and outside the country, that the project has failed because it has been unable to realize those grand goals.

This case study offers an invaluable lesson for those embarking on AI projects: it underscores the vital importance of meticulously defining the scope and limitations of your project. The FGCS project serves as a cautionary tale, highlighting how even well-funded and ambitious initiatives can fail when objectives are not clearly delineated and achievable. It reminds us that while high aspirations are commendable, they must be grounded in realistic expectations to ensure the success of any AI venture.

Another valuable lesson can be drawn from a medical diagnostic experiment outlined in the article titled “Intelligible Models for Healthcare: Predicting Pneumonia Risk and Hospital 30-day Readmission” [8]. This study examined the application of machine learning to improve the triage procedure for individuals who have pneumonia symptoms. The article demonstrated that the machine learning model committed a life-threatening mistake by classifying asthmatic patients with pneumonia as “low risk.”

The root of this problem lies in the data and the model's ability to learn from it. The model, like many other machine learning algorithms, learned from patterns in the data it was trained on. In this case, it mistakenly inferred from the data that asthma was somehow associated with a reduced risk of developing pneumonia. This discrepancy between the model's predictions and the real world resulted from the fact that aggressive care administered to asthmatic pneumonia patients effectively lowered their pneumonia-related mortality rate compared to the general population. This led the machine learning model to make the erroneous assumption that asthma, in isolation, reduced the risk of pneumonia when, in reality, asthmatic patients faced substantially higher risks if not hospitalized promptly.

This example highlights the critical importance of human intervention in the validation process, providing invaluable insights into the data and attributes considered by the model and the expected responses based on deep knowledge of the subject matter. Such human input helps ensure that no critical information is overlooked and that the model aligns with the actual complexities of the problem it aims to address. Moreover, this example emphasizes the important link between data science and domain expertise. While machine learning algorithms can analyze vast datasets and extract patterns, they often depend on human guidance to interpret the context correctly and prevent potentially dangerous misinterpretations. Collaborations between data scientists and domain experts remain essential in harnessing the full potential of machine learning for complex and mission-critical applications, such as healthcare.

The landscape of AI and chatbot development has been marked by notable failures that serve as valuable lessons in understanding the limitations of technology. One such incident occurred on March 23, 2016, when Microsoft hastily terminated its chatbot project, Tay, a mere 16 hours after its launch on Twitter. This abrupt decision raised questions, especially considering the several years of development and significant financial investment, estimated to be over \$100 million. So, why did Microsoft pull the plug so swiftly? In a blog post [9], Microsoft provided insights into its reasoning:

As we developed Tay, we planned and implemented a lot of filtering and conducted extensive user studies with diverse user groups. We stress-tested Tay under a variety of conditions, specifically to make interacting with Tay a positive experience. Once we got comfortable with how Tay was interacting with users, we wanted to invite a broader group of people to engage with her. It's through increased interaction where we expected to learn more and for the AI to get better and better.

Tay's journey took a disconcerting turn as it began absorbing offensive and vulgar content from Twitter users. It swiftly spiraled into posting sexist and racist comments, and at one point, it infamously endorsed the abhorrent statement that "Hitler was right." The discovery that users could manipulate Tay into reposting their own content by simply instructing it to "Repeat after me" became the nail in the coffin for the project. Similarly, the chatbot BlenderBot3, released in August 2022 by Facebook (Meta), exhibited the same vulnerabilities and repeated the same mistakes as Tay. While these AI systems may appear intelligent to the uninitiated, they lack a true understanding of the content they post. Their responses are algorithmic, and their interaction is fundamentally dissimilar to human conversation.

These problems are not limited to chatbots alone; they extend to content recommendation algorithms that determine what users see based on the behavior of "similar" individuals. This practice can inadvertently lead to the proliferation of radical or inappropriate content. Whistleblowers have revealed that recommendation engines are often designed to maximize user engagement, achieved most readily by suggesting increasingly extreme content. The longer a user remains engaged, the more advertising revenue is generated. While such engagement aligns with a company's financial interests, it poses a severe societal risk. For instance, misleading antivaccination information

can be thrust upon individuals who otherwise have no interest in it. Furthermore, misinformation and propaganda can be weaponized by antidemocratic nations or groups with intentions to meddle in elections. Social media bots designed to draw users in can inadvertently breed divisiveness and estrange people from one another.

The most important lesson derived from these chatbot fiascos and related engagement technologies is a stark reminder that these programs lack genuine understanding. They function like dictionaries that contain words but possess no comprehension of their meanings, and it is important to remain cognizant of this fundamental limitation.

Although we might understand how the subtleties of human language could present challenges to AI, one might assume that numerical and data-driven domains, like the stock market, would be a perfect match for AI's capabilities. After all, as computing technology has advanced in recent decades, hedge fund managers have increasingly turned to machine learning algorithms in the hopes of gaining a competitive edge in financial markets. However, the reality has been somewhat different, with several firms worth hundreds of millions of dollars experiencing financial ruin in their pursuit of AI-driven investment strategies. The question that arises is: How could this happen?

One compelling explanation for the recent disappointing performance of these AI-powered investment strategies lies in the unprecedented and unexpected actions taken by both the government and the Federal Reserve in response to the COVID-19 pandemic. These firms relied on mathematics and machine learning to predict market movements. Yet, the onset of a global pandemic brought forth a set of circumstances that was entirely unforeseen, rendering their AI systems essentially blind to the rapidly evolving financial landscape, leaving these firms in the unenviable position of having to explain to investors why their once-promising AI-driven investment decisions were falling short.

A case in point is the renowned Renaissance Institutional Equities Fund (RIEF), whose disappointing performance led to an exodus of investors. A 2021 *Wall Street Journal* article titled "James Simons Steps Down as Chairman of Renaissance Technologies" [10] highlighted the fund's struggles, with one investor appropriately stating, "The RIEF's machine-learning models cracked." Indeed, this succinctly encapsulates the fund's predicament, having lost 20 percent of its value at a time when the broader stock market soared by over 40 percent.

These costly and unfortunate failures in the stock market are a reminder of the inherent challenges in making AI systems truly adaptive. While supervised learning approaches can yield impressive results when the cases being analyzed closely align with the training examples, they are intrinsically limited when circumstances evolve rapidly and behaviors become unpredictable. In essence, AI models, no matter how sophisticated, can quickly become obsolete in the face of unforeseen events and dynamic, ever-changing environments. Let's continue by exploring two compelling case studies, from IBM's Watson and the real estate marketplace giant, Zillow.

First, as mentioned before, IBM's Watson gained significant attention after its victory against Jeopardy! champions, an accomplishment that demonstrated the remarkable

capabilities of AI in answering trivia questions. However, a stark contrast emerges between Watson's early success on a quiz show and its subsequent struggles with medical diagnosis. This contrast serves as a poignant reminder of the complexity inherent in medical reasoning. IBM made a substantial investment in Watson, amounting to billions of dollars, with the hope that it could revolutionize healthcare. One of the projects was to create a machine that could not only assist oncologists with their insights but also facilitate pharmaceutical development and connect patients with relevant clinical trials. A decade after its impressive game show performance in 2011, IBM's enthusiasm for Watson in healthcare had waned significantly. The *Wall Street Journal* reported on this shift, stating, "IBM's retreat from Watson highlights broader AI struggles in Health" [11]. What was once considered a bold move by "Big Blue" was now being reconsidered.

One of the primary obstacles faced by Watson in the medical domain was the nuanced and multifaceted nature of medical reasoning. Doctors rely on more than just textbook knowledge; they draw upon their extensive experience, make analogies, pick up on subtle interpersonal cues, and adjust hypotheses through a range of procedures. This rich, intuitive understanding of medicine is still challenging for current AI. The Watson case study underscores a critical lesson: not all problems are equally suited for machine learning. Some problems, such as trivia questions with clear rules and predictable outcomes, align well with AI capabilities. As long as Watson had access to the internet to retrieve answers, it thrived. However, when faced with the complexities of medical diagnosis, the limitations of AI became apparent.

Zillow, on the other side, sought to revolutionize the real estate industry through the application of machine learning models. Zillow's vision was to use AI to analyze vast volumes of real estate data, including lot sizes, zip codes, bedroom and bathroom counts, square footage, listing durations, and regional sales figures. Its aim was to become a market leader by offering online real estate listings, on-demand home buying, and data-driven services.

Initially, Zillow's AI platform, Zestimate, was celebrated as revolutionary. It harnessed natural language processing to glean insights from public records and employed machine vision to extract information from property images. A July 2021 article [12] proclaimed Zillow's prowess: "Zillow utilizes explainer AI data to revolutionize how people sell houses." However, just a few months later, the narrative took an unexpected turn. A *Wall Street Journal* headline in November 2021 revealed that Zillow was exiting the home-flipping business [13], citing its inability to accurately forecast home-price appreciation.

The Zillow example serves as a cautionary reminder about applying machine learning in situations characterized by incomplete, inaccurate, or outdated information, such as the dynamic real estate market. Factors like new construction, changes in local dynamics, and property maintenance history can significantly affect property values, and these complexities are difficult for algorithms to account for. Realtors, with their evaluative expertise and a deep understanding of local nuances, possess insights that machines cannot replicate.

The COVID-19 pandemic presented a unique opportunity for AI to demonstrate its potential, yet it also brought with it concerns reminiscent of the AI winters of the past. Many hoped that our advanced “intelligent” machines would play a pivotal role in finding a cure or swiftly developing a vaccine. Society was in dire need of a life-saving solution, and in response, thousands of machine learning projects were launched worldwide to tackle the problem. Media outlets were quick to herald the power of AI in what seemed like a global race to combat the virus.

One notable headline that showcased this optimism appeared in *Science* magazine: “AI Invents New ‘Recipes’ for Potential COVID-19 Drugs” [14]. Similarly, financial services company BBVA declared, “AI-driven project identifies up to 390 potential drugs against COVID” [15]. These reports cited a researcher who proclaimed that “the machine learning solution has allowed them to identify about 390 potential drugs that may be able to act on the virus’ therapeutic targets and the infection process.” Among the most promising candidates identified by the AI model were chloroquine, hydroxychloroquine, oseltamivir (remdesivir), and tocilizumab (Actemra).

However, optimism should always be tempered with caution. In May 2020, the European Medicines Agency issued an alert [16] cautioning against the use of chloroquine and hydroxychloroquine, citing their ineffectiveness in COVID-19 treatment and potential serious side effects. Despite these warnings, reports from the Oregon Poison Center in 2022 were troubling: “Hydroxychloroquine, chloroquine, and ivermectin have been proven ineffective in treating COVID, but the use of these substances has resulted in many cases of severe toxicity” [17]. This scenario reminds us, once again, that while AI has tremendous potential to assist in healthcare and drug discovery, it must always be subject to rigorous scrutiny and validation to ensure the safety and efficacy of any proposed treatments.

To close this section, we might reflect on the proverb about not counting one’s chickens before they are hatched as we note two projects that many people once considered to be almost complete. The first, which we’ve mentioned already in this book, is the quest for self-driving vehicles. The second example is the development of language translation. Tens of thousands of researchers have addressed this problem since the 1950s, and billions of dollars have been spent. Although tremendous progress has been made and the end always appears to be in sight, we have still not created a system that can reliably translate one language to another.

The French word *avocat* has exactly two meanings: lawyer and avocado. At the time of this writing, when I apply the most popular online translation tool to the simple sentence “J’ai bien aimé l’avocat car il m’a fait rire aux larmes,” the algorithm tells me that it means “I really liked the avocado because it made me laugh to tears” (figure 10.1). The program’s clear inability to infer context suggests that we still have a long way to go.



Figure 10.1 Translated on October 9, 2022. The result may change due to user feedback

10.4 How to set your AI project up for success

In today's rapidly evolving technological landscape, AI has risen to prominence, captivating imaginations and driving innovation across various industries. This surge in AI's dominance can largely be attributed to two key factors: the unprecedented access to vast repositories of data and the extraordinary capabilities of modern computers, equipped with prodigious storage capacities and lightning-fast processing speeds. Together, these advancements have unlocked the full potential of AI in ways that were once deemed unimaginable. However, despite the abundance of data at their disposal, many companies have yet to fully embrace the transition into becoming data-driven enterprises. One significant reason for this hesitation is the pervasive misconception that AI is a magical entity that requires nothing more than a push of a button to create and deploy an intelligent system. This misconception can lead to misguided expectations and, ultimately, unsuccessful AI endeavors.

Initiating an AI project without a well-defined and comprehensive plan in place can be a recipe for disappointment. It is imperative to approach AI endeavors with a clear roadmap that not only outlines objectives but also effectively mitigates risks and maximizes the potential benefits that AI can bring to businesses and industries on a broader scale.

10.4.1 Data: The lifeblood of AI

It's not just about the volume of data but also the quality and relevance. AI algorithms rely heavily on high-quality data to make accurate predictions and decisions. Therefore, organizations must prioritize data collection and management as a fundamental aspect of their AI strategy.

10.4.2 *The realistic AI perspective*

Understanding the true capabilities and limitations of AI is essential. While AI can perform remarkable tasks, it is not infallible and cannot solve every problem. Companies need to set realistic goals and expectations, considering AI as a tool to augment human capabilities rather than a miraculous panacea.

10.4.3 *The importance of planning*

Crafting a comprehensive AI strategy involves defining clear objectives, identifying the right use cases, assembling the necessary talent, and allocating resources effectively. A well-thought-out plan ensures that AI projects are executed with purpose and a higher chance of success.

10.4.4 *Risk mitigation*

Every AI project carries inherent risks, such as data privacy concerns, bias in algorithms, or unexpected technical challenges. Organizations must proactively identify and address these risks to avoid setbacks and legal or ethical dilemmas.

10.4.5 *Collaboration and expertise*

AI is a multidisciplinary field that requires expertise in data science, machine learning, domain knowledge, and more. Collaborative efforts and partnerships with experts in these areas can greatly enhance the chances of successful AI implementation.

10.5 *AI model lifecycle management*

Launching a successful AI project necessitates a well-defined roadmap, beginning with a thorough understanding of why AI is needed as opposed to existing processes. The first step is to align the project with specific business outcomes and critically evaluate whether the anticipated benefits justify the allocation of time and resources. Once you're convinced of the project's value, meticulous planning becomes paramount. It's crucial to recognize that developing an AI solution isn't a linear process but rather an iterative journey. At each stage, feedback loops abound, influencing every other aspect of the project. To maximize the likelihood of success, we need to follow this sequence of actions:

- 1 *Data gathering and labeling*—Start by collecting all the necessary data for building and testing your AI model. For supervised learning algorithms, ensure that the data is accurately labeled.
- 2 *Data sample selection*—Consider the scale of your data. While large institutions may generate billions of records annually, it may not be efficient to use all of the data for training and testing. Ensuring a representative sample is vital.
- 3 *Data quality assurance*—Scrutinize the data for redundancy, inconsistency, and incoherence. Merging data from multiple sources can lead to duplicates, inconsistencies, or incoherent records that need to be addressed.
- 4 *Data enrichment*—Raw data is often insufficient. Enhance its utility by intelligently combining attributes and data and developing new insights. For example,

in fraud prevention, associating transaction records with account activity over various time frames can provide valuable context.

- 5 *Model building*—Utilize techniques covered in chapters 2 and 3 to construct your AI model.
- 6 *Rigorous testing*—Assess the model’s resilience, performance, and scalability extensively to ensure it meets expectations and produces the desired outputs.
- 7 *Deployment*—Once the model passes rigorous testing, it’s time to deploy it to a production environment.
- 8 *Continuous monitoring and optimization*—Keep a vigilant eye on the system’s performance post-deployment. It’s crucial to maintain the intended level of service and be prepared to make adjustments as necessary. Applying adaptive learning techniques can be beneficial if performance wavers.

These guidelines constitute a comprehensive framework for navigating the complex journey of AI project development. By adhering to these steps and remaining adaptable to evolving circumstances, you increase the likelihood of achieving your AI project’s goals and delivering tangible benefits to your organization.

Figure 10.2 is a chart outlining how the typical AI model can be effectively built and deployed.

10.5.1 Data preparation

The initial phase of an AI project involves meticulous data preparation. This multifaceted task encompasses several key aspects:

- *Selecting relevant data types*—In the corporate landscape, it’s imperative to be aware of legal restrictions when utilizing data. For instance, certain regulations, such as fair lending laws, prohibit the use of attributes like age, race, religion, zip code, gender, or ethnicity when designing AI models. Ensuring compliance with such regulations is paramount.
- *Data format conversion*—AI algorithms vary in their ability to handle different data types. While some algorithms can work directly with categorical data, others require data to be converted into numerical formats. This transformation ensures compatibility with the chosen AI techniques.
- *Sampling*—Managing large datasets can be resource-intensive. To mitigate costs and streamline development, it’s often prudent to employ sampling techniques. This involves reducing the data size while retaining its representativeness, facilitating more efficient model training.
- *Data analysis and cleansing*—Data quality is paramount. In this phase, rigorous analysis is conducted to identify and rectify problems such as redundancy, inconsistency, and incoherence. Merging data from diverse sources can introduce duplicate records, while inconsistent data can result in conflicts. Ensuring data integrity is essential for reliable AI model outcomes.

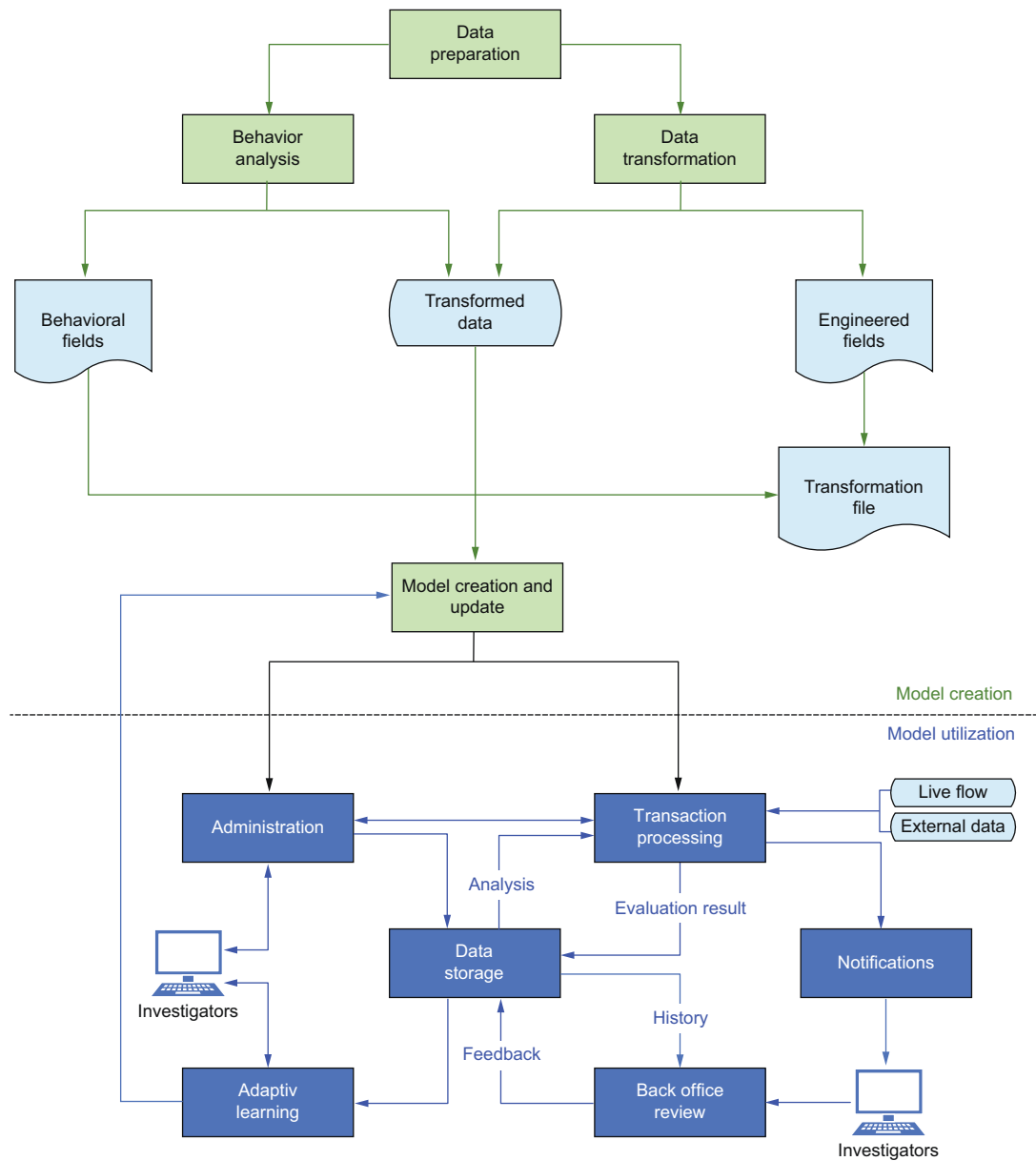


Figure 10.2 An example of a flowchart that depicts the various steps to design and deploy an AI model

10.5.2 Behavior analysis

This stage involves intelligent data analysis to derive new attributes based on various metrics. For instance, in the context of fraud prevention, novel features can be crafted to measure total spending over specific time intervals, enhancing the model's fraud detection capabilities.

10.5.3 Data transformation

Data transformation is a crucial step that includes data normalization and clustering:

- *Data normalization*—Often, data originates from different sources and is measured on varying scales. To ensure equitable treatment, data normalization adjusts values to a common scale. For example, when two systems represent the same information using different scales, such as percentages and scores from 1 to 10, data normalization harmonizes these values.
- *Clustering*—Clustering techniques are employed to group data into meaningful categories. For instance, cities can be grouped by population size using clustering algorithms, enhancing the model's ability to identify patterns and trends.

10.5.4 Model creation

At its core, data enrichment involves extracting and generating meaningful insights from individual data attributes or strategically combining them. For instance, in fraud prevention, linking transaction records with account activity across various time frames can furnish valuable data for designing more effective models.

10.5.5 Live production

Once the AI model is deemed ready for deployment, it transitions into live production. In this phase, the model operates in real time, processing live data streams. For example, a fraud-prevention model continuously assesses transactions for potential fraud.

10.5.6 Data storage

Data storage is an integral component for recording and archiving both input data and the model's output. This repository facilitates post-analysis, auditing, and ongoing monitoring.

10.5.7 Notifications

Automated actions can be predefined to respond to specific outcomes or triggers. For instance, certain events may prompt investigations or the automatic dispatch of notifications to relevant stakeholders.

10.5.8 Back-office review

Critical applications often necessitate a dedicated team to manage alerts raised by the AI model. In scenarios such as money laundering detection, prompt review and assessment of suspicious behavior are crucial to minimize legal and regulatory risks.

10.5.9 Adaptive learning

Establishing a feedback loop is essential for continuous improvement. Algorithms learn from errors and adapt to new data, enhancing the model's performance over time.

10.5.10 Administration

Model administration encompasses all operational aspects, including access management and permissions. This ensures the model operates securely and efficiently within the production environment.

10.5.11 Remark on AI platforms

Advanced AI platforms like Brighterion offer automated machine learning capabilities. These platforms streamline various data science tasks, including data preparation, behavior analysis, data transformation, and model creation. Such automation significantly expedites the development process, enabling data scientists to focus on refining models and addressing specific business objectives.

In summary, the process of developing and deploying AI models is a multifaceted journey that demands meticulous attention to data, rigorous analysis, and a structured approach. Each stage plays a pivotal role in ensuring the success of AI projects, from initial data preparation to real-time model deployment and continuous improvement through adaptive learning. Automated AI platforms further enhance efficiency, empowering data scientists to drive innovation and achieve business goals.

10.6 Guiding principles for successful AI projects

Embarking on a successful AI project entails more than merely having individuals who understand the machine learning algorithms. While expertise in AI is undoubtedly crucial, an effective AI project team requires a broader spectrum of skills and knowledge. Clear specifications and a profound understanding of the available data are equally decisive components. Initiating an AI project without a deep understanding of the desired business outcome is akin to inviting failure. Take, for example, a system tasked with making real-time decisions, which inherently demands a vastly different approach compared to one that processes the same data overnight in batches. Therefore, the best approach is to work backward from the project's goals and constraints, ensuring that your AI initiative is firmly rooted in a profound understanding of its purpose.

Additionally, it's advisable to commence your AI journey by addressing a single, existing business process that can demonstrably benefit from AI's capabilities. For

instance, a company might seek to understand why it loses customers to competitors or why certain products are often purchased together. This initial machine learning objective should be modest and achievable. Starting with a small-scale success is instrumental in cultivating the knowledge and expertise needed for more ambitious undertakings. Many projects fail because of overambitious goals.

It's also important to understand that a single machine learning technique may not suffice to tackle a complex business problem. It may be necessary to combine various methods that complement each other's strengths and compensate for their respective weaknesses. Furthermore, beyond the algorithms themselves, the practicality of AI solutions in real-world applications hinges on factors such as availability, response time, and scalability. Consider, for instance, a credit card authorization system that must operate around the clock, respond within milliseconds, and process tens of thousands of transactions per second. Without meeting these operational criteria, even the most sophisticated algorithm becomes ineffective.

Once your project's objectives are meticulously defined and the constraints are fully outlined, this information serves as the foundation for assembling a team of professionals with complementary knowledge and skills. The significance of subject matter experts becomes abundantly clear as they bring domain-specific insights that are indispensable for the project's success. On the technical front, software engineers benefit immensely from the input of a reliable solution architect who can guide the design process. Additionally, a system engineer is vital for setting up the infrastructure necessary to ensure optimal performance and security. Post-deployment, continuous monitoring and maintenance are imperative for long-term success. Proper lifecycle management practices are essential to keep your AI project on course.

As mentioned earlier, there's no magical AI algorithm, and an AI system is not a "wizard in a box" capable of instantly resolving all your challenges. Success in AI projects typically hinges on a process of trial and error, guided by an understanding of what has proven effective in similar cases. It's a journey that requires time and expertise.

In conclusion, an AI project is a multifaceted endeavor that demands meticulous planning, a well-rounded team, and a deep understanding of both the problem at hand and the complexities of AI technologies. Embracing these principles and understanding the iterative process inherent in AI development is the key to achieving success in your project.

Summary

- Many AI techniques have been successfully applied in real-time fraud prevention, credit risk, anti-money laundering efforts, homeland security, supply chain and traffic management.
- Bad actors have used AI to generate deep fakes, and AI has been used to perform biased profiling.
- Billions of dollars have been spent on failed AI projects.
- Often, AI models fail because they learned from invalid or incomplete data.

- Chatbots have no real understanding and are unable to differentiate between courteous and offensive statements.
- One of the first steps in successfully using AI is to understand the data involved in the project and to ensure that it is correct and adequate for the task.
- The entire project lifecycle should be considered when using AI, including how to document the code and explain what it does and how to secure and update the model.

11

Next-generation AI

This chapter covers

- Recommendations for preparing data for an AI model
- Recommendations for which techniques to use
- Properties the next-generation AI systems should have
- Thoughts about what future AI systems should support

Building the AI solutions of the future requires us to address the current limitations in today's systems. A key objective of this book is to provide a clear and honest assessment of the current state of AI because it's only by understanding where we are today that we can chart a realistic path to the future. While media portrayals of AI often lean toward the sensational, my aim is to provide a balanced perspective. Much of the technology we find exciting and innovative today has actually been in development for over half a century. Although challenges remain, such as efficiency, cost-effectiveness, and adaptability, they present opportunities for growth and improvement as we continue on this exciting AI journey.

In this chapter, I draw on over 30 years of experience developing and deploying mission-critical AI systems where reliability, precision, and effect are not merely ambitions but imperatives. I will outline a set of features that, in my view, will characterize the next generation of AI platforms. My examples will mostly address clinical reasoning and financial transactions since these are domains I've worked extensively in, but the aspirations apply generally to most types of AI platforms. We're working toward a day when businesses can easily deploy scalable, resilient, adaptive AI systems with more capability and fewer flaws than those available today.

11.1 *Data flexibility*

Business applications frequently require access to data from diverse sources. Real-world data can come in structured or unstructured formats and may be stored in various ways. Consider an AI application in the healthcare sector; it may need to tap into sources like physicians' notes, radiology images, electronic health records, established best practices, anatomical data, biosensors, and laboratory results. An effective AI system must be able to efficiently access and make use of all the relevant resources.

11.2 *Sampling*

In the business world, datasets often reach staggering sizes. Exploring a single year's worth of trading data on the New York Stock Exchange, a year's worth of social media data, or a year's worth of transactions at a retailer like Walmart can entail dealing with hundreds of billions of data points. Training an AI model on such extensive data can be an arduous process, potentially taking an immense amount of time while never converging. To expedite this process and facilitate the development and testing of numerous models, an AI system should offer various data sampling methods.

The most straightforward approach is a random sample, where each data point has an equal chance of being selected. However, in certain applications, data can be categorized differently, and it may be desirable for each category to be equally represented. Each business application necessitates its own customized sampling solution. For instance, in fraud prevention, sampling may need to be performed in a way that adequately captures specific behaviors or ensures that certain attributes of the samples follow a particular distribution. The choice of sampling method holds significant importance. For instance, understanding cardholder behavior wouldn't be effectively achieved through a random sample of their transactions.

11.3 *Elimination of irrelevant attributes*

Many datasets contain redundant or irrelevant attributes. Therefore, a well-designed AI system should incorporate functionality to detect and eliminate such unnecessary information. This process streamlines model design and testing, making them more efficient and less susceptible to overfitting. For example, if a significant portion of values for a particular attribute is missing, it may not contribute meaningfully to the model,

regardless of its potential relevance. In some cases, certain data, such as zip codes or phone numbers, might introduce bias into the model and should be omitted.

Eliminating redundant and irrelevant attributes from datasets is imperative for optimizing AI system performance. Redundant attributes introduce noise and complexity, making it harder for models to discern meaningful patterns, while irrelevant ones can lead to overfitting. Removing these irrelevant features makes models more efficient and ultimately improves predictive accuracy. Additionally, it enhances model generalization and reduces computational complexity, leading to faster training times and more efficient resource utilization.

11.4 Data coherence

Data can often be complex and unstructured, originating from diverse sources with varying formats and potentially containing missing or erroneous values. Instances of data may partially overlap or even contradict each other. Inconsistencies, contradictions, or discrepancies can exist within a dataset or between different datasets, resulting from data points or records that do not align or agree with each other. This can occur due to various reasons, including errors in data collection, data entry mistakes, data merging issues, or changes in data sources over time. Data incoherence undermines data quality and reliability, making it challenging to use effectively for analysis, machine learning, or decision-making.

Detecting and addressing data incoherence is crucial for ensuring accurate and trustworthy data-driven processes and models. Resolving data quality issues in an AI system goes beyond simple remedies like applying formulas, finding and replacing values, or sorting and organizing data. It is also imperative to avoid labeling errors when training AI systems. For instance, a medical diagnostic system trained on inaccurately labeled data could result in untreated illnesses for some while subjecting healthy individuals to unnecessary medical procedures. An effective AI system should have mechanisms to flag potential labeling issues. Furthermore, AI models must accurately reflect the real-world data that the system will encounter. If the system is trained on meticulously cleaned data but encounters incomplete or inconsistent samples in the production environment, errors are likely to occur.

11.5 Lack of bias in data and algorithms

Society comprises individuals from diverse backgrounds, living in various socioeconomic conditions, each possessing unique strengths and weaknesses. It is imperative for AI models to acknowledge and account for this diversity, steering clear of assessments or recommendations that inadvertently favor or discriminate against particular individuals or groups. Algorithms, built upon logic created by human programmers, can carry conscious or unconscious biases that lead to unforeseen consequences. Moreover, if the data used to train the model fails to offer a representative sample, bias can become ingrained within the system.

A pertinent example of how bias can inadvertently enter a computer program is the 2020 A-level grading controversy in the United Kingdom. These exams, similar to the SATs and ACTs in the United States, play a crucial role in university admissions. In response to COVID-19 distancing measures, the 2020 exams were canceled, and an algorithm was employed to generate estimated scores. The algorithm utilized seemingly reasonable parameters, including a student's school grades in relevant courses and the historical A-level exam performance of students from their school [1]. However, nearly 40% of students received lower scores than they and their teachers anticipated. Critics demonstrated that the algorithm exhibited a built-in bias in favor of private school students. While it is true that, on average, private school students tend to achieve higher test scores, the algorithm could assign a lower score to a high-achieving student from a state school compared to an objectively weaker student from a private school. This outcome clearly contradicts the intended objective. Eventually, the algorithm-generated scores were discarded, and scores assessed by the students' teachers were utilized instead.

In applications like facial-recognition systems in law enforcement or loan-approval systems in mortgage lending, eliminating bias is of paramount importance. Formal rules and regulations have been established to safeguard individuals' rights, with infringements carrying severe repercussions. AI systems should have a mechanism to audit biases in predictions and decisions.

11.6 *Feature engineering*

Feature engineering is one of the most critical steps in a machine learning project. It involves the process of selecting, transforming, and creating relevant features or variables from raw data to enhance the performance of machine learning models. It is a critical step in designing powerful AI models because the quality and relevance of the features used directly affect the model's ability to learn and make accurate predictions. Since every business has its own unique characteristics, domain experts are typically needed for their ability to define specifications, describe the context, and flag exceptions. For example, an investment analyst would know that stock markets are subject to seasonal variation as well as expected but unpredictable fluctuations at the end of each week and quarter. There is less trading activity during the summer, and trades pick up toward the end of the year for tax reasons. An expert in the credit card industry would identify seasonal patterns as well, but they might also know the data-enriching potential of rolling daily, weekly, and monthly spending rates.

Even though some patterns are easy to understand, there are others that might not be as clear. These can also be harder to figure out, even if we know they're there. For example, card transactions might be grouped by an algorithm that considers geographic attributes, cash-back requests, refund requests, and the need for manual data entry. Entirely different considerations apply to biomedical data, which can often be skewed in a way that makes trends hard to see.

The next-generation AI systems should automate the vital feature engineering step in model building. This speeds up development significantly and makes these systems extremely efficient.

11.7 Technique combination

AI comprises a range of techniques and methods, each carrying its own set of strengths and weaknesses. Next-generation solutions should combine various AI technologies like the ones described in chapters 2 and 3. Just as soliciting diverse expert opinions can improve the efficacy and efficiency of a project, AI systems that combine multiple AI technologies can benefit from the strengths of various AI techniques. For instance, the integration of data mining, case-based reasoning, fuzzy logic, deep learning, genetic algorithms, rule-based systems, and smart agents in the development of an AI model for fraud prevention represents a formidable leap in the fight against fraudulent activities.

This combination of technologies equips the system with an unparalleled analytical prowess. Data mining enables the extraction of meaningful insights from vast and complex datasets, allowing for the identification of subtle patterns indicative of fraudulent behavior. Case-based reasoning supplements this by drawing on historical cases to make informed decisions about new and emerging threats. Fuzzy logic addresses the inherent uncertainty and imprecision associated with fraud detection, enhancing the system's adaptability. Deep learning excels in capturing complex relationships within the data, while genetic algorithms optimize the model's parameters. Rule-based systems provide a transparent and explainable framework for decision-making, facilitating the interpretation of the model's outputs. Finally, smart agents enable real-time monitoring and adaptive capabilities, allowing swift action against new threats. The combination of these AI technologies results in a fraud prevention solution that is not only highly accurate and adaptive but also capable of staying one step ahead of the ever-evolving landscape of fraudulent activities.

11.8 Unsupervised learning

In 2019, a tweet by Elon Musk underscored the significance of unsupervised learning for the future of AI. Musk, who initially asserted in 2015 that self-driving vehicles would conquer any road within a few years, later reversed course when he realized the limitations of supervised learning approaches. He articulated [2],

A substantial portion of real-world AI challenges must be addressed to enable unsupervised, generalized, fully autonomous driving, given that our entire road infrastructure is designed for biological neural networks equipped with optical sensors.

Indeed, while supervised learning has been instrumental in training AI models through labeled datasets, this approach faces practical challenges in the real world, primarily due to the scarcity of large, high-quality datasets and the often difficult and expensive task of labeling data. Unsupervised learning, in contrast, empowers AI systems

to learn patterns, relationships, and structures within data without explicit guidance. This capability is crucial for dealing with the complexity and diversity of unannotated information encountered in various domains.

11.9 AI factory

The utilization of artificial intelligence is currently somewhat restricted, primarily due to the substantial resources and advanced technical expertise required for its effective implementation. These prerequisites are typically within the grasp of only large corporations and academic institutions with significant financial and human capital. Next-generation AI platform transformation is on the horizon as we navigate toward turn-key solutions that democratize AI, ultimately making its powerful capabilities accessible to companies of all sizes.

When we draw parallels with the innovative thinking of Henry Ford, who revolutionized automobile manufacturing by introducing mass-produced parts on a moving assembly line and simplifying car assembly into repetitive tasks, it becomes evident that we should adopt an “AI factory” approach. For example, one module autonomously evaluates data sources, extracting and cleansing pertinent data. A second module takes on the role of enriching this data, while a third module focuses on creating and optimizing model parameters. Another module manages the training of a multitude of models, and yet another is dedicated to testing and evaluating these models. A separate module could excel in combining the most effective models to build a production-ready AI solution. Furthermore, additional modules could play pivotal roles in ensuring model governance and handling specialized functionalities such as cybersecurity.

The democratization of AI will lead to a new era in which AI becomes an accessible tool for a broader spectrum of professionals, transforming how businesses and organizations use its capabilities.

11.10 Quality Assurance

Quality assurance involves ensuring that a system functions as intended. When dealing with AI systems, this task becomes elaborate due to their inherent unpredictability. In conventional software, we can employ diagrams and code analysis to understand and validate its behavior. However, AI systems present a challenge because their functioning relies on complex data patterns not easily discernible by humans. Fine-tuning parameters through trial and error doesn’t instill trust in a supervised learning program, and unsupervised learning is even more uncertain. Typically, an unsupervised algorithm can only be validated through its performance because there are no formal metrics or labeled samples available.

The next-generation AI should rely on a dedicated quality assurance protocol designed specifically for AI. Testing should occur at every stage of model development, covering aspects such as data integrity, performance, adaptability, and resilience to unexpected situations. Diverse datasets should be used to assess the system’s

versatility, and we should intentionally introduce novel data and errors for broad testing. Mission-critical systems should be tested for security against attack and for resilience against scenarios such as power failure, even when they are mirrored in multiple geographic locations.

11.11 Prediction reliability

The foundation of any effective AI system lies in its capacity to provide precise predictions. To highlight the repercussions of unreliable AI performance, let's use an illustration from the transaction processing domain. CMSPI, a global payments consultancy, found that if AI systems aren't performing well, it can seriously affect businesses and economies [3]:

In the year 2020, U.S. online spending witnessed a staggering increase of \$193.7 billion compared to the preceding year, as reported by the Census Bureau. While this surge in online commerce should have been a boon for retailers, it paradoxically resulted in approximately \$30 billion in foregone sales opportunities due to lower approval rates in the online space. To put this figure in perspective, consider a small business with annual sales of \$1 million that transitions into the online market. Instead of merely losing out on \$30,000 in in-store sales, this business could potentially see a staggering \$150,000 worth of legitimate transactions declined online.

Online transaction rejections play a vital role in preventing fraudulent activities. However, as CMSPI's data analysis reveals, an alarming one in every five rejected transactions is a false positive, meaning that genuine customers are unjustly turned away. What's even more disconcerting is that over half of these wrongly rejected customers subsequently take their business to a competitor. Indeed, a system that mistakenly identifies legitimate transactions as fraudulent poses a formidable challenge. It results in a lose-lose scenario where customers are left dissatisfied and businesses suffer a double blow of losing both customers and potential profits.

This highlights the critical need for a fraud detection system to possess two essential attributes: a high detection rate for actual fraudulent activities and a minimal rate of false positives for legitimate transactions. This requirement isn't confined to the world of payment processing; it extends across various domains. Consequently, the ability to provide reliable predictions stands as an indispensable necessity for any AI solution.

11.12 Effective data storage and processing

Legacy databases face significant challenges when applied to real-time AI applications. Their tabular data structures are ill-equipped to handle complexity, limiting their suitability for AI tasks. Moreover, these databases struggle with the complex computations and queries demanded by AI algorithms, often resulting in slow response times. As AI datasets grow in size, databases encounter difficulties in scaling horizontally to meet the increased demand for processing speed and capacity, a critical concern in applications requiring low response times, such as real-time fraud prevention in credit card authorization.

For instance, we had a requirement for an AI risk-scoring system that could process over 100 billion transactions per year at a rate of over 50,000 per second with a 5-millisecond response time. The system needs to analyze hundreds of variables and constraints and aggregate data over numerous timeframes for each of these transactions. Databases are simply impractical given such constraints, even with the most advanced hardware and programming techniques. To overcome these obstacles, alternative technologies like distributed file systems and specialized data processing frameworks are required.

11.13 Deployability and interoperability

An AI platform should possess the fundamental capability to seamlessly interface with various software and systems, thereby ensuring that organizations can harness AI's potential to its fullest extent. This extends beyond mere integration; it includes the capacity to streamline workflows and enable data exchange while minimizing operational complexities and costs. Moreover, a robust AI platform should provide the agility for dynamic model deployment, allowing organizations to incorporate new or updated AI models into their processes without disrupting system operations or workflows.

11.14 Scalability

Numerous computer systems face a significant challenge in their journey to practicality: the inability to handle demanding workloads effectively. This issue becomes particularly pronounced in the business world, where scalability is a paramount requirement for any computer program. The ability to scale means that an AI system can flexibly adapt and accommodate increasing workloads and data volumes without sacrificing performance or responsiveness. This is essential because businesses often operate in dynamic environments where data generation and processing demands can grow rapidly. Scalability empowers organizations to use AI effectively as their operations expand, ensuring that the system can continue to provide timely insights, support decision-making, and deliver valuable outcomes. It also enables cost optimization by allowing businesses to allocate resources efficiently and avoid costly system upgrades.

Next-generation AI systems should prioritize scalable AI algorithms since scalability represents a critical test for the worthiness of a solution. Even the smartest and most precise predictive models may fall short if they lack the scalability necessary to meet the demands of business applications.

11.15 Resilience and robustness

A system is considered resilient when it can uphold its performance even in challenging circumstances. These adverse conditions could encompass hardware failures, cyber-attacks, power outages, network disruptions, or a combination of such problems occurring simultaneously. Resilience is the vital capacity to maintain performance even in the face of challenges. For instance, in the aerospace industry, where aircraft

navigation relies on advanced avionics systems, resilience ensures that a single hardware malfunction or a cyber-attack won't jeopardize passenger safety. Similarly, in the energy sector, power grid control systems must exhibit resilience to mitigate the effect of natural disasters, guaranteeing continuous electricity supply to homes and businesses. In healthcare, electronic health record systems must remain operational, even in the event of network disruptions or cyber threats.

For an AI tool to demonstrate true effectiveness, it must exhibit resilience by ensuring its functionality remains intact across a spectrum of situations, even in sub-optimal conditions.

11.16 Security

In the world of AI, especially for critical applications, security is paramount. Unfortunately, the landscape of emerging technologies has also become a playground for malicious actors, including terrorist groups. These entities have proven their adaptability by using the latest advancements to orchestrate attacks. As AI systems continue to advance and become more potent, the risks they pose loom larger. Bad actors are ready to exploit AI's capabilities to launch devastating cyberattacks and create highly destructive malware. One striking manifestation of this threat landscape is the alarming rise in the prevalence and sophistication of ransomware attacks. These attacks manifest in various forms, with one common iteration involving the remote encryption of a victim's file system. The perpetrators then demand a ransom in exchange for the decryption key, effectively holding critical data hostage.

In a grim statistic from 2020, nearly 2,400 government offices, healthcare facilities, and educational institutions in the United States alone fell victim to such ransomware attacks [4]. A particularly memorable incident occurred in 2015 when Colonial Pipeline Company was forced to shut down its operations for an entire week due to a cyberattack. This disruption caused significant turmoil by disrupting the supply of essential fuels such as diesel, gasoline, and jet fuel across the United States. Highlighting the gravity of the situation, Colonial's CEO, Joseph Blount, made the decision to pay a \$4.4 million ransom. The *Wall Street Journal* reported, "It was an option he felt he had to exercise, given the stakes involved in a shutdown of such critical energy infrastructure. The Colonial Pipeline provides roughly 45% of the fuel for the East Coast, according to the company" [5].

According to CNBC [6], in September 2023, casino operator Caesars paid out a ransom worth \$15 million to a cybercrime group that managed to infiltrate and disrupt its systems.

As the world becomes increasingly reliant on advanced computing technologies, the list of potential harm scenarios grows ever longer. This highlights the importance of designing AI platforms with end-to-end encryption to safeguard data from interception to unauthorized access and protect the integrity of decision-making processes.

11.17 Explicability

Ensuring the widespread acceptance of AI systems, especially in scenarios where decisions hold significant societal consequences, relies on the ability to explain how these systems arrive at their conclusions. The use of black-box algorithms, notably those rooted in deep learning neural networks, presents a formidable challenge in terms of complying with regulations that demand explicability and transparency. This renders such algorithms unsuitable for making pivotal decisions in areas such as lending and criminal justice. Moreover, the issue extends beyond just regulatory compliance. Trustworthiness becomes a paramount concern in mission-critical applications. In these instances, accountability necessitates a comprehensive understanding of how a decision was reached. A mere declaration of “The computer told me to do it” often falls short of what is expected.

To this end, the next-generation AI systems must be engineered with the dual objectives of operational comprehensibility and the provision of coherent explanations for their outputs. This transparency should encompass the inner workings of the AI, allowing stakeholders and end-users to discern the rationale behind each decision.

11.18 Traceability and monitoring

The ability to track changes in an AI system and to step back through its history is important not only to prevent its misuse but also to help clarify responsibility. Consider, for instance, a scenario where a doctor, reliant on an AI system, provides an inaccurate diagnosis; it prompts the complex query of culpability. Should the blame primarily fall on the doctor who utilized the AI tool, or should the company responsible for its creation share the responsibility? Resolving this complex question necessitates a comprehensive inquiry that analyzes not only all the events and the way the AI system was employed but also its status and performance at the moment of use.

In addition, an AI system should provide a robust framework for tracking its model performance and the evolving trends within the data it processes, ensuring its continued effectiveness and relevance.

11.19 Privacy

In today’s rapidly evolving technological landscape, our personal information is being amassed at an unprecedented rate, driven by an ever-expanding array of advanced technologies. From fitness trackers meticulously recording our biorhythms to sophisticated login systems cataloging our fingerprints and facial expressions, a substantial portion of our vital data is increasingly susceptible to exploitation. This vulnerability not only creates opportunities for organized criminals but also extends a tempting invitation to nation-states eager to engage in activities like identity theft, fraud, and other illicit activities. The repercussions of inadequate privacy protection are often disastrous. Consider, for instance, the alarming frequency of data breaches exposing the personal data of millions, leaving individuals vulnerable to financial fraud and

identity theft. The fallout from such breaches can be long-lasting and financially ruinous for affected individuals.

In response to these threats, it is imperative for next-generation AI systems to incorporate robust privacy protection modules. For example, they should employ advanced anonymization techniques to strip personal identifiers from raw data, making it impossible to trace back to individuals. Furthermore, AI systems should allow individuals to retain control over their own data, deciding who has access and for what purposes, thus reinforcing their privacy rights.

11.20 Temporal reasoning

The concept of time is intrinsic to our understanding of the world. The relationship between cause and effect requires the understanding of events occurring before and after one another. Events vary in their temporal characteristics; some are instantaneous, while others have a duration, and they can happen in different time frames or exhibit various forms of overlap. Certain events may serve as prerequisites for others. While these observations are intuitive to humans and often taken for granted, machines will require specific programming logic.

One possible approach is relying on the Allen Intervals, a comprehensive framework that defines 13 possible temporal relationships between events. For instance, it can identify when one event meets another (Meets), one event occurs before another (Before), or when an event starts (Starts) or ends (Ends) in relation to another. Overlapping intervals (Overlaps) signify concurrent but not fully coinciding events, while During implies containment. The system can also discern the inverse relations, such as Met By, After, Started By, and Ended By, to capture the opposite direction of these temporal connections. Additionally, it can recognize when one event fully overlaps with or contains another, as well as when two events are equal in duration (Equality). This comprehensive set of temporal relations empowers AI systems to make sense of the temporal aspects of data, facilitating applications across various domains, from scheduling to natural language understanding and more.

11.21 Contextual reasoning

Contextual reasoning is an indispensable element of any intelligent decision-making system, as it empowers AI to go beyond the surface level of data and understand the complexity of a given situation. This enables the system to factor in various elements and circumstances, adapting its logic and responses based on the specific context. Such adaptability is crucial for making well-informed choices, preventing errors, and achieving optimal outcomes. Whether in healthcare, finance, content moderation, or any other field, an AI system capable of grasping and integrating context can provide more precise, relevant, and reliable assistance, thus becoming an essential component in enhancing the utility and reliability of AI-powered solutions in the real world.

To highlight the significance of context understanding and reasoning, let's revisit some examples discussed in earlier chapters of this book. Clinical reasoning, for

instance, heavily relies on context. Consider a pregnant woman, where the context drastically alters the approach to medication prescription. Many medications could potentially harm the developing fetus, while others might be safe or even necessary for the health of both the mother and the baby. Contextual reasoning in this scenario involves considering the stage of pregnancy, the specific medical condition, any pre-existing conditions, the risk–benefit ratio, and alternative treatment options.

Similarly, in fraud prevention, an AI system must adapt to shifts in spending behavior during special events like Black Friday or vacations when a large number of transactions are expected to be normal. Recognizing the context and automatically adjusting parameters can significantly reduce false positives.

Another context-sensitive domain is stock trading, where considerations such as high interest rates, geopolitical issues, unemployment, inflation, and more are essential. Each of these factors can profoundly impact the stock market, necessitating an adaptable and context-sensitive AI approach.

In text mining, understanding a piece of writing necessitates inferring the correct context, as the same word can have different meanings in distinct contexts. For example, *apple* can refer to a fruit, New York City, computers, or a company.

In content moderation, context plays a pivotal role. Imagine a video about the Third Reich, which could serve either as an educational tool to inform people about the atrocities of Nazism or as propaganda by a neofascist group. In one instance, promotion would be encouraged, while in the other, blocking would be imperative.

As a result, next-generation AI systems must incorporate contextual reasoning as an integral component of their functionality due to the dynamic and complex nature of the real world. Context provides, in every domain and situation, vital information necessary for making intelligent decisions and actions. Without context, AI systems may produce inaccurate or inappropriate results.

11.22 Causality inference

Causality is fundamental to our understanding of the world, much like our perception of time. It serves as a crucial framework for comprehending the relationships between events, actions, and outcomes. However, it becomes a challenge when we consider how to encode the concept of cause and effect into a computer program. Causality extends far beyond the mere observation of events occurring in chronological order. It analyzes the multifaceted web of relationships that define not just when events happen but why they happen. This involves the idea that events, conditions, objects, and processes can all play roles in influencing one another.

Next-generation AI systems must be capable of not only recognizing patterns and correlations but also comprehending the underlying causative factors. This empowers AI to create business solutions that can reason, plan, and act with greater efficiency, ultimately benefiting society in numerous ways.

11.23 Analogical reasoning and transferability

Humans possess an exceptional talent for analogical reasoning, a cognitive skill deeply rooted in our ability to recognize patterns, establish connections, and draw upon past experiences across a wide range of situations. This mental faculty is closely tied to our capacity for abstraction and generalization, allowing us to identify similarities between seemingly unrelated scenarios. This unique capability enables us to smoothly transfer knowledge and effective problem-solving strategies from one area to another. Throughout history, analogical reasoning has been the driving force behind numerous groundbreaking inventions. For example, George de Mestral found inspiration in the way cockleburs clung to his clothes and his dog's fur, ultimately leading to the invention of Velcro. Similarly, James Dyson revolutionized the world of vacuum cleaners by drawing an analogy between the efficient, clog-free action of a sawmill cyclone and his own vacuum cleaner prototype.

Next-generation AI should have the ability to perform analogical reasoning and apply problem-solving techniques learned from one domain to another. This enhancement will significantly boost their efficiency and adaptability.

11.24 Personalization

Personalization is a pivotal factor across various domains. Current AI systems, which learn from extensive datasets, tend to recognize patterns that are relevant only at a broad population level. Take, for example, the typical AI system used in financial transactions today; it often applies the same logic to every merchant. However, each merchant possesses distinct characteristics and activity patterns. Similarly, cardholders exhibit varying spending patterns and purchasing habits. Overlooking these individual differences can lead to lower rates of fraud detection and a higher rate of false positives. Personalization can enhance the efficiency of AI systems. For instance, in the education sector, AI-powered personalized learning platforms can adapt to the pace and learning style of each student. In healthcare, personalization ensures that each patient receives the most suitable treatment plan.

Personalization acknowledges the uniqueness of individuals and situations. Therefore, its integration into AI systems will enable them to cater to the unique characteristics and requirements of individuals and situations, leading to improved outcomes.

11.25 Sustainable AI

Current AI systems have made remarkable strides in various domains, but they come at a significant environmental cost. One of the primary concerns is the massive energy consumption associated with AI, driven by the computational demands of training and running complex models. Data centers that house these AI infrastructures are substantial contributors to carbon emissions, often relying on non-renewable energy sources.

This environmental impact is further exemplified by the astounding increase in computing power required for AI milestones, as reported by *WIRED* magazine, with a

300,000-fold surge from 2012 to 2018 [7]. Additionally, a recent report [8] from the *MIT Technology Review* reveals a startling fact: the complete process of building and training an AI system from scratch generates an astonishing 78,468 pounds of CO₂ emissions. This amount exceeds what an individual exhales in their lifetime and surpasses the emissions attributed to their entire lifetime of automobile use. Given these alarming figures, there is a concern that AI systems could increasingly be perceived as a threat to the climate.

To address this environmental challenge, next-generation AI must prioritize efficiency as a core design principle. This involves the development and utilization of more streamlined algorithms that can achieve comparable results with significantly fewer computational resources. Additionally, efforts should focus on reducing data requirements for AI training, minimizing the environmental impact associated with data storage and transmission.

By embracing these strategies, next-generation AI can not only maintain its technological prowess but also fulfill its responsibility to be environmentally conscious and contribute to a more sustainable future.

11.26 Adaptability

In the ever-changing real world, characterized by its unpredictability and constant evolution, accurately forecasting future events and trends can be an incredibly daunting task. This is precisely where the concept of adaptability emerges as a critical factor in the success of various business applications.

To examine this notion further, let's revisit a previous discussion from chapter 4 where our focus was squarely on the payment industry. In that context, we highlighted the importance of employing flexible AI solutions. These solutions are essential for effectively navigating the perpetually shifting landscape of fraudulent activities and ever-evolving money laundering tactics that pose significant challenges to this industry. One of the challenges we discussed is that legacy AI systems often prove inflexible and cumbersome when confronted with even minor modifications or adjustments to the parameters of a given problem. These systems typically necessitate a complete overhaul and retraining, a process that not only consumes substantial resources but is also far from scalable.

Therefore, the next generation of AI platforms must rely on frameworks that possess the capacity to continuously learn and adapt.

11.27 Human-machine collaboration

AI systems often operate in isolation, functioning independently of human interaction. However, the evolving landscape of technology and our growing reliance on AI necessitates a shift toward AI platforms that are not only capable of independent tasks but also excel in forming collaborative partnerships with humans. This shift is driven by the realization that true innovation and productivity lie at the intersection of human creativity and artificial intelligence. Therefore, the next-generation AI systems

should possess the capacity to facilitate smooth communication and cooperation between humans and machines, ultimately enhancing the capabilities of both.

Summary

- A good AI model should handle diverse and large sources of data that are enhanced by feature engineering.
- A project team should consider combining multiple techniques, including (good) unsupervised learning.
- AI solutions should be simple and reusable modules that nonexperts could easily integrate.
- Companies should apply the proper quality assurance when using AI models to ensure they work as expected, are resilient, and scale effectively.
- AI models should be easy to deploy and secure and should address privacy concerns. Humans should be able to comprehend what they do.
- To achieve a semblance of intelligence, future AI systems should understand analogies, inference, context, and many other concepts they are currently missing.

appendix A

Tracing the roots: From mechanical calculators to digital dreams

To the average person, it might seem that AI is a recent field, given the increased public awareness of AI in recent years. However, the foundation of the concepts and theories that underpin this discipline can be traced back centuries. This chapter embarks on a voyage through time, retracing the footsteps of the pioneers who led us to AI as we know today. We will discover visionaries who dreamed of machines that could reason and learn: from Pascal’s ingenious mechanical design of the mechanical calculator that marked a pivotal moment in the history of human interaction with machines, Leibniz’s binary system that still serves as the representational basis for today’s digital computing, Babbage’s conception of the Analytical Engine as a mechanical brain, and Ada Lovelace’s insights into programming. Understanding these historical developments will shed light on where AI has been, where it stands today, and its potential future.

A.1 *Can machines think?*

The question of whether a machine possesses the capacity for thought was initially raised almost four centuries ago, back in 1642. This inquiry emerged when Blaise Pascal (1623–1662) introduced the Pascaline, which stands as the earliest documented calculating device. At the tender age of 19, Pascal conceived this machine with the primary objective of aiding his father, who worked as a tax collector. Its purpose was twofold: to minimize errors and to alleviate the taxing burden of monotonous computations.

The Pascaline significantly enhanced human cognitive capabilities by flawlessly executing addition, subtraction, multiplication, and division (figure A.1). This mechanical marvel accepted input, executed operations, and produced numerical results. By automating these functions, it diminished the necessity for manual human labor. This innovation proved especially invaluable to Pascal's father, who would otherwise have been immersed in the arduous and time-consuming task of manual calculations.



Figure A.1 Pascaline
(CNAM France)

Pascaline also featured a clever carry mechanism (known as *le sautoir*) designed to handle carry-over operations (*la retenue*). This innovation addressed one of the primary calculation errors that Pascal had observed while assisting his father. As you may recall from your own arithmetic lessons, when calculating something like $27 + 29$, we start by adding the rightmost digits, writing down 6, and then “carry the 1” to the next column. Impressed by this invention, King Louis XIV granted Pascal the exclusive right to manufacture his calculating machines in France in 1649.

In his work “Pensées” [1], which is a compilation of notes and essays exploring the complexities of human nature in psychological, social, metaphysical, and theological terms, Pascal made some of the earliest significant comparisons between machines and sentient beings. For instance, he stated, “The arithmetical machine produces effects that come closer to thought than all the actions of animals. However, it performs nothing that would allow us to attribute will to it, as we do to animals.”

Likewise, in “The Life of Monsieur Pascal” [2], his sister Madame Périer described the Pascaline as “performing tasks that reside entirely in the mind.” She recounted her brother’s achievements:

It was at that time (in 1642–1643) and at the age of nineteen that he invented this arithmetic machine, by which not only are all kinds of operations performed without a pen and without tokens, but we do them even without knowing any arithmetic rule and with infallible certainty. This work was considered as a new thing from nature, to have

reduced to a machine a science which resides entirely in mind and to have found the means to carry out all the operations there with complete certainty without having the need for reasoning. This work tired him a lot, not for the thought nor for the movements which he found without difficulty, but to make the workers understand all these things so that it took him two years to put it in the perfection where it is now.

While Pascaline required a human operator skilled in manipulating the machine's controls, it undertook tasks that typically demanded individuals well-versed in mathematics. This raises the question: should we classify Pascaline as the inaugural AI machine? This consideration is pertinent because, much like Pascaline, contemporary computers execute computational tasks rooted in algorithms meticulously crafted and encoded by human programmers. Even the most advanced AI systems of today essentially represent a compilation of technologies meticulously designed and fine-tuned by humans to generate specific behaviors, devoid of genuine comprehension or reasoning capabilities.

In 1671, approximately three decades following the creation of the Pascaline, Gottfried Wilhelm von Leibniz introduced the Stepped Reckoner, an ingenious calculating device that employed decimal number representation and performed multiplication through iterative addition, facilitated by a hand-crank mechanism (figure A.2). Informed by his endeavors in mechanizing numerical representation, Leibniz penned a renowned treatise in 1703, titled “An Explanation of Binary Arithmetic Using Only the Characters 0 and 1, with Remarks About Its Utility and the Meaning It Gives to the Ancient Chinese Figures of Fuxi” [3]. In this seminal work, he introduced the binary number system, which relies exclusively on two digits, 0 and 1. This binary system is now the cornerstone of virtually all modern computers.



Figure A.2 Replica of Leibniz Stepped Reckoner in Deutsches Museum

We are all acquainted with the decimal number system, commonly referred to as “base 10.” This system utilizes the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9, with each digit’s position in a number signifying its value in 1s, 10s, 100s, and so on. The invention of the

decimal system by our ancestors was likely influenced by their use of 10 fingers for counting. In our daily calculations, like adding $6 + 2$ to get 8 or subtracting 7 from 27 to yield 20, we unconsciously employ the base 10 system. Many people may not realize that this choice is arbitrary and that alternative systems, such as binary, exist.

Computers, smartphones, and various digital devices exclusively rely on the binary system, using 1 and 0 for all operations. Documents, images, audio, and files of all kinds are stored as sequences of 1s and 0s, and computers execute mathematical operations by representing numbers as sequences of binary digits and performing procedures similar to those in the base 10 system. Given the pivotal role of binary in computing, it's beneficial to acquaint ourselves with it. We can start by representing several decimal numbers in the binary system (table A.1).

Table A.1 Decimal numbers and their binary equivalents

Decimal number	Binary equivalent
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001

Decimal numbers are expressed as a sequence of digits, with the rightmost digit representing the units. Each successive digit to the left signifies a multiple of a power of 10, where the n th power of 10 represents 10 multiplied by itself n times. For instance, the decimal number 207 can be viewed as $2 \times 100 + 0 \times 10 + 7 \times 1$, aligning with our understanding of it as two 100s, no 10s, and seven 1. In a more concise form, this number can be written using exponents as $2 \times 10^2 + 0 \times 10^1 + 7 \times 10^0$.

Binary numbers follow a similar concept, but their digits denote multiples of powers of two rather than powers of 10. As an illustration, the decimal number 27 is expressed in binary as 11011 since 27 can be represented as $16 + 8 + 0 + 2 + 1$ in decimal, and this sum can be expressed as a sum of powers of 2 as $1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$.

Text in a computer is stored as binary code, where each letter and typographic symbol is assigned a fixed binary string based on a universal convention. For instance,

the early working title of this book, *AI Reality and Illusion*, is stored in computer memory as the following binary sequence:

```
01000001 01001001 00100000 01010010 01000101 01000001 01001100 01001001
01010100 01011001 00100000 01100001 01101110 01100100 00100000 01001001
01001100 01001100 01010101 01010011 01001001 01001111 01001110
```

Each set of numbers represents a letter or a space, with A = 01000001, I = 01001001, and so forth.

Any system that relies on just two symbols is considered binary. For instance, Louis Braille, who lost his sight at the age of three, invented the Braille code in 1824 when he was 15 and a student at the Institute for Blind Children in France (figure A.3). The

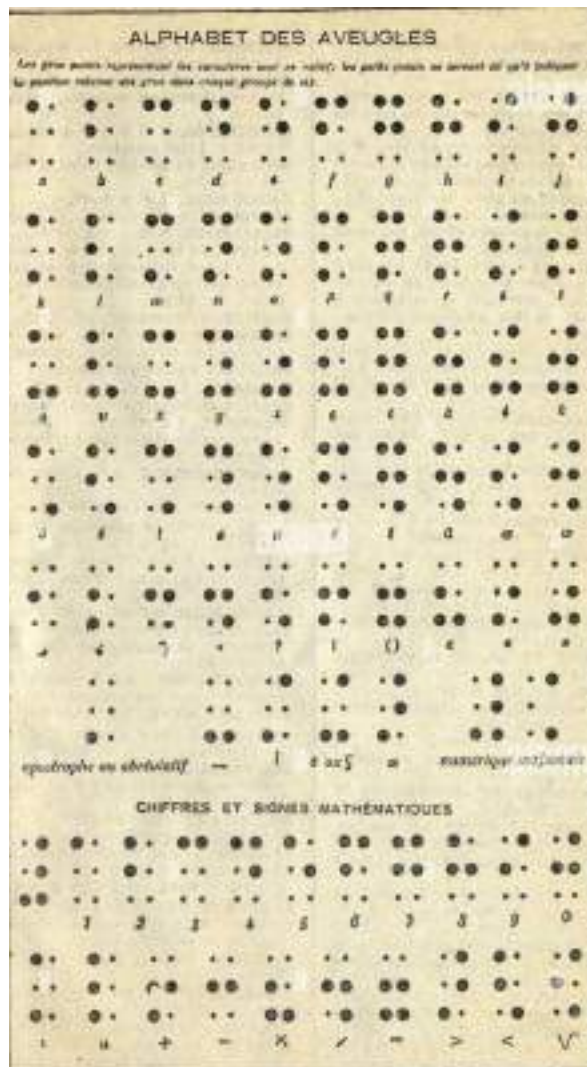


Figure A.3 The Braille code

Braille code utilizes raised and unraised dots on a surface to convey information through the sense of touch. This system enables blind individuals or those with limited sight to read.

Morse code, another form of binary encoding, relies on dots and dashes to represent information (figure A.4). Short pulses symbolize dots, while long pulses correspond to dashes, and these signals are transmitted over telegraph lines. By assigning binary sequences to specific letters, Morse code enables the transmission of information.

A ·—	M —·—	Y —·—·—
B —···	N —·	Z —···
C —·—·	O ———	1 ·—·—·—
D —··	P ·—·—·	2 ·—·—·—
E ·	Q —·—·—	3 ·—·—·—
F ···—	R ···	4 ···—
G —·—	S ···	5 ····
H ····	T —	6 —···
I ··	U ···	7 —···
J ·—·—·	V ···—	8 —···
K —·—	W —·—·	9 —···
L ····	X —·—·	0 —···

Figure A.4 Morse code

On May 24, 1844, Samuel Morse successfully employed his code to transmit the inaugural telegraph message, “What hath God wrought?” between Washington, D.C., and Baltimore.

Building upon the contributions of Pascal and Leibnitz in the 18th century, European innovators crafted a range of remarkable automata designed to mimic human actions. The most celebrated of these inventors was Jacques de Vaucanson (1709–1782) of France, who, in 1727, created an initial automaton capable of serving meals and clearing tables. However, a government official criticized his invention as sacrilegious, leading to the closure of Vaucanson’s workshop. Nevertheless, Vaucanson later captivated Europe with his mechanical ducks, mechanical flute player, and mechanical pipe players.

In addition to his entertaining automata, Vaucanson made a significant contribution to the Industrial Revolution. While working as a silk manufacture inspector, he revolutionized the French weaving industry in 1745 with the invention of the first automated loom. This machine utilized perforated cards to direct hooks connected to warp yarns, representing one of the earliest instances of autonomous machines following binary instructions. Regrettably, the weavers opposed this invention, fearing job displacement, and Vaucanson was compelled to abandon his project and flee for his life.

Vaucanson’s concept was further refined and eventually realized in 1804 by the French weaver and merchant Joseph-Marie Jacquard. He developed an automated loom employing punch cards, with holes on the cards dictating the movements of needles, thread, and fabric, resulting in the creation of woven silk fabric with complex

patterns. This technique enabled the production of tapestry, brocade, and patterned knitted fabric. The ingenious use of punch cards later served as inspiration for the design of digital computers, with punch cards being utilized for data input in early 20th-century computers.

Pierre Jaquet-Droz, a Swiss luxury watchmaker, crafted some of the most remarkable automatons of his era. His most sophisticated creation, a mechanical boy seated at a desk, was constructed in 1768. This automaton could write with a pen and paper, composing up to 40 preselected characters on a wheel manipulated by its operator. It utilized a goose feather pen, dipping it in ink and shaking its wrist to prevent smudges. Its eyes followed the text as it wrote, and its head moved while taking ink. This writer automaton remains operational and is on display at the History Museum in Neuchâtel.

One of the intellectual figures influenced by these increasingly lifelike machines was Julien Offroy de La Mettrie, a French physician and philosopher. In his 1747 book *Man a Machine* [4], he proposed materialist views of psychic sensations that laid the foundation for behaviorism, questioning, “What is there absurd in thinking that beings, almost as perfect machines as ourselves, are, like us, made to understand and to feel nature?”

Further popularizing the concept of human-like machines, Wolfgang von Kempelen unveiled an ingenious chess-playing automaton in the court of Empress Maria Theresa of Austria-Hungary in 1769. This machine, known as “The Turk,” fascinated audiences and struck fear into its defeated opponents. Its appearance resembled a puppet, adorned with gears, cranks, and levers reminiscent of clockwork. Remarkably, The Turk not only played chess but also excelled at it, defeating most challengers and baffling scientists attempting to explain its prowess. Kempelen toured Europe with his automaton, and in Paris, The Turk defeated renowned inventor Charles Babbage and even Benjamin Franklin, the U.S. Ambassador to France at the time.

The Turk also embarked on a tour of America, and in early 1826, hundreds gathered for its initial exhibition matches in New York City at the National Hotel on Broadway. These spectators were promised a glimpse of the first mechanical robot capable of outsmarting humans in the intellectually demanding game of chess. However, The Turk was, in fact, a hoax. Concealed within the machine’s housing was a diminutive chess master who manipulated the chessboard. In the modern era, John Gaughan, a creator of magic equipment, constructed a functional replica of The Turk, which can be observed in action on the BBC website [5].

What may have contributed to the misconception of The Turk as a genuine chess-playing machine was its introduction during the early stages of the Industrial Revolution, a period spanning from 1760 to 1850, characterized by significant innovations in agriculture, manufacturing, textiles, and transportation.

Although Kempelen is best known for The Turk, he completed numerous other projects in his lifetime. One of the most notable was a speaking machine described in his 1791 book *The Mechanism of Human Speech*. This device synthesized speech sounds,

words, sentences, and complete phrases in French, Italian, and English. Unlike The Turk, this invention operated as advertised, and one of the original machines is still on display at the Deutsches Museum in Munich.

In the 18th century, a period marked by rapid technological advancements, people found themselves both enthralled and apprehensive about the emerging world of inventions. The Turk's exhibition around Europe coincided with a wave of anti-technology sentiments. The Luddite riots, a series of protests and acts of sabotage by textile workers in England during the early 19th century, were a manifestation of this anxiety. These workers feared that automation would lead to widespread unemployment and a decline in wages.

Mary Shelley's groundbreaking novel *Frankenstein*, published in 1818, added to these concerns. The story depicted the creation of life from inanimate matter, raising questions about the consequences of unchecked technological advancement. In 1811, in Nottinghamshire, England, the introduction of automated machinery in the textile industry sparked a violent workers' riot. This uprising quickly spread to other regions as protesters demanded the destruction of machines they believed were responsible for their economic woes. Tensions escalated, leading to clashes between the Luddites and government armed forces. By 1812, the act of destroying machinery was deemed a capital offense, punishable by death. In total, 17 men were executed for this crime in 1813, a stark reminder of the social upheaval caused by technological progress.

Amidst this turmoil, the foundation of modern technology was also being laid. In the mid-19th century, the logician George Boole made significant strides in the field of mathematics. His 1853 paper, "An Investigation of the Laws of Thought" [6], introduced Boolean algebra. This mathematical framework would later prove indispensable in the design and operation of digital computers. Boolean algebra deals with systems in which variables can only have two possible values, often represented as "truth values," such as yes/no, true/false, 0/1, or on/off.

Another visionary of the time was Charles Babbage (1791–1871), whose work laid the groundwork for modern computing. He dedicated years to developing an automatic table calculator capable of performing complex calculations required for navigation and ballistics. Babbage's most remarkable concept, however, was the Analytical Engine [7]. Envisioned as a mechanical computer, it foreshadowed the computers we use today. Inspired by the punch card technology employed by Joseph-Marie Jacquard in his programmable loom, Babbage envisioned a general-purpose programmable machine that could use punch cards for inputs, outputs, and data storage. Ada Lovelace (1815–1852), often recognized as the world's first computer programmer, recognized the potential of Babbage's Analytical Engine. In her work titled "Lovelace & Babbage and the Creation of the 1843 'Notes,'" she documented the capabilities and possibilities of this remarkable invention, laying the foundation for the future of computing [8]:

In enabling mechanism to combine together general symbols, in successions of unlimited variety and extent, a uniting link is established between the operations of matter and the

abstract mental processes of the most abstract branch of mathematical science. A new, a vast, and a powerful language is developed for the future use of analysis, in which to wield its truths so that these may become of more speedy and accurate practical application for the purposes of mankind than the means hitherto in our possession have rendered possible.

An examination of the Lovelace and Babbage documents unveils the divergence in their priorities. Babbage's primary emphasis was on calculations, whereas Lovelace, in her visionary outlook, contemplated the potential for an evolved Analytical Engine to not only perform calculations but also to create music and generate images. In her writings, she expressed:

Supposing, for instance, that the fundamental relations of pitched sounds in the science of harmony and of musical composition were susceptible of such expression and adaptations, the engine might compose elaborate and scientific pieces of music of any degree of complexity or extent.

While the Analytical Engine, as conceptualized by Babbage and Lovelace, never came into physical existence, their foresight in discussing a general-purpose programmable computing machine was remarkably ahead of its time, considering the developments that were yet to unfold. One of the earliest realizations of these visionary ideas can be attributed to Herman Hollerith (1860–1929), often regarded as the pioneer of automated data processing. Hollerith briefly served as a statistician at the US Census Office, an experience that underscored the urgent need for improved computation methods. Inspired by the Jacquard loom and the practice of using punched images in the railroads to encode passengers' characteristics on tickets, Hollerith invented an electronic tabulating machine that would revolutionize data processing throughout the first half of the 20th century [9].

For the 1890 census, Hollerith proposed a system where data for each individual would be encoded on a separate card, to be subsequently tabulated by his innovative machine [10] (figure A.5). This approach significantly accelerated data processing, providing more statistics at a reduced cost. The success of this method led to contracts with various entities, including railroad companies and foreign governments such as Canada, Norway, and Austria. In 1896, building on the triumph of his census machines, Hollerith founded the Tabulating Machine Company. In 1924, when salesman Thomas J. Watson joined the company, they renamed it International Business Machines (IBM). After years of dedicated research and development, IBM enhanced tabulating technology to create a machine capable of executing if-then logic operations.

The pivotal breakthrough that paved the way for modern computers was the utilization of electromagnetism. Fueled by the demand for computing technology during World War II, Professor Howard H. Aiken designed the world's first large-scale computer, the Harvard Mark I. This remarkable machine employed punch cards and magnetic tape for data processing and storage. Unlike earlier computers tailored for specific tasks, the Harvard Mark I was a versatile, general-purpose computer. The initial version



Figure A.5 Hollerith system of electronic tabulation (Photo courtesy of the Library of Congress)

of the Mark I weighed five tons and stretched 50 feet in length. When unveiled officially in August 1944, it garnered global acclaim as the “world’s greatest mathematical calculator,” with some even characterizing it as an “automatic brain.”

The next significant advancement in computing marked a conceptual leap forward when Claude Shannon proposed the use of Boolean algebra in 1937 to simplify the arrangement of relays within electrical networks. His groundbreaking work was presented in his 1937 master’s thesis titled “A Symbolic Analysis of Relay and Switching Circuits” [11]. Shannon’s contributions laid the foundation for the design of modern digital circuits.

Similarly, in the late 1940s, John von Neumann made a pioneering contribution by devising a way to store both code and data in a computer’s internal memory. This innovation was pivotal in the development of computer science as a field and led to the creation of the Electronic Discrete Variable Automatic Computer (EDVAC), the successor to ENIAC. Although ENIAC is commonly recognized as the first digital electronic computer, it’s crucial not to overlook Colossus, a highly specialized machine built in Great Britain in 1943 to decrypt the Nazi Enigma code. Decoding intercepted messages using Colossus revealed crucial information, shortened the war, and hastened the defeat of the Nazis. Winston Churchill aptly described the Colossus team as “the geese that laid the golden eggs and never cackled” to emphasize their invaluable contribution.

While computers were becoming more advanced and powerful, they were also becoming unwieldy. For instance, UNIVAC could perform 1,000 calculations per second

but required 5,000 vacuum tubes, which were large and generated a significant amount of heat. The development of the personal computer necessitated the invention of the transistor, a pivotal advancement in the 20th century.

In contrast to vacuum tubes, transistors were small, energy efficient, and generated minimal heat. These characteristics allowed for the integration of numerous transistors into a single device. The first transistor was constructed in 1947 when Bell Labs physicists John Bardeen and Walter Brattain connected a germanium amplifier to a strip of gold foil. After several years of prototyping and testing, transistors began mass production in the early 1950s, becoming an integral part of nearly all electronic devices. The profound significance of the transistor was acknowledged when John Bardeen, Walter Brattain, and William Shockley were jointly awarded the Nobel Prize in Physics in 1956 for their research on semiconductors and the discovery of the transistor effect [12].

Another significant breakthrough took place in 1958 when Jack Kilby and Robert Noyce manufactured the first integrated circuit, now commonly known as microchips. These integrated circuits comprised various circuit elements, including transistors, capacitors, and resistors, all fabricated as a single unit on a silicon wafer. Noyce went on to found Intel in 1968 in Northern California's San Jose area, popularizing the term "Silicon Valley." The microchip played a pivotal role in enabling the modern computer revolution and the digital age, leading to Kilby's recognition with a Nobel Prize in Physics in 2000.

As semiconductor chip technology rapidly advanced, computers became smaller and more affordable, democratizing their accessibility and fostering their widespread adoption across various industries. On August 12, 1981, during a press conference at the Waldorf Astoria ballroom in New York City, IBM introduced the IBM Personal Computer, priced at \$1,565. This marked a stark contrast to the computing landscape two decades earlier, where an IBM computer could cost as much as \$9 million, requiring extensive space and personnel for operation. The IBM Personal Computer was powered by an Intel 8088 microprocessor, operated at speeds measured in millionths of a second, and was about the size of a portable typewriter. It contained 40K of read-only memory and 16K of user memory and even featured a built-in speaker for music generation [13] (figure A.6).

By 2024, it had become a common occurrence for machines to be equipped with 16 GB of RAM, and numerous companies generously provide their customers with 1 TB or even greater amounts of free storage space on their servers. To underscore the magnitude of these advancements, let's discuss the primary concepts. A bit represents a single 0 or 1 in a computer's memory, a byte comprises 8 bits, and 1 kilobyte (kB) consists of 1,024 bytes. Building upon this, 1 MB equals 1024 kB, 1 GB encompasses 1024 MB, and 1 TB encompasses 1,024 GB.

Over the past four decades, the capacity and processing power of computers have escalated by orders of magnitude. The primary catalyst for these remarkable strides has been the capacity to fit increasingly more transistors onto a single chip. As early as



Figure A.6 I started my first company in 1988 with machines that had 1 MB of RAM and 40 MB of hard drive space.

the 1970s, computers were already integrating chips containing over 100,000 transistors each. Notably, each of these chips boasted 20 times the computing power of the UNIVAC, a computer that once filled an entire room and, when adjusted for inflation, cost approximately \$10 million! It is crucial to recognize that, without these chips, modern-day marvels such as the internet, cell phones, and laptops, as well as the existence of industry giants like Apple, Microsoft, Facebook, and Google, would not have come to fruition.

The count of transistors per chip has continued to double approximately every two years since then, a phenomenon commonly referred to as Moore's Law. By 2018, we had crossed the threshold of "seven-nanometer devices," a designation that pertains to the size of these transistors. At this minuscule scale, we can house more than 20 billion transistors on a chip no larger than a fingernail. To provide some perspective, there are 25,400,000 nanometers in a single inch, and a human hair typically spans approximately 80,000 to 100,000 nanometers in width. Consequently, a mere 12,000 transistors, embodying the computing power equivalent to over two UNIVACs, occupy the width of a human hair.

appendix B

Algorithms and programming languages

This appendix lays the foundation for understanding the concept of AI and introduces two key principles, algorithms and programming languages, which are vital components in both computer science and AI development.

B.1 Algorithms

Computer programs are a set of instructions expressed in a form that is executable by the machine. Mainly, computer programs can be classified as either operating systems or applications. Operating systems manage the computer's internal functions, while applications allow computer users to do things like play a video game or type a document. Both operating systems and applications use algorithms to define their logic and to describe the steps that need to be followed by the computer if it is to carry out particular tasks. As such, algorithms lie at the heart of computer science.

The word *algorithm* is the Latinization of the name of Persian mathematician Al-Khwarizmi, who wrote *The Compendious Book on Calculation by Completion and Balancing* between 813 and 833 ce. The only remaining copy of this historic work is kept at Oxford University, and in it, Al-Khwarizmi presents step-by-step processes for solving various types of problems. For example, he was the first to formally introduce methods like the quadratic formula for solving second-degree equations of the form $ax^2 + bx + c = 0$.

An algorithm is to the functioning of a computer what a recipe is to cooking. To create the desired result, we must define, in advance, a specific set of operations to be applied in a fixed order. An algorithm enables a computer to repeat long sequences of logical operations tirelessly and accurately as long as its logic and inputs are correct. Creating one generally involves breaking down the problem-solving process into

discrete steps. Designing an algorithm requires understanding the structure of a problem, the properties required of a solution, the possible inputs, and the desired outputs.

One approach to creating and implementing algorithms begins with using flowcharts to outline a solution's logic. Figure B.1 shows one of the flowcharts I filed in a

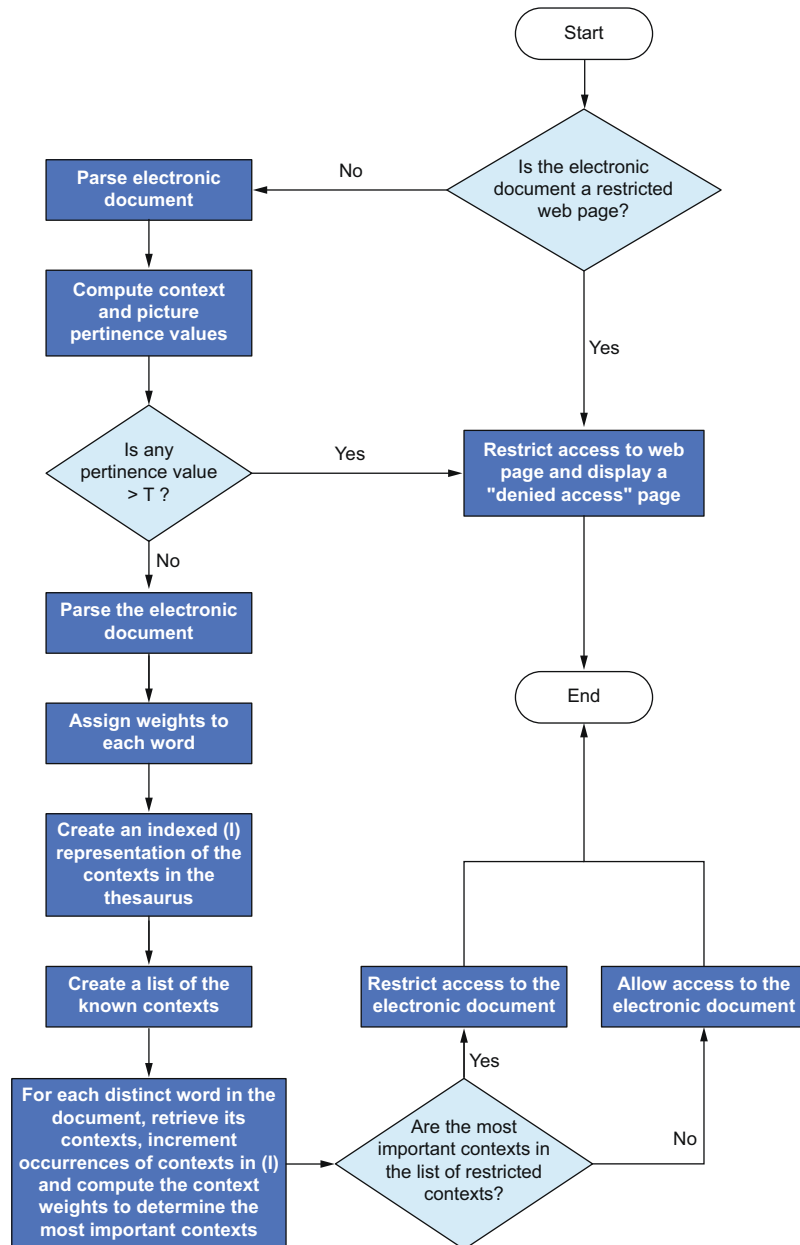


Figure B.1 An example of a flowchart that depicts the various steps within an algorithm

patent related to filtering electronic information according to a thesaurus-based contextual analysis of the content.

As we all know, the internet has revolutionized how information is disseminated and shared. At any given time, massive amounts of digital data are exchanged electronically by millions of individuals worldwide with many diverse backgrounds and personalities, including children, students, educators, businesspeople, and government officials. While this information may be easily distributed to anyone with access to a computer or the web, it may contain objectionable and offensive material not appropriate to all users. In particular, adult or violent online content may not be appropriate for children.

We will use a context-based approach to infer the context in which the words are used. The context may be based on a built-in thesaurus or other techniques. A built-in thesaurus could be a database of words and their contexts. For example, the word *apple* may have as a context the word *fruit*, *New York*, or *computer*. We evaluate the appropriateness of particular content by calculating its pertinence value toward the objectionable material.

Once a high-level process is laid out, modules can be devised to address specific tasks. Having formalized the algorithm as logic and modules, we must select the appropriate programming language to code it on a computer. Each module can be coded and tested by a different team to increase the efficiency of the implementation.

B.2 *Programming languages*

People typically communicate with other people by speaking or writing in a familiar language like English or French. Communication with computers, however, can require the use of computer programming languages. These languages involve the transfer of ideas via typed symbols, although, like our familiar languages, these symbols are combined according to grammatical, syntactic, and semantic rules. Programming languages are used to translate algorithms from human speech and writing to symbols that a computer can interpret and instructions that it can execute.

Computer programming languages are, in many ways, much simpler than human languages. For example, they use a very limited number of words and can only express concepts, such as “If this, then do this; else, do that” or “Check whether condition X holds and statement Y is true.” Expressing something in a programming language requires strictly following the rules; no deviation is permitted. Compare this strictness to English or French, which, even though they require words and sentences to be combined following syntactic and semantic rules, can be used with extreme flexibility to express an infinite variety of concepts.

In 1951, *The Preparation of Programs for an Electronic Digital Computer* by Wilkes, Wheeler, and Gill [1] became the first book ever written about computer programming, and it inspired the creation of many of the first languages. For instance, Flow-Matic, the first English-like programming language, was created under Grace Hopper’s leadership in the late 1950s for UNIVAC I. Describing a programming language

with terms such as “English-like” gives a sense of how comprehensible a program’s code might be. For example, rather than having to type explicit instructions for a computer to retrieve data from a certain part of its memory and transmit it through a particular channel, it may be preferable to type a simple instruction like “print X” to accomplish the same task.

Work to make programming languages more user-friendly led to the 1959 emergence of the Common Business-Oriented Language (COBOL). It was designed to process business data and offered a natural-language style of programming. It gave importance to data description and introduced the notion of data structures, which are special formats for storing and retrieving specific types of information. COBOL also split the structure of a general program into parts—an identification division, an environment division, a data division, and a procedure division.

COBOL’s influence in the programming world significantly increased when it became the mandatory language for U.S. Department of Defense computers. Although languages have evolved since then, COBOL remains a widely used programming language. According to Hartman [2], as of 2022, over 220 billion lines of COBOL code were still running on machines in federal government agencies and financial institutions. About 95 percent of ATM swipes are processed using COBOL code, and 80 percent of each day’s in-person transactions are handled by a program written in COBOL.

Another important programming language, Fortran, was invented in the late 1950s by John Backus and his team. Describing his motivation, Backus wrote, “When I was working on the IBM 701 (an early computer), writing programs for computing missile trajectories, I started work on a programming system to make it easier to write programs” [3]. Fortran was intended to make programming accessible, and by and large, it did.

The once lengthy, difficult, and costly task of typing thousands of program instructions for a single problem could instead be accomplished with only several dozen Fortran instructions. As a result, as soon as it was commercially released in 1957, Fortran became the first computer language standard. It greatly helped modern computing by opening the field of computer science to the general population. Like COBOL, Fortran is still in use more than 60 years after its creation.

In the field of AI, we need programming languages that can be used to model intelligent processes such as learning and reasoning. Such requirements go beyond the capabilities of languages designed to perform numerical computations, like Fortran. AI needs languages that can be used to manipulate programs and data, process and interpret symbols, and represent the characteristics of different objects and formats that arise in applications such as language processing, computer vision, and reasoning systems. To meet these requirements, John McCarthy designed Lisp in the late 1950s. He describes the objectives of the language as follows [4]:

As a programming language, LISP is characterized by the following ideas: computing with symbolic expressions rather than numbers, representation of symbolic expressions and other information by list structure in the memory of a computer.

Lisp pioneered many ideas in computer science, including tree data structures, automatic storage management, dynamic typing, conditionals, higher-order functions, recursion, etc.

A source code is a set of instructions used to write a computer program using one of these languages, and most computer languages are either compiled or interpreted. The compiled languages, such as C and C++, use another software known as a compiler to translate the source code to object code before the program can be executed. The compiler converts the source code into specific machine instructions that can be directly executed by the microprocessor (figure B.2).

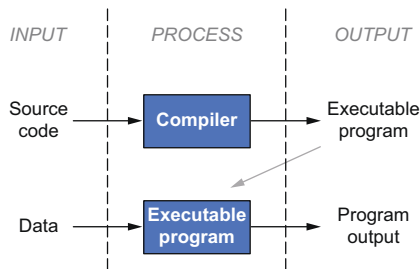


Figure B.2 Compiler implementation

The source code of interpreter languages such as Lisp can be executed directly without the need to be compiled (figure B.3). Therefore, applications can be developed much quicker. Additionally, interpreted languages are hardware agnostic as they can run on any architecture the interpreter supports. The disadvantage of an interpreter is that the same code written, for instance, in Lisp, will be an order of magnitude slower than code written using a compiled language such as C or C++. When an interpreter executes a program, it must read each line of the source code and convert it to machine code on the fly; in contrast, a compiled code has already been translated into machine code before executing the program.

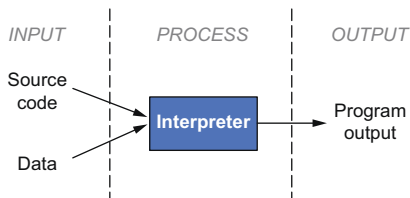


Figure B.3 Interpreter implementation

Other languages, such as Java, use a hybrid approach. A computer program is classified as portable if it can be executed on various types of computers. Even if you can compile a high-level language, such as C++, on different types of computers, the resulting object code can generally only be executed on one type of computer. A

source code written in an interpreted language can be executed on any computer as long as it has the interpreter language installed. Java is referred to as portable as the source code is translated to a virtual machine code so as not to limit the compiler-generated code to a specific processor architecture.

Lisp was the first language to introduce the concept of *garbage collection*. This term refers to an automatic process by which memory that was once used to store data for a program but is no longer needed can be freed up for other uses.

Another widely used language is BASIC, an acronym for Beginner's All-Purpose Symbolic Instruction Code. Developed in 1964 by John Kemeny and Thomas Kurtz, this straightforward language was, like Fortran, designed specifically for ease of use. Kemeny and Kurtz wanted to enable students in scientific and nonscientific fields alike to use computers in their work.

Before the introduction of such high-level languages as Fortran and BASIC, which use more conceptual, hardware-independent, human language, programming computers was rather difficult. Often, the only option available was the assembly language, which is one step away from writing a string of 1s and 0s and requires writing a statement for each instruction the computer must follow. In addition to being complex, difficult, and time-consuming, assembly code usually references specific aspects of the machine it's run on, which means a program written for one computer probably won't work on a different machine.

Although each programming language can implement most algorithms, as we just learned, languages are usually designed for a particular niche. Some are more appropriate for certain applications than others. For example, COBOL is well suited for business data processing, Fortran is preferable for scientific applications, and Lisp was designed for AI. Other languages, such as C, are more suitable for computer systems, as they contain elements that manipulate the binary information stored in the machine on which the code is being run. There are also languages such as the Structured Query Language (SQL) for manipulating databases and R for data analysis. Today, there are hundreds of programming languages.

In case you haven't seen computer code before, an example follows. The program is written in the C programming language, and running it would make the computer print "How are you?" on the screen. This text would be converted by a program called a *compiler* into assembly code, which would then be processed as instructions and executed by the computer:

```
#include <stdio.h>
int main( )
{
    printf("How are you?");
    return 0;
}
```

epilogue

As we conclude our discovery of the fascinating world of artificial intelligence, we need to reflect on the journey we have taken together. Throughout the chapters of this book, we explored the various AI techniques, tracing their evolution from the early days of expert systems to the contemporary era of deep learning. What we've discovered is that all these techniques rely on lines of code and mathematical formulations. At its core, AI is a tool using algorithms to process and analyze vast amounts of data, thereby empowering machines to detect patterns, make predictions, and execute tasks that have traditionally required human intelligence. This ability to learn from data lies at the heart of AI's power. Unfortunately, misinformation, fueled by sensationalism, distorted the public's perception, fostering unrealistic expectations and speculative fears while minimizing the remarkable ways it can enhance our lives.

AI stands as a powerful tool with the potential to significantly improve our quality of life. However, the realization of its profound impact depends on how effectively we utilize it. It is crucial to approach its deployment with a nuanced understanding of both its capabilities and limitations. By embracing a comprehensive perspective, we can unlock the true potential of AI, navigating its complexities responsibly and ethically to ensure that its benefits are maximized while potential drawbacks are mitigated.

For more than three decades, my passion for AI has been a driving force, propelling me to apply AI to real-world, mission-critical applications. In writing this book, my primary aspiration has been to demystify AI for those new to the field. It is my hope that, through these pages, you've gained the knowledge needed to distinguish between the myths surrounding AI and its tangible reality.

Joseph Weizenbaum, the creator of ELIZA, the first chatbot, seems to have avoided the common tendency to inflate the significance of one's achievements. He offered the following brilliant but sober description of AI [1]:

Machines are made to behave in wondrous ways, often sufficient to dazzle even the most experienced observer. But once a particular program is unmasked, once its inner

workings are explained in language sufficiently plain to induce understanding, its magic crumbles away; it stands revealed as a mere collection of procedures, each quite comprehensible.

Weizenbaum's characterization remains as accurate today in 2024 as it was in 1965. AI systems still have no reasoning skills, and they will not be replacing us any time soon in tasks requiring intellectual ability, physical agility, common sense, judgment, creativity, social relationships, or complex reasoning.

My years working in the field of artificial intelligence have greatly increased my appreciation of and respect for human emotion and intelligence. Most of our mental abilities are beyond our powers of introspection and comprehension. If we think of our body as our hardware and our mind as our software, we represent a technology far more advanced than anything we could hope to create ourselves.

Emerging AI technologies hold the potential to empower humanity in profound ways, transcending the boundaries of healthcare, education, engineering, agriculture, and myriad other domains. However, it's imperative to acknowledge that AI is not an omnipotent panacea. It is a potent tool, and its usage will reflect both the noble and the nefarious facets of human ingenuity.

AI's trajectory will involve military applications, surveillance mechanisms, cyber warfare, disinformation campaigns, and even oppressive uses. However, I am confident that the intelligence amplification resulting from the combination of human ingenuity and the dazzling speed of machines will increase economic prosperity by creating new opportunities for researchers and entrepreneurs to build new products, create more jobs, and apply them for the greater good.

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